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Master's Thesis

## **A Vision Transformer Model for EEG-Based Motor Imagery: enhancing generalization through Self-Supervised Pre-Training**

**Machine Learning**

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*“L’atto è l’oblio dell’azione  
nello smemoramento di sè.”*  
— C.B.

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# 1 Introduction

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A brain computer interface (BCI) is a system that decodes the user's intention and maps such information to the action of a device that interacts with the surrounding environment [1]. Possible interactions include simple actions, such as answering "yes" or "no" to a question, or more complex ones, such as moving a leg or a hand through the aid of external devices like a robotic arm. For this reason, BCIs are primarily used by individuals with neuromuscular impairments such as amyotrophic lateral sclerosis (ALS), spinal cord injury (SCI) or stroke but whose cognitive function remains intact. There are various methods available to acquire neural signals, including both non-invasive and invasive approaches: invasive BCIs provide more precise and high-resolution signals; however, they are difficult to integrate into daily life, as they require direct access to the brain. Even though they often do not penetrate neural tissue, their installation still involves complex surgical procedures. Non-invasive BCIs, on the other hand, are easier to use in daily life as they have minimal impact on the user. However, they offer limited performance because the signal must be transmitted from brain tissues, through the skull and to the scalp, this results in a limited signal-to-noise ratio which makes it more difficult to decode the signal. Non-invasive BCI technology primarily utilizes EEG recordings that measure electrical potentials from the scalp, as EEG is completely non-invasive and easy to set up, it can be employed by large studies to obtain robust statistical conclusions. The signals collected through this process are robust and widely used for solving tasks related to BCIs, such as predicting epileptic seizures in a patient or performing motor imagery.

Motor imagery (MI) is one of the most widely studied paradigms in the development of non-invasive Brain-Computer Interfaces (BCIs). It refers to the mental rehearsal of a motor action, such as moving a hand or a leg, without its actual execution. The primary objective is to decode the user's intentional state by analyzing the electroencephalographic (EEG) signals generated during the imagined movement. One of the most widely used procedures for collecting useful data is to ask a subject, after receiving an appropriate auditory or visual cue, to perform kinesthetic motor imagery, i.e., to mentally simulate a specific movement (e.g., hand or foot movement) while remaining physically still. This task elicits characteristic patterns of neural activity, notably event-related desynchronization/synchronization (ERD/ERS) in the sensori-

motor cortex, which can be captured through EEG. Using an EEG cap equipped with multiple electrodes placed according to the 10–20 international system, the BCI system records the brain activity. If the system successfully detects and classifies the EEG patterns associated with the motor imagery task, the corresponding command can be translated into control signals for external devices, such as a robotic arm or prosthetic limb, enabling the user to interact with the environment solely through neural activity.

Over time, numerous approaches have been developed for motor imagery-EEG classification, starting from traditional machine learning methods based on handcrafted features, progressing through shallow neural networks, evolving into deep learning architectures, and more recently incorporating attention-based models such as Transformers[2]. These models aim to improve classification performance by tackling key challenges inherent to MI-EEG data, including low signal-to-noise ratio, limited availability of labeled training data, and the effective utilization of spatial and temporal information across EEG channels.

This work aims to develop a neural network model based on a specific variant of Transformers: Vision Transformers[3], leveraging the capability of EEG signals to be represented as "images" through the time-frequency wavelet transform. Furthermore, the study investigates the impact on classification performance by introducing a model that processes each EEG channel individually: the EEG Multi-Channel Vision Transformer. Finally, the model's ability to enhance performance will be evaluated by exploiting neural networks with a proposed pre-training strategy on EEG signals that do not only relate to the motor imagery task, but represent any type of neural activity of a subject, with the goal of moving toward a more generalizable BCI.

In the remainder of the **chapter 1**, the problem of decoding EEG signals for motor imagery tasks will be defined within the context of computer engineering, highlighting the main challenges that affect performance. **Chapter 2** will present a comprehensive review of the state of the art, with a particular focus on recent Transformer-based models. To support this, a dedicated section will introduce Transformers and Vision Transformers, clarifying the rationale behind the proposed model. Additionally, an analysis of domains related to motor imagery, such as the audio domain, will be provided to emphasize the advantages of representing signals through time-frequency transforms. **Chapter 3** will formally introduce the EEG Multi-Channel Vision Transformer model designed for MI classification, discussing how this architecture can benefit EEG signal processing in motor imagery tasks. Furthermore, the curriculum masked spectrogram pre-train will be presented, a novel pre-training method for Vision Transformers that aims to address the lack of inductive bias. Finally, **chapter 4** will present and discuss the experimental results.

## 1.1 Formal definition of the problem

MI refers to the mental rehearsal of a motor action, such as moving a hand or a leg, without its actual execution, as explained in the introduction. At its core, the problem can be formulated as follows: given an electroencephalographic (EEG) signal recorded from a patient performing a motor imagery task, the goal is to pass this signal through a dedicated decoding system such that the output accurately identifies the specific task the subject was asked to imagine. However, this process is far from straightforward. Since non-invasive brain–computer interfaces (BCIs) are used, the signal is often degraded. As EEG signals travel from their source in the brain to the scalp surface through the low-conductive skull, they encounter various forms of noise caused by both biological and non-biological factors [1]. Also, oscillatory EEG data, such as those related to motor imagery or emotion, lack stationary patterns. Moreover, as is common in this type of application, the amount of data available for training and analysis is limited. This is due to the fact that data recording sessions are particularly demanding: they require a high level of concentration from the subject, and since high-quality data is essential, the sessions must be kept relatively short (30-60 minutes). An additional challenge arises from the strong subject-specific nature of EEG signals. The neurophysiological patterns associated with MI varies significantly from one individual to another, making it extremely difficult to extract consistent, generalizable features that would be useful for decoding motor imagery tasks.

## 1.2 Relevance of the problem in the context of computer engineering

To effectively address the challenges outlined above, it is essential to design a robust EEG decoding pipeline tailored to motor imagery (MI) tasks. This pipeline typically includes the following stages: signal preprocessing, feature extraction, classification, and application-level interfacing, such as controlling a robotic arm or prosthetic device. Each stage plays a critical role and must be carefully designed. In particular, signal preprocessing is fundamental to mitigate the inherently low signal-to-noise ratio (SNR) of EEG signals: brain signals are attenuated and distorted as they traverse the skull and scalp, this leads to contamination by both endogenous (e.g., eye blinks, muscle activity) and exogenous (e.g., environmental electromagnetic interference) noise sources. Therefore, selecting appropriate bandpass filters is crucial for isolating frequency components relevant to motor imagery, typically within the  $\theta$  (4–8 Hz),  $\alpha$  (8–13 Hz) and  $\beta$  (13–30 Hz) bands, while attenuating noise-related components. As will be further detailed in **Chapter 2**, the most commonly used classification systems include traditional machine learning algorithms and, more recently, deep neural networks. that offer the significant advantage of being capable of learning effective signal representations directly from raw input. However, neural networks, especially architectures based on Transformers, as will be discussed, require large amounts of training data. Given the limited data availability in MI-BCI scenarios, it becomes necessary to either make the most of the available data through efficient modeling strategies or apply data augmentation techniques. A neural network can be fed with raw EEG signals, which typically consist of a multichannel time series where the channels correspond to the electrodes used in signal acquisition. The strategy adopted for handling the spatial dimension of the signal namely, the selection and integration of the most informative channels for the motor imagery task, is a crucial design choice in the construction of the model’s input representation; this aspect plays a pivotal role in the performance of EEG-based decoding systems and will be examined in depth in the following chapters of this thesis.

Still, given the scarcity of data, transforming the signal into a more informative representation can enhance classification performance: time-frequency transforms, such as the Short-Time Fourier Transform (STFT) or the wavelet transform, can help highlight discriminative features relevant to the task. Techniques such as data augmentation, transfer learning, and domain adaptation become essential to address data scarcity: transfer learning strategies, including fine-tuning pretrained models or adopting few-shot learning paradigms, can leverage prior knowledge learned from the same or related domains (e.g., audio or time-series classification) to improve generalization on MI-EEG tasks. These approaches aim to bridge the gap caused by inter-subject variability, one of the most prominent challenges in EEG decoding, as individual brain dynamics differ significantly across users. Lastly, while deep models with large parameter counts may pose deployment challenges in real-world BCI systems, ongoing advances in em-

bedded hardware and model compression techniques (e.g., pruning, quantization, knowledge distillation) are making it increasingly feasible to embed complex architectures into wearable or portable neurotechnology platforms.

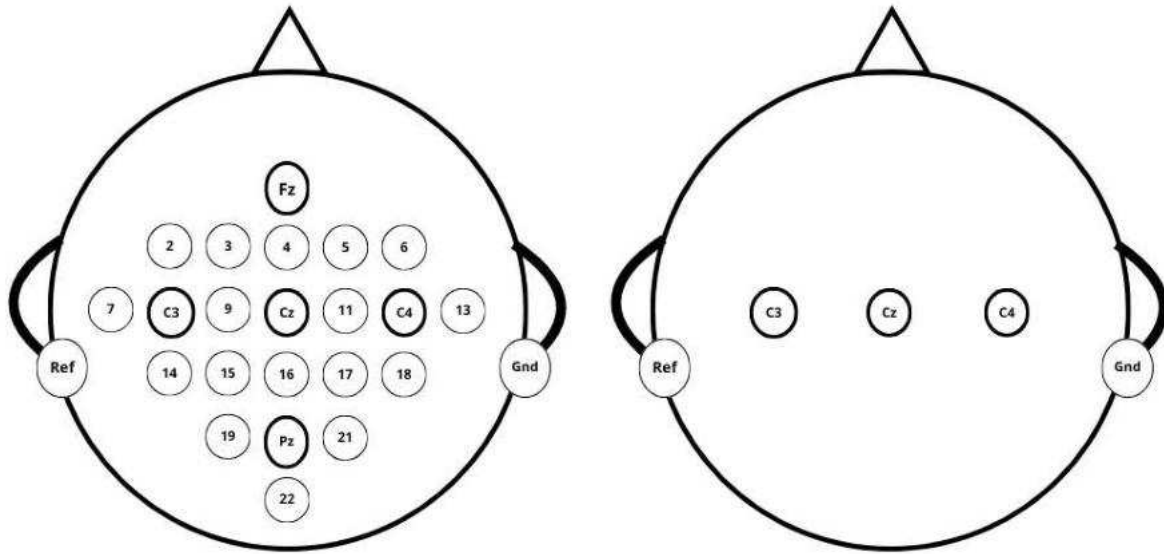


Figure 1.1: Electrodes used to collect the EEG signal, with each electrode corresponding to one signal channel. The figure on the left shows a cap with 22 electrodes and thus 22 channels, while the figure on the right shows a cap with 3 electrodes corresponding to channels C3, Cz, and C4.

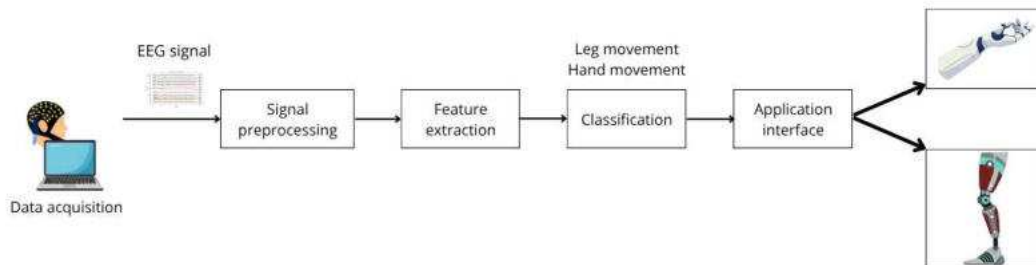


Figure 1.2: This is the system pipeline: the signal is first collected through specific tasks across different recording sessions. Subsequently, it undergoes preprocessing to optimally extract features through a neural network or a machine learning algorithm. Once the features are obtained, classification is performed, and the result is used to move a robotic limb or to control an external device that interacts with the environment.

## 2 State of the art

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This chapter presents the state of the art concerning the motor imagery (MI) task. It will cover EEG signal preprocessing methods, the most widely used neural network architectures, the most common evaluation metrics, and the rationale behind these choices. Subsequently, the architectures of Transformers and Vision Transformers will be described in detail to clarify the motivations behind the architecture proposed in this thesis. Similar domains to motor imagery, particularly the audio domain, will then be analyzed to justify the use of time-frequency transforms on EEG signals and to provide insights into the potential generalization of the task. Finally, considerations on possible advancements to the current state of the art will be discussed, some of which will be applied in the following chapters.

### 2.1 Input transformations

Decoding EEG signals, particularly for motor imagery tasks, is a complex challenge due to several factors: EEG signals are oscillatory and often contaminated by noise; the data acquisition process is laborious, resulting in limited available datasets; prolonged recording sessions may lead to signal degradation due to participant fatigue; and models are difficult to generalize, as EEG data vary significantly across individuals, as previously discussed in earlier chapters. To address these challenges and enhance model performance, various preprocessing strategies can be applied. The quality of the resulting classification depends on the formulation of the input, which can generally be categorized into three types: raw data, transformations that preserve the spatial topology of electrode placement, and signal representations that emphasize temporal, spatial, or time-frequency components (such as Fourier Transform, Short-Time Fourier Transform (STFT), or Wavelet Transform).

Raw data are often used because they do not require additional processing steps. Typically, the input is represented as a 2D matrix where rows correspond to EEG channels and columns represent time points.

Transformations that preserve spatial topology aim to address a limitation of raw data: they ignore the spatial relationships between EEG channels. This type of input can be represented as

a sequence of 2D segments that follow the spatial distribution of EEG electrodes, for example, according to the electrode layout of the acquisition cap. However, this representation is not widely adopted, as constructing the appropriate input structure is time-consuming and requires prior knowledge of the electrode arrangement on the EEG cap. These constraints also limit the generalizability of the approach.

Time-frequency component extraction transformations are among the most widely used, as CNNs or Vision Transformers perform very well on data structured similarly to images. Moreover, this approach is not as time-consuming as preserving spatial topology and since it mainly originates from the audio signal domain, there already exist demonstrations of its performance, making it worth considering for adaptation to the EEG signal domain. However, the main challenge arises when processing EEG channels, because applying the transformation to each individual channel requires careful design of the model’s input structure. For this reason, it is also necessary to choose which transformation to apply in order to make the most of the signal’s characteristics: among time–frequency transformation methods, a Wavelet Transform or a Short-Time Fourier Transform may be applied; Common Spatial Patterns are suitable when focusing on spatial components, whereas the Power Spectral Density or the Fast Fourier Transform can be adopted to extract temporal components. According to the survey on deep learning methods for motor imagery [4], among the proposed models, preprocessing methods that extract features through time-frequency transformations are the most popular, followed by the use of raw data, and lastly by transformations that preserve the spatial topology.

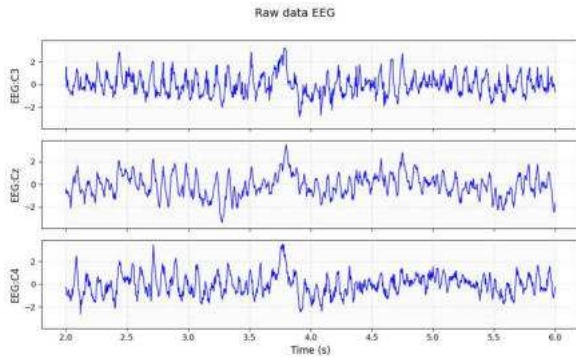


Figure 2.1: Raw EEG signal with 3 channels.

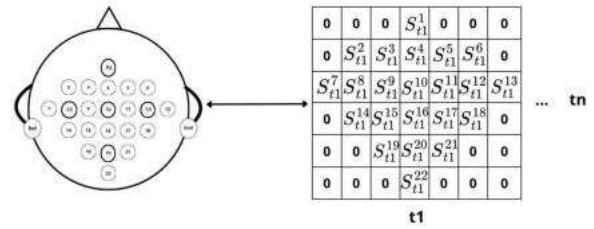


Figure 2.2: Spatial topology preserved representation of EEG signal with 22 channels.

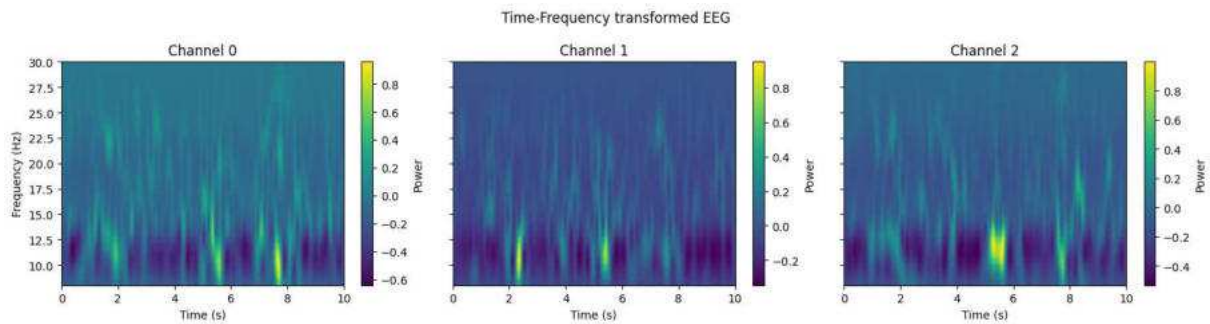


Figure 2.3: Time-frequency transform of EEG signal with 3 channels.

## 2.2 Normalization and filtering

Another important preprocessing step is data normalization. One of the most widely used techniques is Z-normalization, which is particularly useful given the high variability of EEG signals across different recording sessions in the dataset.

$$x_{norm} = \frac{x - \mu}{\sigma}$$

Where  $x$  is the input of a model,  $\mu$  and  $\sigma$  represent the mean and the standard deviation calculated over  $x$  respectively.

Since EEG signals can contain noise caused by ocular or muscular activity, as well as electromagnetic interferences, the input is typically filtered using band-pass filters to isolate the information related to motor imagery, usually contained within the  $\theta$  (4–8 Hz),  $\alpha$  (8–13 Hz) and  $\beta$  (13–30 Hz) frequency bands. For instance the 6th-order Type II Chebyshev filter, used in [5], filters the signal in the [4, 40] Hz range, and the 5th-order Butterworth IIR filter, used in [6] or [7], operates in the [8, 30] Hz frequency band.

## 2.3 Datasets

A system designed to address a motor imagery task is typically trained on EEG datasets that vary in terms of the channels considered, the number of classes, and the number of subjects involved. The most widely used datasets include:

- **BCI Competition IV 2a** that includes EEG signals collected from 9 subjects, each performing 4 distinct motor imagery tasks: imagining the movement of the left hand, right hand, both feet, and the tongue. Each subject participated in two separate recording sessions on different days, resulting in 288 trials per session. EEG signals were recorded using 22 electrodes at a sampling rate of 250 Hz, and each signal has a duration of 7 seconds [8].
- **BCI Competition IV 2b** that includes EEG data from 9 subjects. There are two motor imagery tasks, covering the imagination of moving left and right hand. Five sessions were collected with three bipolar electrodes (C3, Cz, and C4), 3 channels at a sampling rate of 250 Hz and each session contained 120 trials [8].
- **PhysioNet** comprises EEG recordings from 109 healthy volunteers. In each trial, a visual target appears on either the left or right side of a display. The participant then mentally rehearses opening and closing the corresponding hand until the target disappears, and subsequently relaxes. Data were acquired using the BCI2000 system with 64 EEG channels, sampled at 160 Hz.

The most commonly used datasets, which are also the ones primarily considered in this thesis, are BCI Competition IV 2a and BCI Competition IV 2b with the latter being preferred because of its larger data availability. Physionet is less frequently employed, as despite having many subjects, it offers fewer data samples and more channels, leading to higher inter-subject variability and making the implementation more challenging. Training can follow two main approaches: intra-subject or cross-subject with leave one subject out, depending on whether the goal is to generalize across patients. Intra-subject training refers to a hold-out procedure in which a separate model is built for each subject, and the recording sessions are split between training and testing. For example, in BCI Competition IV 2a, nine models are trained, each using the first session for training and the second for testing. For cross-subject generalization, the most common approach is leave one subject out where the model is trained using data from every subject except the one on which it is tested, in order to analyze its generalization capability. In this case, datasets like PhysioNet are also used, as they contain more subjects.

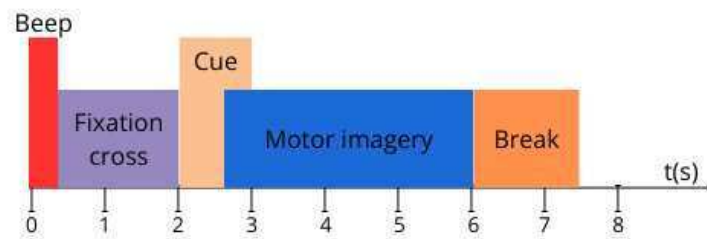


Figure 2.4: Timing scheme of the BCI Competition IV 2a.

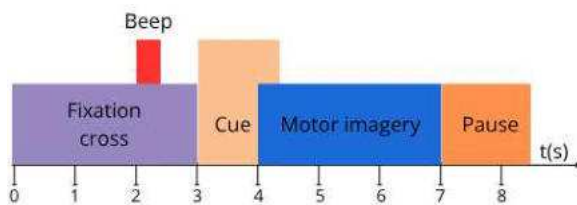


Figure 2.5: Timing scheme of the BCI Competition IV 2b, first two sessions without feedback.

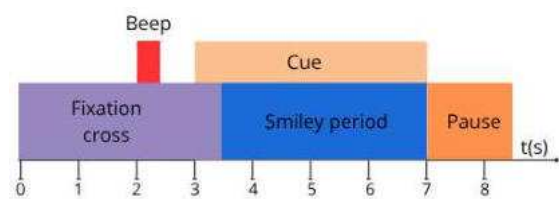


Figure 2.6: Timing scheme of the BCI Competition IV 2b, last three sessions with smiley feedback.

## 2.4 Detailed Analysis of the State of the Art

In this section, the literature on motor imagery models will be reviewed, starting with early approaches based on handcrafted features and moving toward models employing CNN and LSTM neural networks. The analysis will then shift to more modern, state-of-the-art models primarily based on Transformer architectures. Finally, the focus will be on models that attempt to solve the motor imagery task using Vision Transformer architectures which are much less common than standalone Transformer or hybrid CNN+Transformer architectures, with the goal of gaining a deeper understanding of the patch embedding layer and the structure of the EEG signal used as input, as this represents the central theme of the thesis. To this end, the mechanisms of both the Transformer and the Vision Transformer will be described in detail within this section.

### 2.4.1 From handcrafted feature to CNN-based models

Initially, the most widely used methods relied on the extraction of handcrafted features, mainly based on Common Spatial Pattern (CSP), in which features are extracted per class by minimizing the variance of the other classes through the construction of spatial filters to extract spatially discriminative features. Later, methods derived from CSP were proposed, following the same principle but improving feature extraction, such as FBCSP [9] and R-CSP [10] that automatically select subject-specific frequency bands. The first classification algorithms were then introduced: Support Vector Machines (SVM), Linear Discriminant Analysis (LDA) and their combinations. These methods relied on the combination of neurophysiological knowledge and machine learning, but they were still based on handcrafted features, which made them complex since they required domain-specific knowledge and time-consuming feature extraction, which was also performed prior to the classification process. Moreover, the presence of noise in EEG signals greatly limited performance, as did the difficulty of traditional ML algorithms in dealing with nonlinear processes such as those involved in Motor Imagery. Thus, the extraction of handcrafted features is replaced by neural networks. The first neural network models to achieve good performance on the motor imagery task are Shallow ConvNet2017 [11] and DeepConvNet2017 [11], which achieved good results on the BCI Competition IV 2a dataset by leveraging convolution and deep learning, although they focus only on local components of the signal due to their convolutional nature. Shallow ConvNet first performs a temporal convolution and a spatial convolution to simulate the bandpass and CSP filters used in FBCSP. Next, a series of operations like squaring nonlinear transformation or mean pooling are adopted to simulate other steps in FBCSP. In 2018, EEGNet [12] was proposed, a compact CNN architecture that employs depth-wise and separable convolution for EEG-specific tasks. In this network, an efficient extraction of frequency-specific spatial filters is made possible by depthwise convolution, which offers a direct method of learning spatial filters for each temporal filter. The main architectures include single-stream CNNs, multi-stream CNNs, LSTMs, hybrid CNN-LSTM

combinations, and multitask networks. CNNs are certainly the most widely used, but LSTMs are also employed due to their ability to capture temporal dependencies in EEG signals. A 2023 survey on MI Classification [4], comparing various models, highlights some key findings. The performance, evaluated using accuracy, F1 score, AUC, and Cohen’s Kappa, shows that a multi-stream architecture does not outperform a single-stream model. LSTM models capture temporal correlations but not spatial ones, which are important for EEG channels. A fully connected layer increases the number of parameters but does not improve performance (except if in the classification layer), and dropout layers, in this specific context, do not enhance performance either. Following EEGNet, the C-LSTM model [13] achieved excellent performance, reaching an accuracy of 69% on 2a by adding an LSTM network to also capture long-range temporal dependencies. This model highlights how important temporal dependencies are in EEG signals, and how performance can be improved by combining them with an architecture like a CNN, which captures local features. However, the accuracy is still not optimal: the previously mentioned issues remain, and although the addition of an LSTM improves temporal analysis, it is still not sufficient.

## 2.4.2 Transformers

The Transformer has become the foundation of every neural network architecture. In the following, its advantages and disadvantages will be presented, together with its working mechanism, with reference to the original paper [2], in order to clarify the motivation for its use in BCI architectures for the motor imagery task, both on its own and in combination with convolutional networks. The Transformer was introduced to address the need for parallelizing recurrent networks. It exploits the attention mechanism, which had already been implemented and used in the same recurrent networks, while proposing a new and simpler architecture that relies exclusively on it. Originally designed for the translation task, this explains its diffusion mainly in natural language processing, although today it is employed in all types of neural network models. On GPUs, the number of computational operations required for attention is an issue. For ConvS2S [14], the number of operations needed to compute attention between two distant points grows linearly with the position, while for ByteNet [15] it grows logarithmically. One of the advantages of the Transformer, instead, is that this number remains constant, since it computes a weighted average influenced by the resolution and balanced by MultiHeadAttention. The architecture of a Transformer is divided into an encoder, a decoder, and a classification head.

**Encoder:** the encoder consists of  $N$  identical layers, which are 6 in the original paper. Each layer is composed of two sub-layers: a feedforward layer and a multi-head attention layer. Residual connections are implemented around each sub-layer, followed by layer normalization:

$$\text{LayerNorm}(x + \text{Sublayer}(x))$$

The outputs have the same dimension:

$$d_{\text{model}} = 512$$

**Decoder:** the decoder has the same structure as the encoder with  $N = 6$ , but a Masked Multi-Head Attention is added so that the prediction at position  $i$  can only depend on previous positions. The multi-head attention, instead, depends on the outputs of the encoder.

**Attention:** this is the fundamental mechanism of the Transformer, which maps a set of queries, keys, and values to an output. The output is a weighted sum of the values, where the weights assigned to each value are computed by a compatibility function between the corresponding query and key. The input consists of queries and keys of dimension  $d_k$  and values of dimension  $d_v$ , while  $d_{\text{model}}$  is the embedding dimension of the model:

$$\text{Attention}(Q, K, V) = \text{softmax} \left( \frac{QK^T}{\sqrt{d_k}} \right) V$$

It can be seen as a set of weights that are multiplied by  $V$ . The  $1/\sqrt{d_k}$  term is the addition compared to additive attention. This type of attention is much faster than additive attention, but for large values of  $d_k$  it results in small gradients; for this reason, it is scaled by  $d_k$ .

**Multi-Head Attention:** The attention mechanism is repeated  $h$  times, concatenating the final results and projecting them linearly. This allows the model to capture different representations of the subspace at different positions. In the original paper,  $h = 8$ ,  $d_k = d_v = d_{\text{model}}/h = 64$ , so that the computational cost is equivalent to a single full-dimension head.

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_h) W^O$$

where

$$\text{head}_i = \text{Attention}(QW_i^Q, KW_i^K, VW_i^V)$$

and the projections are parameter matrices

$$W_i^Q \in \mathbb{R}^{d_{\text{model}} \times d_k}, \quad W_i^K \in \mathbb{R}^{d_{\text{model}} \times d_k}, \quad W_i^V \in \mathbb{R}^{d_{\text{model}} \times d_v}, \quad W^O \in \mathbb{R}^{hd_v \times d_{\text{model}}}.$$

In the model, there are three different types of attention:

- **Encoder-Decoder Attention:** Queries come from the decoder, while keys and values come from the encoder, allowing each element of the decoder to attend to elements of the original sequence.
- **Encoder Self-Attention:** Standard self-attention applied within the encoder.

- **Decoder Self-Attention:** Similar to encoder self-attention, but future values are set to  $-\infty$  to preserve the autoregressive property.

In addition to the attention layers, there are fully connected feed-forward layers:

$$\text{FFN}(x) = \max(0, xW_1 + b_1)W_2 + b_2$$

**Positional Encoding:** it is introduced due to the absence of convolutions, so that the model can recognize the order in a sequence. Consequently, the relative or absolute position of the token is encoded. It has the same dimension as  $d_{\text{model}}$  so that it can be added to the input embeddings. In the original paper, sine and cosine functions at different frequencies are used. They compare it with learned positional embeddings and find no significant differences. They still use the cosine formulation because they believe the model can extrapolate sequences longer than those seen during training and generalize better.

$$\text{PE}(\text{pos}, 2i) = \sin\left(\frac{\text{pos}}{10000^{2i/d_{\text{model}}}}\right), \quad \text{PE}(\text{pos}, 2i+1) = \cos\left(\frac{\text{pos}}{10000^{2i/d_{\text{model}}}}\right)$$

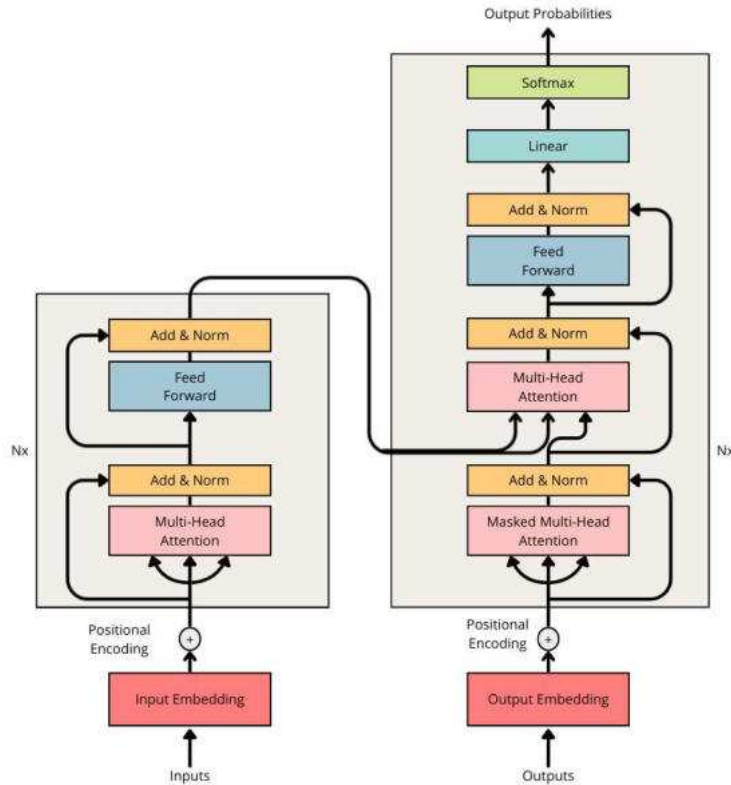


Figure 2.7: Transformer architecture

To motivate the use of self-attention, they compare: the total complexity per layer, the number of computations that can be parallelized, and the path length between long-range dependencies in the network. From the computational complexities, it is shown that a Transformer is much more efficient than an RNN or a convolutional network. As a result, the Transformer is the first model to use attention and can be trained much faster than models based on RNNs or CNNs.

### 2.4.3 State of the art models based on Transformers

The most recent models are based on the attention mechanism, or more precisely, on Transformers. Transformers, as explained in **subsection 2.4.2**, allow capturing global long-term dependencies through the attention mechanism. This improves performance on the MI-task and enables them to replace LSTM networks, as they do not suffer from the vanishing gradient problem. Based on the attention mechanism, TSFF-Net [16] is a network composed of four main components: a time-frequency representation module, spectral feature extraction layers, temporal feature extraction layers, and a module for feature fusion and classification. For pre-processing, a 200th-order Blackman bandpass filter is applied to the raw data to keep the signal within the [4, 38] Hz range. Each trial is normalized to the [-1, 1] range, rather than normalizing per channel, as the study primarily focuses on the analysis of the three channels C3, C4, and Cz. This approach avoids biases due to the characteristics of individual channels. Finally, Euclidean alignment is applied to create a representation invariant between the training and test data. The method aims to construct a covariance matrix aligned across the N trials of both train and test sets, reducing the discrepancy between the datasets. TSFF-img is the component of the network handling time-frequency feature extraction and it is composed of 3 feature extraction layers, each including regularization operations. The time-space feature extraction layer is divided into four modules: channel attention, temporal convolutional layer, depth attention mechanism, and spatial convolutional layer. This architecture is commonly adopted in state-of-the-art models. The feature fusion layer, on the other hand, consists of two main components: weighted fusion of the output features from both branches and alignment in a high-dimensional space using Maximum Mean Discrepancy (MMD) loss, assuming that even when working with time-frequency and time-space features, if represented in higher-dimensional spaces they may have common elements.

$$\mathcal{D}_k^2(p, q) := \left\| \mathbf{E}_p[\phi(\mathbf{S})] - \mathbf{E}_q[\phi(\mathbf{F})] \right\|_{\mathcal{H}_k}^2$$

Where  $p$  and  $q$  are the distributions of the time-frequency and time-space features respectively. The two outputs are normalized with a softmax before applying the MMD loss. The fully connected layer takes as input a weighted average of the outputs from both branches, and the

cross-entropy is computed on this result. The final loss function is a weighted loss:

$$L = L_c + \lambda \cdot L_{\text{MMD}}$$

Where  $L_c$  is the cross-entropy loss,  $L_{\text{MMD}}$  is the MMD loss, and  $\lambda$  is a hyperparameter that balances the two losses. For dataset BCI Competition IV 2b, classification results are: TSFF-img ( $78.5 \pm 13.5\%$ ), TSFF-Net ( $86.4 \pm 11.8\%$ ). For Quadruple BCI Competition IV 2a classification with 3 channels: TSFF-img ( $58.3 \pm 13.5\%$ ), TSFF-Net ( $65.2 \pm 13.6\%$ ).

In 2023, the EEG-Conformer [5] was proposed, which successfully exploits the attention component of Transformers in combination with convolutional networks. The architecture is quite simple and consists of three main modules: a convolutional module that takes the raw 2D EEG signal as input (preprocessed with a 6th-order Chebyshev band-pass filter and Z-normalization), along with a pooling layer to prevent overfitting. The resulting representation is then passed to a multi-head self-attention module to extract long-term dependencies, and finally to a classifier with two fully connected layers. The convolutional module applies 1D convolutions along both the spatial and temporal dimensions. The model was tested on the BCI Competition IV 2a dataset, using the first session for training and the second for testing (standard procedure), and employed a data augmentation method based on segmentation and reconstruction, which will be further discussed. The EEG-Conformer achieves an accuracy of 78.66%, outperforming previous state-of-the-art models such as FBCSP (67.75%) [9] and EEGNet (74.50%) [12] on BCI Competition IV 2a, demonstrating the effectiveness of Transformers for EEG signal decoding. In fact, removing the attention component leads to an average performance drop of 6% across subjects, highlighting the importance of global feature extraction. This model exemplifies not only the importance of using Transformers for EEG decoding but also the effectiveness of a very common architecture among state-of-the-art models: the combination of CNNs and Transformers. This is because while Transformers can compute global dependencies, they typically require a large amount of data to learn local patterns. Hence, the so-called inductive bias is compensated by the introduction of convolutional layers. Since the introduction of the EEG-Conformer, most recent networks have adopted either a Transformer or an attention mechanism, such as EEGVTTCNET [17]. This model features an architecture in which the signal is first encoded using convolutional blocks inspired by EEGNet, applying both temporal and spatial convolutions. The architecture then splits into two separate branches. The first branch, called the TCN Residual Block, encodes the signal with the goal of focusing only on current and past inputs, avoiding future information. To achieve this, it uses causal convolutions, for instance, instead of applying a kernel over  $x_t, x_{t-1}$ , it might use  $x_t, x_{t-2}$ , and so on. In particular, it employs dilated convolutions to expand the receptive field exponentially. The second branch consists of a Vision Transformer with overlapping windows and 1D convolution for patch embedding. Both branches continue to a classification stage, and the final loss is computed as a cross-entropy loss, resulting from the weighted combination of the losses from both branches, controlled by a

lambda factor. By adjusting lambda, it is possible to give more or less importance to either the TCN or the ViT branch, allowing the model to balance between convolutional and attention-based features. The objective is to combine the global feature extraction of the Transformer with the temporal modeling of the TCN, starting from a shared feature representation. Accuracy analysis shows that the ViT-only branch performs worse than the full model, while the TCN-only branch achieves even lower performance. This demonstrates both the effectiveness of the hybrid architecture and the importance of incorporating Transformers for EEG decoding. MI-MBFT [18] is based on an architecture divided into three main modules: the first module, called IFE, consists of convolutional blocks combined with an attention mechanism, the second module, MFB, includes four parallel patch embedding blocks, each followed by a Transformer aimed at extracting different frequency components of the signal, this is followed by a fusion attention block and, finally, a classification head. The first module applies two convolutions, one spatial and one temporal, equivalent to a 5 Hz high-pass filter, these are followed by global average pooling and a squeeze-and-excitation attention mechanism, here, an average is computed for each channel, producing a vector  $z$  of size  $C$  that is multiplied by weight matrices, the resulting vector is then applied to the original sample, effectively assigning different importance weights to each channel, which helps to reduce the influence of less relevant channels. The second module applies convolutions with various kernel sizes, followed by four parallel patch embeddings, each processed by a Transformer encoder, based on the original 250-Hz sampling frequency, these temporal convolutions allow the network to focus on specific frequency bands around 25, 12.5, 6.25, and 3.125 Hz, respectively, these correspond to the alpha and lower beta bands of somatosensory rhythms, which are known to be rich in motor imagery information. After concatenating the features, they are passed to a classification layer, the loss function combines the losses from each individual branch along with the loss from the fused features, this model aims to exploit feature extraction with kernels of different sizes, in order to assess the contribution of each important frequency band for the motor imagery task. The accuracy of individual branches is lower than that of the combined branches, which indicates that feature fusion has a beneficial effect in this case. Overall, this architecture improves upon simple CNN+Transformer models, employing convolutions with varying kernel sizes and multiple branches to fully capture both spatial and temporal features at different frequencies.

2StageTransformer [19] employs a two-stage architecture. In the first stage, it uses a parallel convolution based on EEGNet, consisting of three parallel temporal convolutional blocks, each designed to extract information from different frequency bands: the theta band (4–8 Hz) using a kernel size  $K_{10} = 64$ , the alpha band (8–13 Hz) with  $K_{11} = 32$ , and the beta band (13–30 Hz) with  $K_{12} = 16$ , totaling three temporal 2D CNN layers. Here,  $K$  represents the kernel size. The resulting features are then passed through an additional convolutional block followed by a time slicing operation, which slices the features along the temporal dimension with a stride of 1 to learn local temporal relationships. These features are fed into a Multihead Attention block to capture global dependencies. Next, a Separable Temporal Convolution (STCN) block is applied

to capture long-range temporal correlations. This block leverages dilated causal convolutions, depthwise convolution, and pointwise convolution (two separate 1D convolutions). Residual connections link the two dilated convolution layers by adding the input of the first dilated convolution directly to its output, and similarly for the second. Finally, the outputs are averaged and passed through a softmax layer. In the second stage, the Short-Time Fourier Transform (STFT) is applied to the raw features to estimate the time-frequency distribution, thereby enhancing patterns in specific frequency bands relevant to Motor Imagery. This transformed representation is then concatenated with the output from the first stage to improve classification performance. The concatenated features are fed into a TabNet classifier, which encodes data in forms similar to images and text, utilizing a sequential attention mechanism composed of multiple attention layers in sequence. The architecture consists of an encoder and a decoder: the encoder includes a feature Transformer, an attentive Transformer, and a masking module. The masking module selects important features and masks those already processed by the Transformers. The attentive Transformer performs multiple decision steps to guide the masking process. The decoder attempts to mimic a decision tree, with the collection of choices at each decision step deriving the final features for classification. This complex architecture is designed to address the scarcity of data in EEG-MI tasks and to reduce overfitting, even at the cost of increased computational complexity, by combining handcrafted features with deep learning. Incorporating the attention mechanism leads to a 3% performance improvement when the number of heads is set to 2; increasing heads beyond that does not yield further gains.

Regarding subject-dependent performance, EEGVTTCNET achieves an accuracy of 84.58% on BCI Competition IV 2a and 90.9% on 2b. MI-MBFT reaches 86.93% accuracy on 2a, but it has not been tested on 2b. The 2StageTransformer scores 88.5% on the 2a dataset and 88.3% on 2b. For subject-independent performance, EEGVTTCNET obtains 71.21% on 2a, while 2StageTransformer achieves 72.9% on 2a and 77.4% on 2b. These values show significant advancements in the state of the art over the past two years, highlighting the continuous evolution of the Motor Imagery task field using EEG signals. From this review of the state of the art, it emerges that the most widely used models for MI tasks exploit Transformers [2], often in combination with CNNs. However, analyzing domains that use EEG signals for tasks like seizure prediction or emotion recognition reveals that Vision Transformers [3] are becoming increasingly popular. This is because they leverage the EEG signal's potential to be represented with an image-like structure, making them potentially better suited to capture both local and global components. Regarding the MI task, this approach has not yet been extensively explored. Therefore, in the next section, some Vision Transformer models used both for the motor imagery task and for similar tasks will be analyzed to highlight the advantages that could contribute to advancing the state of the art.

## 2.4.4 Vision Transformers

Transformers have become the reference models for Natural Language Processing tasks, the typical approach is to pre-train on a large corpus and then fine-tune on the specific task. However, in computer vision, convolutional architectures remain dominant. Vision Transformers (ViT) aim to create an architecture that completely replaces the use of convolutional networks. Nevertheless, unlike CNNs, Vision Transformers suffer from a lack of inductive bias, such as translation equivariance and locality, and therefore do not generalize well when trained on insufficient amounts of data. The goal of this thesis is precisely to leverage a Vision Transformer model; therefore, citing the problem of inductive bias is necessary for making the appropriate considerations. In this section, the general model will be illustrated following the scheme of the original paper [3]. The first applications of self-attention to images required each pixel to attend to every other pixel, with a quadratic cost relative to the number of pixels. Naturally, this is not feasible as it does not scale well with typical image sizes. A more recent approach, based on Vision Transformers and not relying on CNNs, applies attention to  $2 \times 2$  patches of the image. The Vision Transformer performs a similar task, but its goal is to work with images at a larger scale. The process is straightforward: The image  $x \in \mathbb{R}^{H \times W \times C}$  is divided into patches, i.e., pieces of the image of a fixed size ( $16 \times 16$  in the original work). The patches are then flattened to obtain  $x_p \in \mathbb{R}^{N \times (P^2 \cdot C)}$ , where  $P \times P$  is the resolution of each patch and  $N = HW/P^2$  is the resulting number of patches.

Next, a trainable linear projection is used to map each patch to a dimension  $D$  (latent vector size), which serves as input to the Transformer Encoder. The output of this projection is called the *patch embedding*.

A standard learnable 1D positional embedding and a classification token [class], similar to BERT, are added to the patches. The sequence of flattened patches becomes very similar to the token sequence in a Natural Language Processing task. These are fed into a Transformer Encoder as described in [2], and finally classified with an MLP head consisting of two layers and a GELU activation. During fine-tuning, the pre-trained prediction head is typically removed and replaced with a  $D \times K$  feedforward layer, where  $K$  is the number of classes for the downstream task. Since fine-tuning can be performed on images of larger size than those used in pre-training, the patch embeddings must remain the same, which results in a longer sequence. For this reason, a 2D interpolation of the positional embeddings must be performed; otherwise, the positional information might no longer be meaningful. This is the only point where an inductive bias is introduced. In the original paper [3], the performance of ViT models pre-trained on JFT-300M and ImageNet-21k is compared. The model pre-trained on JFT outperforms the state-of-the-art models considered, although the model pre-trained on the smaller dataset also achieves excellent results. However, this not only highlights the strong performance of ViT, but also shows that pre-training on a larger dataset leads to better results, especially when considering a larger-scale ViT. An interesting test was conducted on smaller portions of the JFT-300M

dataset. In this case, it can be observed that ViT models overfit faster than ResNet models and perform poorly on subsets of 9M samples, but better on subsets of 90M+ samples. This is further evidence that the inductive bias of CNNs is useful for small datasets. These results are sufficient to explain why Vision Transformers are rare among motor imagery models, and when present, they are always accompanied by a CNN backbone for feature extraction. However, it raises the question of whether an alternative Vision Transformer model could work, or whether an alternative representation of the input signal could compensate for the problems caused by the lack of inductive bias, thus allowing the advantages of Vision Transformers to be exploited.

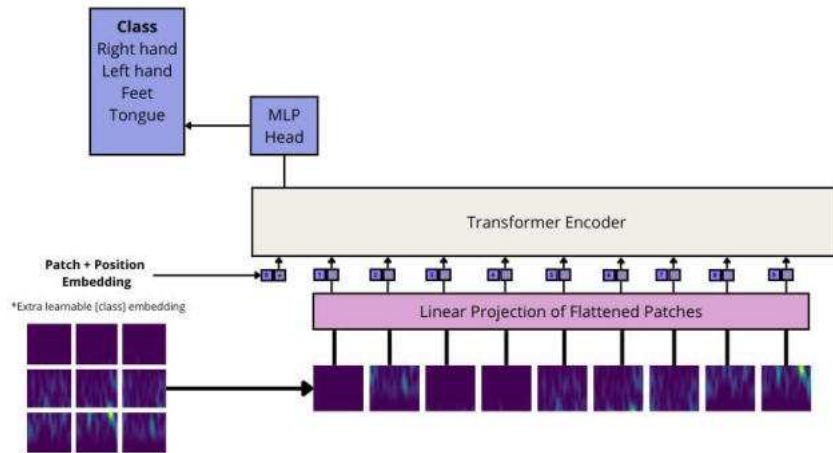


Figure 2.8: Vision Transformer architecture. Here the image is the spectrogram of an EEG signal, which is divided into nine patches. A positional embedding is applied to these patches, and they are then fed into the encoder of a Transformer, which extracts features for classification. In the image the classes are the same as those of the BCI Competition IV 2a dataset.

## 2.4.5 Vision Transformers in motor imagery

As mentioned in **subsection 2.4.4**, Vision Transformers represent an excellent architecture for working with EEG signals, since such signals can take a form similar to that of an image, and consequently a ViT could enhance their characteristics through patch embedding and the attention mechanism. However, they suffer from the lack of inductive bias, meaning they require large amounts of data to achieve good performance: a CNN exhibits inductive bias since convolutions act on local areas of the image, enabling it to detect important patterns like edges or textures that can be leveraged to create more complex features for generalization. Additionally, the same convolutional filter is applied throughout the image, recognizing repeated patterns, such as same objects in different positions. In this sense, the convolution process, being much more local than Transformer attention, helps achieve better generalization with limited data. This limitation makes ViT difficult to train for motor imagery tasks, since the available datasets refer to individual patients and are therefore scarce. For instance, the previously mentioned

dataset in **section 2.3**, BCI Competition IV 2a, contains only 288 trials per subject. In the original model, the scarcity of data is compensated by large-scale pre-training, while in subsequent models it is mitigated by adding a convolutional network for feature pre-extraction. This drawback explains why pure Vision Transformer models are rarely found in the motor imagery literature and, when present, are always accompanied by CNNs. In this section, three models that employ a Vision Transformer to solve the motor imagery task will be presented, all of which achieve excellent performance. One of them, EEGVTTCNET [17], has already been described in the previous section. The reasons behind the strong performance of these models will be investigated, in order to understand how a Vision Transformer should be used to obtain good performance on the motor imagery task under study.

MSVTNet [7] is the first model that will be presented: it employs a Vision Transformer to solve the motor imagery task. It is based on multi-scale filters in order to capture more accurate global and local features, since the ERD/ERS frequency bands that best discriminate between different classes vary from subject to subject. Using a network with multi-scale convolutions helps to cover a wider range of frequencies. The architecture consists of three main blocks:

- **MSST**: multi-scale spatio-temporal convolution blocks,
- **CSGT**: the Transformer block, designed to capture global correlations,
- **ABL**: an auxiliary loss block.

The MSST consists of  $B$  parallel branches at different scales (very similar to EEGNet). Each branch passes through  $F_b$  filters with kernel size  $(1, k_b)$  to perform a form of temporal filtering. Subsequently, in order to extract spatial features, the signals are passed through a convolutional layer with kernel size  $(C, 1)$ . The new features then go through an average pooling layer of size  $(1, 8)$ , which downsamples the signal from 250 Hz to 32 Hz. This is followed by another convolutional layer of size  $(1, 15)$ , which represents 500 ms of EEG activity. Finally, a new average pooling layer of size  $(1, 7)$  is applied. In total, the number of branches is 4. The features extracted by the MSST are concatenated along the feature dimension and then passed as input to a Transformer encoder, after adding learnable positional encodings and a classification token. The ABL block follows the MSST block. Its role is to compute the loss of each individual branch, whose sum defines a loss  $L_a$ , which provides auxiliary supervision to the loss of the [CLS] token at the output of the Transformer encoder.

$$L = \lambda L_c + (1 - \lambda) L_a$$

The subject-dependent analysis is divided into session-dependent and session-independent. In the session-dependent case, session 1 is used for both training and testing. In the session-independent case, session 1 is used for training and session 2 for testing. For evaluation, a 5-fold cross-validation was applied, where the decoding accuracy of each subject was computed as the

average over the 5 folds. For the subject-independent analysis, a leave one subject out strategy was employed. Regarding subject-dependent (intra-subject) accuracy, the session-independent setting achieves 78.50% with a standard deviation of 10.86 on 2a and 86.70% with a standard deviation of 8.35 on 2b. For the cross-subject setting, the accuracy is 64.10% with a standard deviation of 14.47 on 2a and 76.45% with a standard deviation of 7.00 on 2b. The ablation study also shows that removing the CSGT block results in a performance drop of about 5%, demonstrating the importance of the Transformer encoder in processing the extracted features. In addition, S&R is used as a data augmentation method; without it, the accuracy decreases by about 3%. Moreover, using too many branches leads to learning redundant spatio-temporal features and thus overfitting, while too few branches would ignore important information. The parameter  $\lambda$  is set to 0.6, since setting it to 1 (i.e., completely removing the ABL) causes a performance drop of about 2%. This model does not represent a “pure” ViT; however, it shows that the use of different branches, which apply filters very similar to those of patch embedding, can indeed have a beneficial effect on performance, as it allows the model to separately process and exploit the frequency bands that are most relevant for the motor imagery task.

The second model that will be presented is CWT+MVIT [20]: this model focuses on the use of scalograms to address the problems of inter- and cross-subject classification. The architecture is very simple, as it relies on a Vision Transformer modified at certain levels. The considered wavelet transforms are Morlet, Mexican Hat, Shannon, Frequency B-spline, and Gaussian. The model is based on the assumption that by combining scalograms from different wavelets, more precise and easily distinguishable features can be generated, thus facilitating the classification task. The architecture follows that of a standard ViT, but with a modification: the output of the first MLP is fed into the addition layer (residual connection) of each subsequent Transformer encoder. In this way, the patches follow a parallel as well as sequential flow. The preprocessing pipeline includes a 4th-order Butterworth band-pass filter between 8 and 30 Hz, detrending, signal selection in the interval 2.4-5.6 s for each trial, min-max normalization, and a Savitzky-Golay filter for data smoothing. The combination of wavelets that achieves the best accuracy is Morlet + Shannon, with 83.77% (intra-subject) and 76.19% (cross-subject) on the BCI Competition IV 2b dataset. Compared to using only Morlet, this results in an accuracy improvement of 3%, and compared to Shannon alone, the improvement is 9%. Looking at the other results, it can be observed that every set of scalograms consisting of more than one wavelet but fewer than five achieves better performance in terms of accuracy and standard deviation. This model is not significant for its architectural innovations, but rather because it is one of the few that employs a Vision Transformer in combination with a wavelet transform without relying on a convolutional network. The results demonstrate that the model is able to achieve good performance, moreover, the data come from the BCI Competition IV 2b dataset without any form of augmentation.

## 2.5 Analysis of motor imagery related domains

One of the main challenges in developing a BCI for motor imagery is that this field has only begun to grow significantly in recent years. The limited amount of data and the inherent difficulty of working with EEG signals have posed major obstacles. As a result, the number of available models, especially for innovative architectures such as Vision Transformers, remains limited. Nevertheless, EEG signals are also widely used in more established domains, such as epileptic seizure detection, and when represented as spectrograms they can even be regarded as similar to audio signals. These similarities make it possible to draw insights from architectures that perform well in other domains, which could in turn be adapted to achieve strong performance in motor imagery. The domain of epileptic seizures is very similar to that of the motor imagery task. To detect whether a patient is about to experience a seizure, EEG signals are also used, and consequently all the previously discussed challenges re-emerge. Research on epileptic seizures is more extensive than that on motor imagery, both due to the relative ease of data collection and the pressing need to develop systems that can provide real assistance to patients. As a result, a wide variety of architectures has been explored. Given the similarities with motor imagery, these architectures could indeed prove useful for motor imagery classification as well. In this section, we will present the MVIT architecture [21], which achieves excellent performance in epileptic seizure prediction, and subsequently discuss its potential use for the motor imagery task. The audio signal domain is particularly interesting when EEG signals are expressed through a time-frequency transform and represented as spectrograms. As mentioned earlier, this allows EEG signals to be treated as "images" and thus benefit from computer vision models. However, one may also ask whether a model could be trained via transfer learning starting from audio signals. By leveraging the network's ability to interpret spectrograms in a general way and then adapting it to the specific task, it may be possible to improve performance in motor imagery classification. For this reason, in this section we will present the AST model [22] for audio signals and subsequently discuss the potential for the motor imagery classification task.

The paper "Multi-Channel Vision Transformer for Epileptic Seizure Prediction" [21] will be of fundamental importance for this work, which, as stated in the introductory sections, aims to investigate the relevance of EEG channels while simultaneously exploring the most effective signal representation for the motor imagery task. The architecture presented in this paper has not yet been applied to motor imagery tasks, yet it has achieved excellent performance in epileptic seizure recognition. A closely related variant, presented in "Deep learning based automatic seizure prediction with EEG time-frequency representation" [6], achieves 97.57% accuracy on the CHB-MIT dataset (a benchmark for seizure prediction), highlighting the potential of this model. The CHB-MIT dataset is very similar to the BCI Competition IV 2a dataset: it contains EEG signals recorded from 23 channels (22 in BCI Competition IV 2a) at a sampling frequency of 256 Hz (250 Hz in BCI Competition IV 2a), collected from multiple patients. The proposed

architecture is inspired by the success of the ViT [3] and is composed of multiple branches, each operating on a different channel. The preprocessing phase includes segmentation, whose goal is to divide the non-stationary EEG signal into small pseudo-stationary EEG chunks. This procedure also serves as a form of data augmentation. In particular, each 10-minute EEG clip is divided into 60 non-overlapping segments, each with a length of 10 seconds. Subsequently, scalograms are obtained from these signal segments using the Wavelet transform. This step is particularly useful, as it has been shown that methods relying on hand-crafted temporal or spectral EEG features often result in a high number of false positives. Moreover, even visually, spectrograms clearly reveal differences between the preictal and interictal phases that precede epileptic seizures. This is advantageous because it introduces an inductive bias into the network already at the early stages, thereby facilitating classification. At the end of preprocessing, the data have a shape of  $60M \times N \times h \times w$ , where  $60M$  is the new number of samples after augmentation. For each individual channel, the data are passed through a patch embedding layer. Once the patches are obtained and projected to a latent dimension  $D$ , they are fed into a Transformer encoder. Finally, the features output by each encoder for each branch are aggregated and used as input to an MLP for classification. In summary, the architecture consists of one branch per channel, each composed of its corresponding patch embedding layer and Transformer encoder, followed by a classification head applied to the concatenated features from all branches. MVIT was trained and tested on the CHB-MIT dataset using leave one subject out cross-validation, demonstrating the model’s ability to generalize by achieving 99.8% accuracy, at the cost of high computational complexity. This raises the question of whether this approach could be equally effective for the motor imagery task. As previously mentioned, the dataset considered is very similar to BCI Competition IV 2a, which partially addresses concerns about the computational complexity of the architecture. If the model works well with 23 channels for seizure classification, it could potentially perform effectively with 22 channels or fewer for motor imagery. Moreover, this method also provides insights into the use of EEG channels when applying a time-frequency transform to the signal. These considerations form the basis for further discussion in the following chapters.

AST [22] is a well-known Vision Transformer model in the field of audio signal classification. The study starts from the premise that, similar to the motor imagery task, most models combine attention with CNNs; however, it remains unclear whether the CNN is actually necessary, or if a purely attention-based network is sufficient. Consequently, the authors aim to answer this question by introducing a pure ViT model without convolution, and evaluating it on three different audio datasets. A key strength of AST is its ability to handle inputs of varying dimensions, allowing it to be adapted to different audio tasks without the need to retune parameters. This is made possible in part by the absence of a CNN. It also enables knowledge transfer from pre-trained ViTs to the AST ViT, bypassing the time-consuming and critical process of pretraining Vision Transformers. Moreover, it allows for training in a very small number of epochs. The architecture follows the standard ViT [3], with  $16 \times 16$  patches, but with an overlap of 6 in both

time and frequency. This is because the input images are spectrograms computed using a 128-dimensional Mel filterbank. The Transformer encoder layers use an embedding dimension of 768, with 12 layers and 12 heads, to facilitate fine-tuning. The difference between a standard ViT and AST is that ViT operates on 3-channel images, while AST works with 1-channel spectrograms. To adapt a standard ViT, the weights corresponding to the three channels of the patch embedding layer are averaged and used for the AST patch embedding layer. This is equivalent to replicating the single channel three times. Moreover, although a ViT naturally supports inputs of varying dimensions and can therefore handle spectrograms that are not always  $224 \times 224$ , the positional embedding must be adapted if pretraining is required. AST therefore performs a cut and bi-linear interpolation: if AST has patches of size  $12 \times 100$  while ViT uses  $24 \times 24$ , the first dimension of the ViT is cropped and the second dimension is interpolated. Finally, the classification layer is reinitialized. Analyzing the model on AudioSet, it can be observed that AST outperforms models that combine CNN and attention. Specifically, AST achieves a mean Average Precision (mAP) of 0.486, compared to 0.474 for the best CNN+Attention model. Moreover, AST was trained in only 5 epochs using an ensemble procedure. It is interesting to note the performance difference between the model trained from scratch and the one pre-trained on ImageNet. The first achieves a mean Average Precision (mAP) of 0.366, while the latter reaches 0.459, demonstrating the beneficial effect of pretraining. Moreover, a model with the positional embedding fully reinitialized achieves 0.305 mAP, whereas using bilinear interpolation increases the mAP to 0.347, representing a significant performance improvement. It is also noted that using rectangular patches leads to better performance; however, no pre-trained ViT models currently employ this approach. The performance of this model not only demonstrates the beneficial effect of pretraining, but also shows that a ViT can be effective even without the addition of a CNN, as consistently observed so far. If this finding also holds for the motor imagery task, it would represent a breakthrough for the field, which has so far relied exclusively on models combining a convolutional network with a Transformer.

## 2.6 Possible pretraining strategies for Vision Transformers

The inductive bias, as previously noted, plays an important role in the motor imagery task, especially considering the size of the BCI Competition IV 2a and 2b datasets. For this reason, when considering pretraining, a supervised pretrain is excluded since the limited amount of labeled EEG data for the motor imagery task would not be sufficient. As already mentioned in **section 1.2**, subjects need to be trained to ensure a meaningful signal for the task, and the acquisition times are very long. Consequently, the best solution is a self-supervised pretrain, allowing the model to benefit not only from EEG data related to motor imagery but also from EEG data of patients of all types. In this way, the model will learn to understand the structure of the data rather than the specifics of a single task. In Chapter 3 **section 3.4**, the pretraining method used in this work will be introduced. However, it is only one of the possible valid ap-

proaches for pretraining a Vision Transformer. One such method was already mentioned when discussing AST [22], which adapts the weights of a Vision Transformer pretrained on images and then incrementally pretrains it on audio signals. While this can improve performance, it is unclear how much knowledge actually transfers between such distinct domains as images and audio. Therefore, incremental pretraining (pretraining first on images, then on the audio domain) might be required to achieve effective domain transfer. For data in spectrogram form, possible pretraining methods include contrastive loss and masked patch. Since a masked patch method will be applied, it is interesting to examine a model implementing this approach.

MaskSpec [23] aims to improve the performance of self-supervised methods, drawing inspiration from the success of masked autoencoders (MAE). The method masks a certain percentage of patches in the input, which are removed from the encoder. In this way, the pretrained model gains the ability to understand the time-frequency structure of the spectrogram. It is pretrained on AudioSet and evaluated on five downstream tasks. Although the model operates on audio spectrograms, since the method is task-independent, it can be applied to any other task using spectrograms as input. In MaskSpec, the spectrogram is chosen because it can be used directly as input without extra convolutional layers. As mentioned previously, this is necessary to understand the effect of the inductive bias and to support the thesis that a Transformer model, pretrained with an appropriate amount of data, performs better than a CNN. Inspired by Masked Language Modeling (MLM), it uses the same random masking strategy: given a spectrogram  $T$  in  $\mathbb{R}^{n_t \times n_f}$ , a  $p \times p$  sliding window with the same hop size is applied to obtain the patches  $E = \{e_1, \dots, e_n\}$ , where  $n$  is the number of patches and  $N = n \times \alpha$  is the number of masked patches.  $\alpha$  is the masking ratio, with  $\alpha \in [0.05, 0.95]$ . The MaskSpec model consists of 12 Transformer blocks and a learnable linear projection. In this case, a decoder is also added to the main architecture. The decoder has 8 Transformer layers and a linear projection layer. The output of the final FFN must be converted to obtain the masked patches, each with dimension  $p \times p$ . Using the position index of the masked patches, a learnable vector is placed in the masked regions of the encoder output, and the patches are reassembled to match the original number of patches. Before feeding them to the decoder, information about the absolute position of the tokens in the sequence is injected. The loss is computed as the difference between the reconstructed patches and the original ones. Comparing the models reveals that traditional pre-training methods, like ImageNet, increase performance over randomly initialized weights, while the MaskSpec approach provides an additional boost. mAP improves from 0.354 with random weights to 0.397 using ImageNet pre-training, and reaches 0.403 with MaskSpec self-supervised training. These results show that a neural network, through this type of task, can understand the structure of the spectrogram and leverage this knowledge to improve classification.

## 2.7 Identification of potential advancements over the state of the art

Following the analysis of the state of the art, it can be observed that the models remain around an accuracy of 85% on BCI Competition IV 2a dataset and just slightly higher on the 2b dataset. Among those examined, the 2StageTransformer [19] performs best, reaching an accuracy of 88.5% on 2a and 88.3% on 2b. In the previous section, however, it was shown that models related to epileptic seizure detection [21] achieve higher performance when working on the same type of signal but for a different task. This raises the question of what causes such a disparity in performance. The difficulties related to the motor imagery classification task have already been extensively discussed in the previous sections like 1.1 and 1.2: some of these include the limited amount of available data and the non-stationarity of the signal. From the analysis of the models, however, an additional problem emerged: the lack of inductive bias during the training of deep neural networks such as Transformers. This is a natural consequence of the scarcity of data, and therefore makes it necessary to include a convolutional neural network in every model that employs a Transformer. It has been observed that in the presented models this combination appears in different ways: from the concatenation of features to a weighted loss between the Transformer and the convolutional network [18] [17]. The CNN is necessary because the Transformer can capture global correlations, however, with limited data it struggles to compute local relationships (this is precisely the so-called inductive bias), managing to emphasize components that are temporally distant while neglecting the spatial ones. The spatial components, in an EEG signal, refer to the channels which correspond to the electrodes: therefore, it is worth investigating whether working directly on the channels could lead to improved performance.

Moreover, in the analyzed architectures, which often include a CNN followed by a Transformer, no specific representation is used for the input data, or rather the raw data are employed. Using an appropriate data representation, such as a time-frequency transform, could provide an implicit bias to the network that may facilitate the classification task. Furthermore, by applying a transform to the input EEG signal, one can leverage properties from domains that use similar time-frequency transforms, such as audio signals, and consequently mitigate the problem of data scarcity in applications like motor imagery through transfer learning. This would allow the EEG signal to have the same structure as an image, benefit from models and techniques associated with computer vision, and simplify the use of architectures such as Vision Transformers without relying on a CNN upstream.

Another question that arises when looking at the Vision Transformer models analyzed so far concerns the patch embedding layer. To improve performance, it is necessary to understand the actual contribution of this layer, and whether using a specific patch division or employing a neural network composed of different patch embedding layers provides tangible benefits to the model. With different divisions of the signal according to frequency bands [19][7], it has been

observed that performance can increase significantly. It is therefore worth investigating whether such a division could be performed directly at the patch embedding layer.

Furthermore looking at Natural Language Processing models, it is evident that pretraining is a fundamental component for improving task performance. However, due to the lack of data, as well as the complexity of pretraining a hybrid architecture, pretraining is very rare in the motor imagery task. In addition to the lack of data, there is a problem that has already been clearly discussed in the previous chapters: the data distributions vary from patient to patient, meaning that EEG signals, as stated by several neurologists, differ depending on the individual. Consequently, the benefits that a potential pretraining could bring remain uncertain, since the model would be trained on a large dataset coming from distributions different from the one it will actually operate on, and this might not result in any performance improvement; however, pretraining a neural network could provide an interesting insight into this theory. It is therefore worth questioning whether pretraining could improve the model's performance, but above all whether there is actually a difference in performance between a pretrained model and a model trained from scratch. The Vision Transformer architecture is therefore particularly interesting for the classification of EEG signals in the motor imagery task, as previously seen in the models that employ it, providing a significant increase in performance. Answering these questions could further promote the development of BCIs for motor imagery; consequently, in the next chapters, these issues will be investigated by presenting various Vision Transformer models and a new pretrain method for the motor imagery task.

### 3 Original contribution to the solution of the problem

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In this chapter, all the topics discussed above will be further explored. In particular, the main model at the core of this work, namely EEG-MVIT, will be presented in detail, together with its preprocessing and training pipeline, especially on the already mentioned datasets BCI Competition IV 2a and BCI Competition IV 2b. Special attention will be given to the formulation of the input data in the form of a specific spectrogram: the scalogram, obtained through the Continuous Wavelet Transform, which allows for a clear separation between the channels of the EEG signal and makes the input more suitable for a Vision Transformer, which is typically designed to work on images. Subsequently, the architectures used for comparison with the EEG-MVIT model will be presented in order to understand the reasons behind its performance. The analysis starts with how the features obtained from the patch embedding layer are combined, then considers how the channels are concatenated at the input, and finally focuses on the patch embedding layer itself. This will enable an investigation of the presence of a CNN in a Vision Transformer model in relation to the amount of training data, and consequently, the impact of inductive bias, which is an advantage when the amount of data is limited but a significant drawback for generalization. Finally, aiming to address this limitation, and therefore to assess whether a Vision Transformer model can indeed bring benefits to tasks, overcoming the lack of inductive bias, a self supervised pre-training methodology will be presented, leveraging the ability to operate on patches of EEG-MVIT, through the use of independent branches, with the aim of understanding the structure of the EEG scalogram to improve performance. The pretrain is carried out on a dataset containing unlabeled EEG signals from different subjects, not limited to the motor imagery task. The results of this pre-training will make it possible to understand whether EEG signals, and consequently the signals from different subjects, follow distributions that share common features, and thus whether it is feasible to build a general model capable of achieving good performance while also being suitable for practical use in medical contexts. Moreover, if the self supervised pre-training performed on this dataset is successful, it will be possible to carry out pre-training on a larger and more general EEG signal dataset.

### 3.1 Definition of the proposed methodology

From the analysis of the state of the art, it has been observed that most models employ an architecture that combines convolutional networks and Transformers, motivated by the presence of inductive bias, which is not easily compensated by the limited amount of data typically available in motor imagery datasets. Hence, understanding the separate contributions of the convolutional network and the Transformer to performance is necessary. The reason a CNN is placed upstream of a Transformer, aside from the inductive bias, is that it can transform the EEG input signal into a sequence that facilitates token extraction for the Transformer, making the process similar to extracting tokens from text for natural language processing applications. However, when using only the Transformer, this is not possible. For this reason, a Vision Transformer model [3] is chosen, which can analyze the EEG signal by dividing it into patches that reflect the spatial and temporal structure, as explained in **subsection 2.4.4**. It is essential to determine whether the signal, decoded by a ViT, provides benefits to the task compared to a convolutional network, to proceed with potential pre-training. For this purpose, the signal needs to be preprocessed to support effective network learning.

#### 3.1.1 Continuous wavelet transform (CWT)

The raw EEG signal consists of a spatial component, composed of the electrodes or channels used to record the signal, and a temporal component over which the signal extends:

$$X \in \mathbb{R}^{C \times T}$$

where  $C$  denotes the channels and  $T$  the time points.

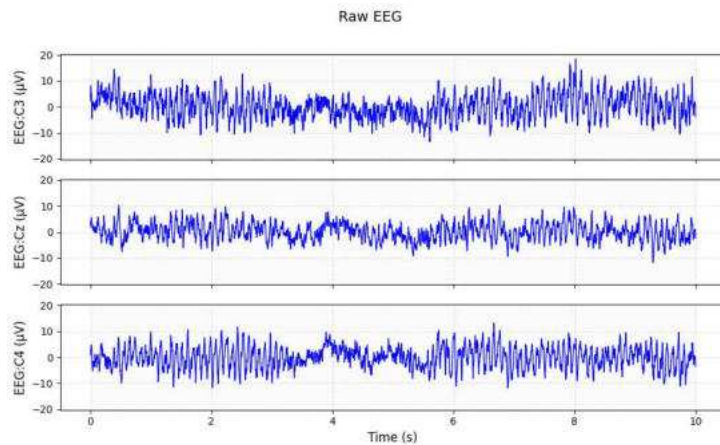


Figure 3.1: Raw signal extracted from the BCI Competition IV 2b dataset; the y-axis shows the amplitudes of the three channels, namely C3, Cz, and C4, while the x-axis represents the temporal direction.

EEG channels, unlike RGB channels in an image that provide depth as a third dimension, cannot directly serve as input for a Vision Transformer. Although it is possible to append a dummy

third dimension, this solution is not valid, as the added dimension does not convey any informative content. For this reason, a time-frequency transform should be applied to the EEG signal to enhance its features.

The MEL transform is not suitable for EEG signals, as it is mainly used for audio signals due to its scaling, which reflects human auditory perception. In practice, the frequency axis of a spectrogram is remapped onto the Mel scale to give greater weight to the frequency bands that are more perceptible to humans. This approach is widely used for audio signals, as explained in **section 2.5** and shown in [22]. However, it is not appropriate for EEG signals, since it applies a logarithmic mapping to high frequencies in the range [20–20000] Hz, which is meaningless for the  $\theta$  (4–8 Hz),  $\alpha$  (8–13 Hz) and  $\beta$  (13–30 Hz) bands used in motor imagery EEG signals. These frequencies are not only much lower than those of audio signals, but such a mapping would also risk losing information relevant to the task.

The Short-Time Fourier Transform (STFT) could be appropriate; however, careful selection of parameters such as the window type, size, and stride is necessary. Moreover, the sliding window over the signal is fixed, so the resulting spectrogram will inevitably suffer from the time-frequency trade-off.

Precisely for this reason, the Wavelet transform is chosen. Here, a Continuous Wavelet Transform (CWT) is applied to transform the signal into a scalogram.

The Wavelet transform decomposes the signal into waves that are scaled in frequency and shifted in time, making the window dynamic rather than fixed as in an STFT. This way, the time-frequency trade-off becomes less restrictive.

$$CWT_x(a, b) = \frac{1}{\sqrt{a}} \int_{-\infty}^{+\infty} x(t) \psi^* \left( \frac{t-b}{a} \right) dt$$

- In this case, the chosen mother wavelet ( $\psi$ ) is the Morlet wavelet, as it is the most commonly used in the literature for EEG signals, for instance, it is employed in [6] and in other models that operate on EEG signals represented as spectrograms, as discussed in **subsection 2.4.5**. Other transforms, such as the Mexican Hat or Shannon wavelet, are rarely applied to EEG.
- $a$  represents the scale: increasing  $a$  elongates the wavelet, making it better suited to capture low frequencies that are less localized in time. Decreasing  $a$  shortens the wavelet, making it more suitable to capture high frequencies that are more localized in time. Since the ultimate goal is to obtain a scalogram, the signal is convolved with the mother wavelet at different scales, channel by channel. This enhances both high- and low-frequency components, mitigating the time-frequency trade-off.
- $b$  is the temporal translation.
- $\psi$  denotes the complex conjugate of the mother wavelet.

Continuous wavelet transform (CWT) is particularly suitable for non stationary signals, like EEGs used in motor imagery tasks, and is frequently employed for denoising. While the discrete wavelet transform (DWT) offers higher computational efficiency, the available amount of data makes CWT feasible. Moreover, this transform facilitates the analysis of ERD/ERS by emphasizing power fluctuations in the alpha and beta frequency bands during motor imagery.

- ERD (Event-Related Desynchronization) is observed in the alpha and beta bands during the execution or imagination of a movement, and reflects a desynchronization of neurons, meaning that they stop operating together and become independent.
- ERS (Event-Related Synchronization) is observed in the beta band after an actual or imagined movement, and reflects a re-synchronization of neurons that return to oscillating together, as if returning to a resting state.

ERD and ERS play a crucial role in BCIs and motor imagery tasks because they allow a clear distinction of real movements and are expected to support the detection of imagined movements as well. This makes the wavelet scalogram transform especially valuable, as it emphasizes the power frequency bands in which ERD and ERS occur. In the spectrogram, ERD manifests as a reduction of power in the alpha and beta bands, whereas ERS is visible as an increase in power in the beta band. A simple Fourier Transform (FFT) could reveal the frequencies present, but not when they occur, and such temporal information is crucial for a classification task.

When the CWT is applied to each channel, the original signal  $X \in \mathbb{R}^{C \times T}$  is converted into a scalogram, which takes on the structure of an image  $X' \in \mathbb{R}^{C \times F \times T}$ .

Here, the signal was compared with a wavelet of size  $32 \times 1008$  for the BCI Competition IV 2a dataset and  $32 \times 1000$  for the BCI Competition IV 2b dataset. The number of bins was set to 32, ensuring that the time–frequency ratio is not overly distorted while providing a frequency-axis dimension suitable for standard patch sizes (e.g., 8 or 16). The wavelet cycles, defining how far the wavelet extends in frequency, were initially set in proportion to the frequency:

$$n_{\text{cycles}} = \frac{\text{freqs}}{2}$$

This approach helps manage the time–frequency trade-off, resulting in fewer cycles at lower frequencies and more cycles at higher frequencies. However, after several experiments, the parameter was fixed as follows:

$$n_{\text{cycles}} = 7$$

since this value yielded better performance. Moreover, the same choice is consistently adopted in several models from the literature. This transform was applied to all the models presented in this work. Next, the signal power was computed as the squared modulus of the complex coefficient, and converted from a linear to a logarithmic scale by applying the power logarithm. This transformation was required since spectrogram values in linear scale may span several orders

of magnitude, potentially destabilizing training and producing excessively large gradients. To address this, a z-score normalization was then applied to the spectrogram, standardizing it to zero mean and unit variance. The resulting input is therefore an EEG scalogram  $\mathbb{R}^{C \times F \times T}$  where  $C$  is the number of channels,  $F$  the number of frequency bins, and  $T$  the number of time points:

- $22 \times 32 \times 1008$ , for BCI Competition IV 2a;
- $3 \times 32 \times 1000$ , for BCI Competition IV 2b.

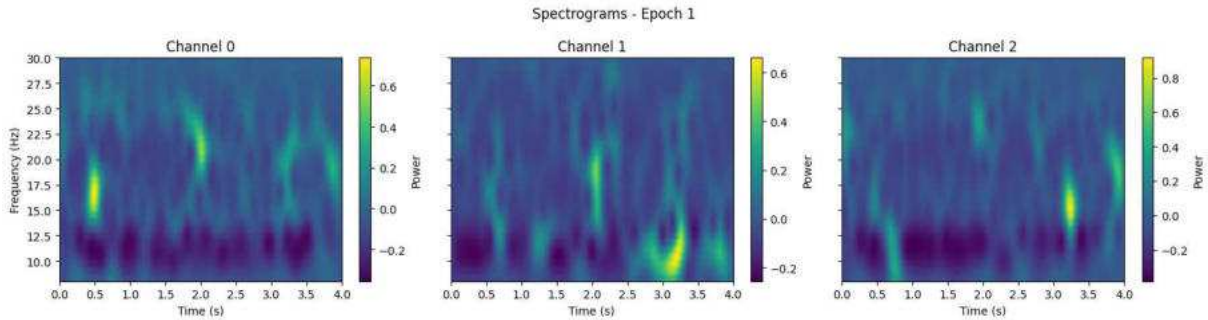


Figure 3.2: Scalograms generated from the CWT of subject 1 from the BCI Competition IV 2b dataset. The figure shows the three channels after applying the CWT.

### 3.1.2 Preprocessing on raw data

Before converting the raw EEG into a scalogram, a preprocessing stage is applied. Such preprocessing is crucial when dealing with small-scale datasets, and becomes even more important in the absence of inductive bias. The operations performed are consistent with state-of-the-art approaches and match those used in previously described models, applied uniformly to both the BCI Competition IV 2a and 2b datasets. The first step involves removing the EOG channels that capture eye movements during task recording, since they are considered noise and must be excluded. Subsequently, a 4-second segment of the signal is selected:  $[2, 6]$  seconds for dataset 2a and  $[3, 7]$  seconds for dataset 2b, in accordance with the dataset documentation [8] or **section 2.3**. This segment corresponds to the period where the motor imagery task is actually performed after the cue, and where ERD activity is typically observed in the spectrogram. With a sampling rate of 250 Hz, the temporal axis contains about 1000 samples.

### 3.1.3 Frequency filtering and normalization

The signal is filtered using a 5th-order Butterworth IIR filter to maintain frequencies within the  $[8, 30]$  Hz range, since the mu (8–12 Hz) and beta (13–30 Hz) bands contain the most informative content for motor imagery tasks. This filtering also removes involuntary movements, such as eye movements, which often appear at higher frequencies and are irrelevant to the task. Furthermore, given the limited amount of data, a priori filtering can guide the network to focus

on the correct frequency bands. Another option is a 6th-order type II Chebyshev filter, restricting the signal to [4,40] Hz, thus including theta and gamma frequency components as well, as explained in **section 2.2**. The Butterworth filter was chosen as it is the most commonly used in the literature, particularly in ViT-based models with CWT as inputs such as [6] and [7].

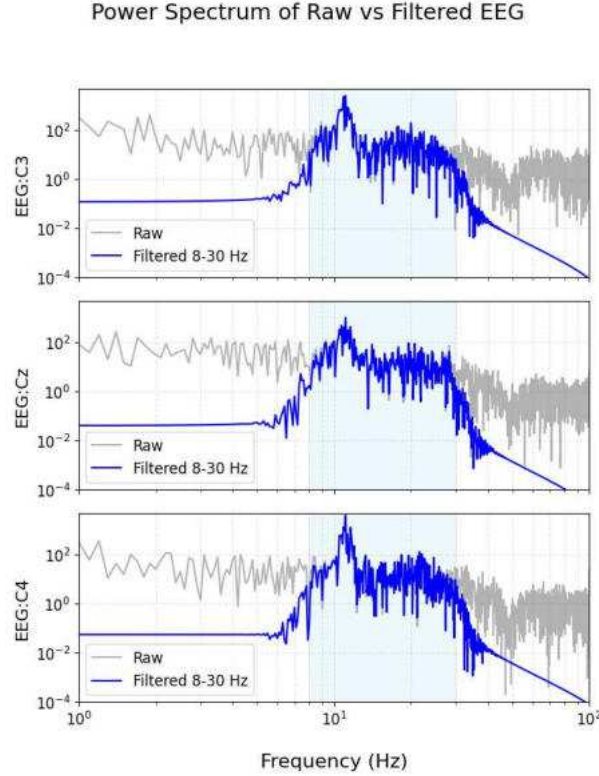


Figure 3.3: Spectrum of the raw filtered signal extracted from the BCI Competition IV 2b dataset using a 5th-order Butterworth IIR filter [8,30]Hz. It can be observed, for all three channels, that the spectrum of the unfiltered signal, shown in gray, extends beyond 40Hz, potentially introducing noise for the motor imagery task.

After filtering, the signal is normalized per channel using Z-Normalization, following the standard practice adopted in the literature, as in the EEG-Conformer [5] and 2StageTransformer [19] models. This allows the network to operate on values on the same scale and converge faster. It also prevents channels with larger values from dominating classification while channels with smaller values could be ignored. Moreover, in physiological data such as EEG, amplitude can vary across subjects, sessions, or even for the same subject at different times (suggesting that EEG signals from different subjects may not follow the same distribution; this will be discussed in **subsection 3.4.2**).

$$x_{\text{norm}} = \frac{x - \mu}{\sigma}$$

Here,  $x_{\text{norm}}$  is the normalized EEG signal,  $\mu$  is the mean of the channel, and  $\sigma$  is the standard deviation calculated per channel using only the training set. The final normalized EEG channels have mean = 0 and variance = 1.

### 3.1.4 Data augmentation

As discussed in the previous chapters, one of the key issues when dealing with medical signals, and particularly EEG signals, is the limited amount of data. For example, the BCI Competition IV 2a training set contains only 288 epochs (288 samples) for 9 patients. When each patient is considered separately, as required for intra-subject training, this leaves just 288 samples for training. This clearly highlights why convolutional networks are often used instead of Transformers. A limited amount of data may be insufficient if the network is too large, and the risk of overfitting increases. Data collection involves long and costly procedures, and datasets for the motor imagery task are even scarcer, as, according to the introductory chapter **section 1.2**, subjects must be trained to perform the task, which requires considerable time. For these reasons, a data augmentation procedure is necessary to increase the number of samples available for training without altering their distribution. In this work, two data augmentation methods were applied: one applied during raw data preprocessing, and the other after converting the signal into scalograms.

The first augmentation method is Segmentation & Reconstruction (S&R), a technique commonly used in the literature, as seen in EEG-Conformer [5]. As the name implies, the method has two stages: segmentation and reconstruction. Segmentation involves splitting the signals in the dataset into "segments," while reconstruction consists of recombining them to form completely new EEG signals. In this work, the number of segments is 8.

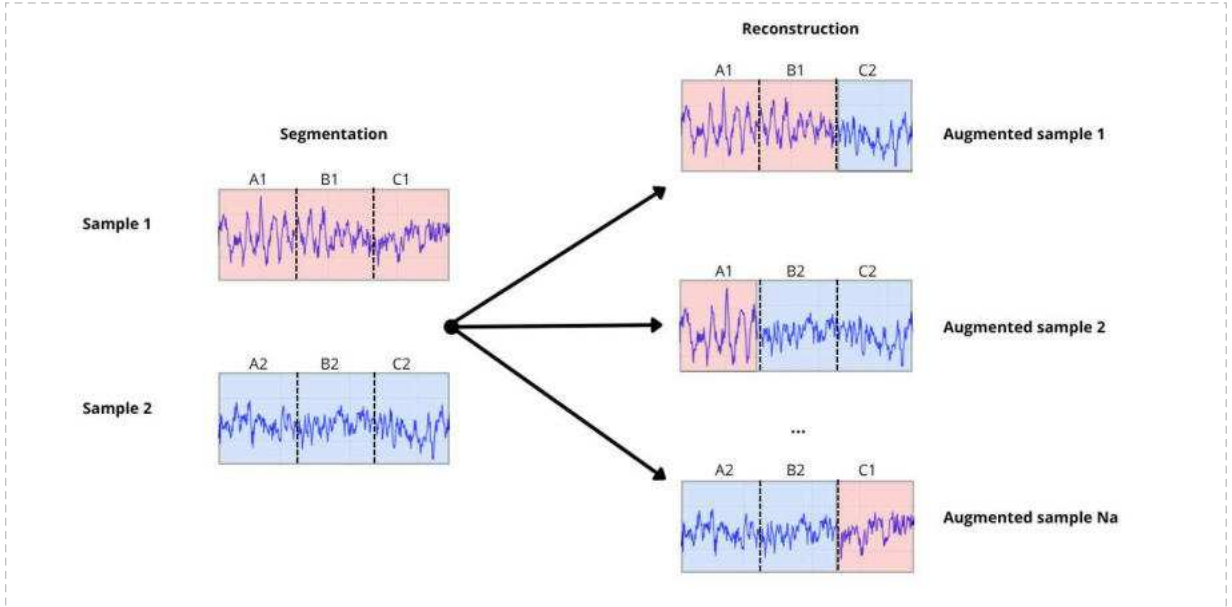


Figure 3.4: **S&R augmentation.** During the segmentation phase, each sample is divided into segments along the time axis; in the image there are  $N_s = 3$  segments, while in this work  $N_s = 8$  will be used. During the reconstruction phase, the segments are randomly recombined while maintaining temporal coherence, as can be seen in the image, in order to generate new samples; the total number of new samples is  $N_a$ , in this case twice the original number.

These segments are randomly selected in temporal order from different epochs of the same class

and then recombined sequentially. Maintaining temporal order is essential to avoid altering the data distribution. This process is repeated for each class. As both datasets do not have imbalanced classes, the augmentation results in a sample set twice the original size, i.e.,  $288 \times 2 = 576$  for each subject in dataset 2a. Consequently, the complete dataset with all subjects will also be doubled. The augmented data preserves the original temporal structure while providing new patterns for the model to learn, thereby maintaining the physiological significance of the EEG signals while increasing their quantity. The second data augmentation method is applied after computing the spectrograms, as it acts on the time-frequency structure. This method, also found in image-based models, consists of applying frequency masking and time masking to the data within a batch. These two transformations, along with their combination (frequency-time masking), were chosen to avoid altering the semantic meaning of the spectrogram, which could have been affected by transformations such as horizontal flip or random rotation. Frequency masking hides a consecutive band of frequencies, simulating the loss of information in certain bands, forcing the model to avoid over-relying on specific frequencies. Time masking conceals a temporal segment, simulating, for example, noise or interruptions in the data. These transformations are applied to batch elements with a certain probability (which means that a sample can also be passed to the network without any transformation). They reduce overfitting and, most importantly, simulate real-world conditions where noise may be present in the data.

### 3.1.5 Total preprocessing pipeline

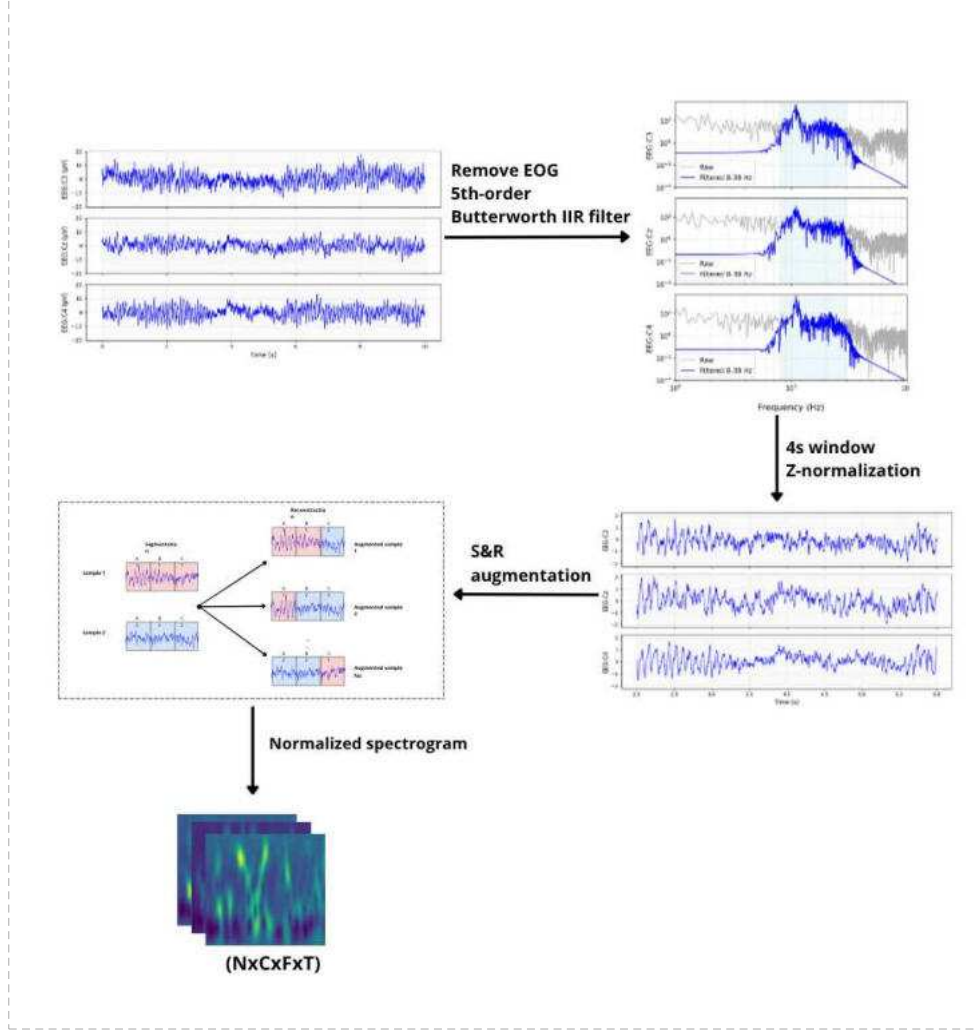


Figure 3.5: **Total preprocessing pipeline.** At the end of preprocessing, each channel is a scalogram. For patient 1 of dataset 2a, for example, the final size will be  $288 \times 22 \times 32 \times 1008$ , where 288 are the scalograms containing the 4 seconds of the motor imagery task signal, 22 the channels, 32 the frequency bins, and 1008 the time points.

The final signal is thus a spectrogram of size  $X_s \in \mathbb{R}^{C \times F \times T}$ , where  $C$  is the number of channels,  $F$  the frequencies, and  $T$  the time steps. The signal is first cleaned of EOG artifacts, then filtered, cropped to a 4-second interval, subsequently normalized, augmented, and finally transformed into a normalized scalogram ready for the model. The resulting spectrogram for the BCI Competition IV 2a dataset has dimensions  $22 \times 32 \times 1008$ , while for the 2b dataset it is  $3 \times 32 \times 1000$

## 3.2 Design of the Proposed System

In order to employ EEG signals as Transformer inputs, it is first necessary to define how to construct the tokens over which the attention mechanism operates. For this reason, the Vision Transformer is chosen as the model and the time-frequency wavelet transform as the representation. As shown in **figure 3.2**, the EEG signal is divided into channels, and each channel corresponds to a wavelet scalogram. Since the wavelet transform is applied to each EEG channel, it is necessary to design an architecture that can effectively exploit the inter-channel relationships, as each channel contains information that may be useful for solving the task. For this reason, an MVIT-inspired architecture was implemented.

### 3.2.1 EEG-MVIT architecture: a multi-channel VIT for EEG classification

The EEG-MVIT model processes EEG channels individually, with each channel handled by a dedicated branch that receives its corresponding scalogram after passing through a patch embedding layer. The generated patches are then provided to the channel-specific encoder, which applies the attention mechanism to extract informative features. This produces a channel-level embedding, from which either the [CLS] token can be taken or, as in this work, an average is computed to obtain a compact channel representation. All channel embeddings are concatenated and passed through a multilayer perceptron (MLP), which outputs a probability for each class. This design was chosen because it has shown strong performance in seizure detection tasks, and because it emphasizes the contribution of each EEG channel through independent processing branches. Furthermore, seizure detection and motor imagery share important similarities, since both involve EEG signals and face the challenge of limited dataset size, as already noted in **section 2.5**. The following section provides a detailed description of the architecture with all its layers. Lastly is presented a fully convolutional network with a spectrogram as input, aimed at verifying whether the use of a Vision Transformer yields higher performance compared to a model that does not employ any attention mechanism.

### 3.2.2 EEG-MVIT Patch embedding layer

EEG signals, after the preprocessing described in **section 3.1**, are represented as 2D scalograms  $x \in \mathbb{R}^{F \times T}$ . These are divided into  $N = \frac{F \cdot T}{H' \times W'}$  non-overlapping 2D patches, where  $(H' \times W')$  denote the patch height and width, respectively. Each patch is thus described as  $x_p \in \mathbb{R}^{N \times (H' \times W')}$ . In this model, an adaptive patch embedding layer is employed: a convolutional layer first generates patches of size  $[B, D, H', W']$ , where  $B$  is the batch size, and  $D$  is both the patch embedding dimension and the latent vector size for the Transformer. A flatten layer then reshapes them into  $[B, D, N]$ , and finally, through a transpose operation, the structure is rearranged into  $[B, N, D]$ ,

ready to serve as the Transformer encoder input. The adaptive patch embedding layer is chosen since it is the most widely adopted in the literature. However, it can be replaced with a linear patch embedding layer without affecting performance.

### 3.2.3 EEG-MVIT Transformer encoder

Once the patches of dimension  $D$  are obtained, they are fed into a Transformer Encoder as described in **subsection 2.4.2**. A  $[CLS]$  token of dimension  $D$  is concatenated on top of the patches, making the input shape  $[B, N + 1, D]$ . To include positional information, a learnable positional embedding is added, which does not change the input dimension. The data are then processed by the Transformer encoder with  $H$  attention heads and  $L$  layers. Each encoder block includes an MLP that doubles the embedding dimension and then projects it back to the original size before passing it to the next encoder block, repeating this for all  $L$  layers. Residual connections after each block pass the output to the next block without going through the non-linear activation. The GELU activation function is used, and two dropout layers are included. The final output is normalized using Layer Normalization. The mean of the  $N$  embeddings (excluding the CLS token) is extracted to obtain a summary representation of each channel. This procedure is applied individually to every channel.

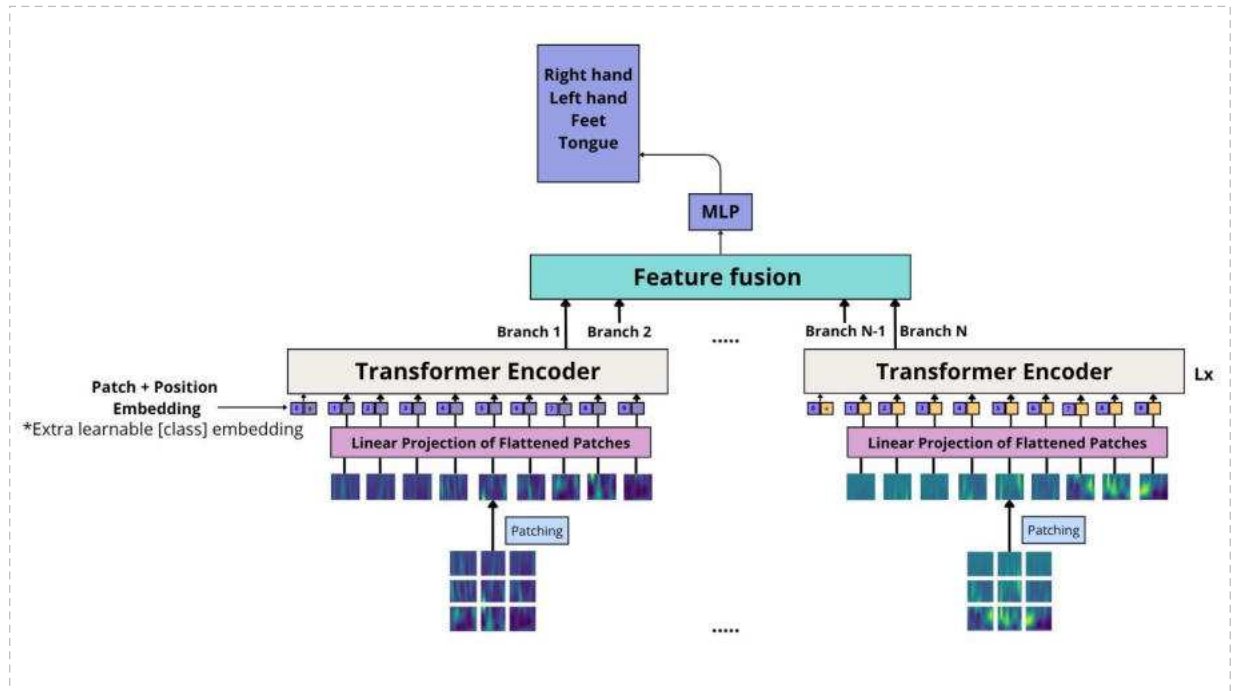


Figure 3.6: **EEG-MVIT architecture**. The model employs  $N$  parallel branches corresponding to the EEG channels. For dataset 2a,  $N = 22$ , while for dataset 2b,  $N = 3$ .

### 3.2.4 TSFF-img: comparison between CNN and ViT

A hybrid CNN+Transformer architecture achieves good performance, however it is necessary to investigate the effect of a standalone Vision Transformer compared to a CNN. For this reason,

an architecture called TSFF [16] was implemented. TSFF-Net is a network composed of four main components: a time-frequency representation module, layers for spectral feature extraction, layers for temporal feature extraction, and a module for feature fusion and classification as described in **section 2.4**. A Continuous Wavelet Transform (CWT) is applied to each channel, after which the obtained scalograms are concatenated along the depth axis, and the resulting matrix is downsampled to  $224 \times 224$ . TSFF-img, the part of the network responsible for time-frequency extraction, consists of three feature extraction layers, each including regularization operations. In standard image processing pipelines, a max pooling layer is typically employed, however, due to the nature of EEG signals, characterized by a low signal-to-noise ratio and possible artifacts even in small portions of the signal, an average pooling layer is adopted instead.

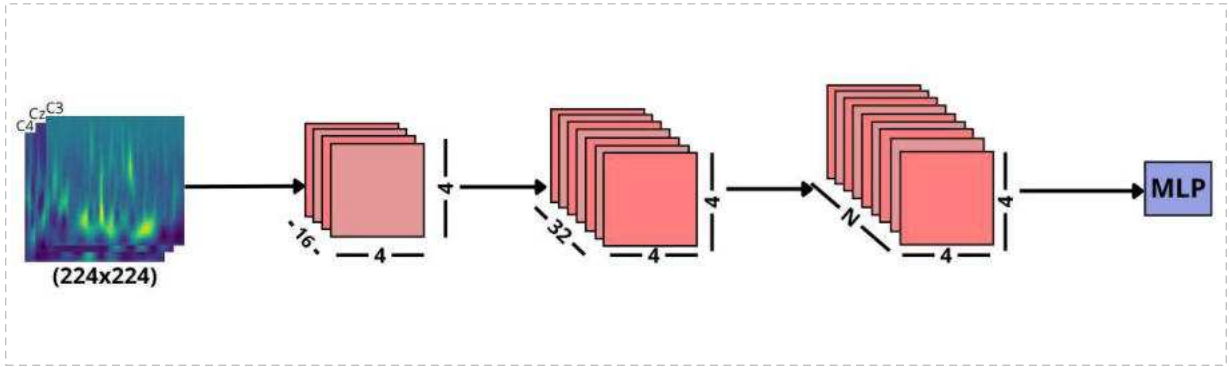


Figure 3.7: **TSFF-img architecture**. The model consists of 3 convolutional layers that take as input the spectrogram with channels concatenated along the depth dimension.

For the feature extraction layers, the parameters are fixed in the first two levels, with values of  $(16, 4, 4)$  and  $(32, 4, 4)$ , whereas the preceding layer must be adjusted according to the number of EEG channels, that is 22 for dataset 2a and 3 for dataset 2b. The TSFF-img module consists of three layers only, as the design goal is to learn shallow information, such as the edges of the time-frequency spectrum, while avoiding more complex information that may introduce noise during the learning process. From the complete architecture, only TSFF-img, namely the branch devoted to time-frequency feature extraction, was retained, since the objective is to compare its performance against EEG-MVIT on datasets 2a and 2b, to investigate whether performance differences arise.

### 3.2.5 EEG-MVIT feature fusion and MLP head

Each channel produces an encoder output of dimension  $D$ . The outputs from all channels are concatenated to create a single, unified representation. This representation is then passed to an MLP, which accepts  $(22 \times D)$  as input, corresponding to the concatenated outputs of all channels' encoders, and maps it to the target number of classes.

### 3.3 Variations of the main architecture for performance analysis

In this section, comparison architectures will be presented to investigate their effects on performance, as discussed at the beginning of the chapter. The focus will be on the way features extracted from the patch embedding layer are concatenated, on the structure of the input channels, and on the design of the patch embedding layer itself. Lastly, a hybrid CNN–VIT architecture will be analyzed to better investigate the role and effects of inductive bias.

#### 3.3.1 ENC-MVIT: feature fusion at the encoder level

In order to investigate the impact of concatenation on EEG-MVIT, a modified version of the architecture is proposed, in which concatenation takes place at the encoder stage. Unlike the original model, which uses  $C$  separate encoders branches (where  $C$  corresponds to the number of spectrogram channels), this variant concatenates the output of each channel’s patch embedding layer, yielding an input  $x$  of dimension  $[N \times C, D]$ , with  $N$  denoting the number of patches per channel and  $N \times C$  the total number of patches across all channels. This concatenated input is then passed to a single Transformer encoder branch composed of  $L$  layers and  $H$  heads, producing a single output, which can be the [CLS] token, but in this case corresponds to the mean of the encoder outputs. This representation is then fed into a classification head that outputs the class probabilities.

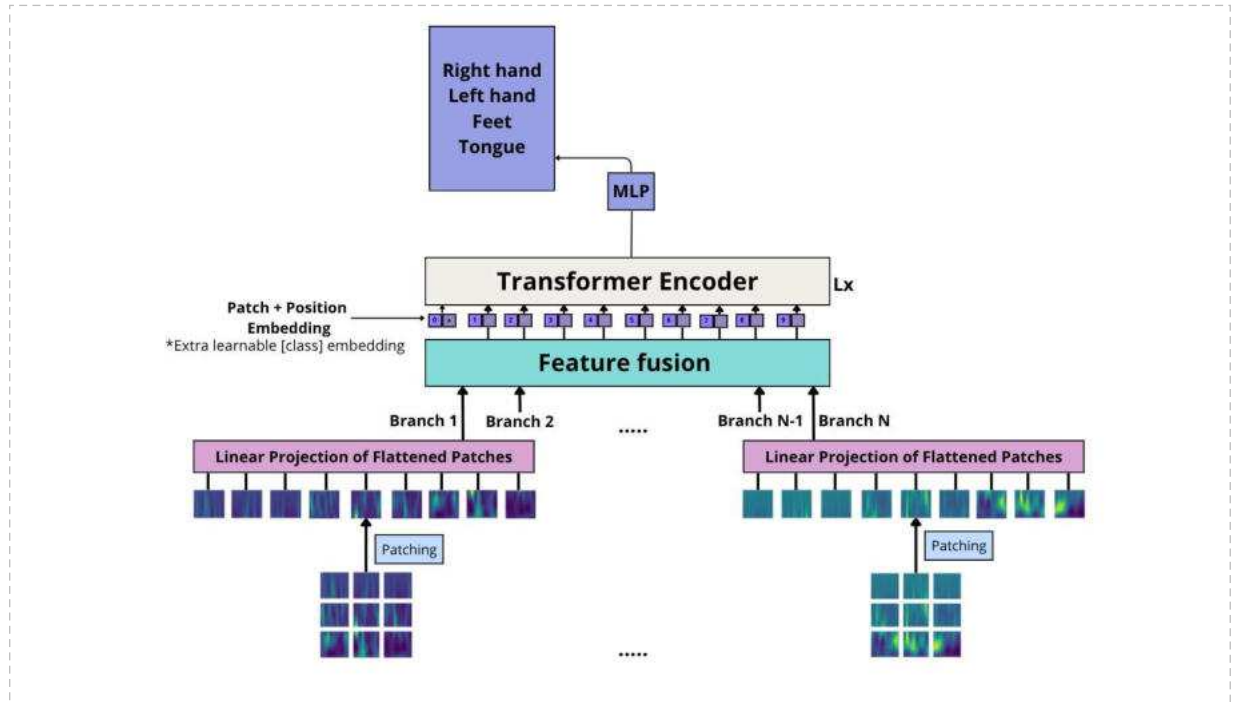


Figure 3.8: **ENC-MVIT architecture.** The feature fusion for each branch occurs immediately after the patch layer, therefore a single Transformer encoder can be used.

With this approach, the channels are initially processed independently at the patch embedding stage, after which the attention mechanism is applied jointly across the patches of all channels. This allows each patch to assess the importance of patches from other channels, contributing to the construction of the final embedding.

### 3.3.2 DEPTH-ViT: channels concatenated along the depth dimension

This variation was implemented to examine the impact of the input channel structure. In MVIT, each branch operates on a single channel, based on the assumption that the channels of an EEG signal are mostly independent. Depth-ViT is the model that most closely resembles the structure of a standard ViT by definition, as presented in [3] and in **subsection 2.4.4**. The structure of the input signals is designed to mimic that of an image, as the EEG channels are concatenated along the depth dimension, resulting in an input  $x_s \in \mathbb{R}^{C \times F \times T}$ . For the BCI Competition IV 2b dataset, the number of channels is 3, which makes the final structure essentially identical to that of a standard image. The resulting signal is first divided into patches, then passed through the Transformer encoder with L layers, and finally classified using an MLP head.

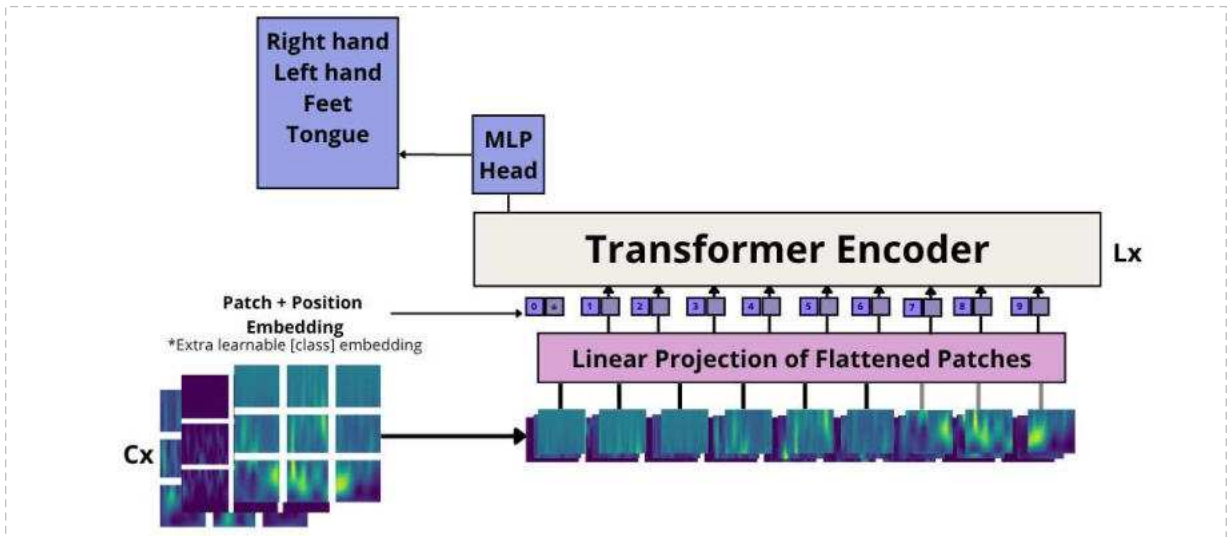


Figure 3.9: **Depth ViT architecture.** All channels are concatenated along the depth dimension.

### 3.3.3 3BRANCH-ViT: patches of different sizes in the patch embedding layer

This model, together with the next one, was implemented to evaluate the effectiveness of dividing EEG signals into patches, and therefore of the patch embedding layer. The 3branch-ViT architecture consists of three branches, each generating patches of different sizes. From state-of-the-art studies, several key observations have emerged:

- good performance can be achieved by implementing convolutional layers that analyze the frequency bands relevant to motor imagery;

- changing the number of heads or layers has little impact on performance, except for the jump from 0 to 1, while increasing these values too much can sometimes degrade results;
- rectangular patches may improve model performance [22], although they are rarely used, since pre-trained models on images, which are typically fine-tuned, use square patch sizes such as  $16 \times 16$  or  $8 \times 8$  during pre-training.

For these reasons, the three branches employ:

- rectangular patches along the frequency axis to emphasize the effect of different bands, focusing more on the local component of the signal;
- rectangular patches along the time axis to highlight the temporal samples, aiming to capture the global component of the signal;
- standard balanced square or rectangular patches.

The patches generated by the three branches are concatenated before being passed to the Transformer encoder with  $L$  layers, and the channels are arranged along the depth dimension, making this architecture a hybrid between ENC-MVIT and Depth-MVIT.

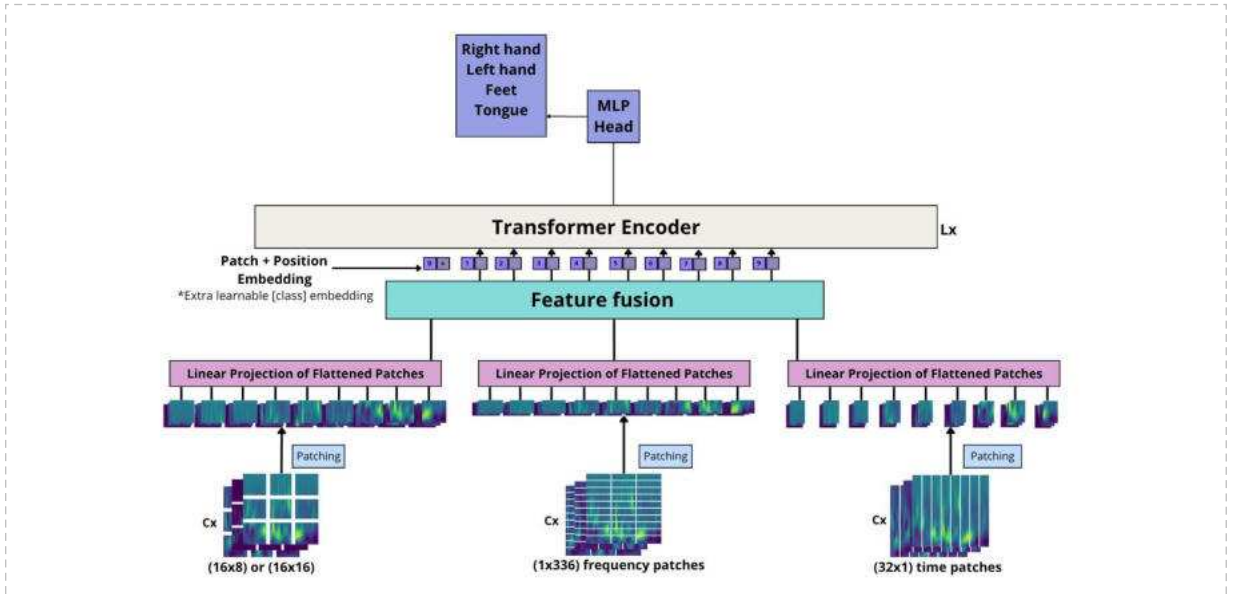


Figure 3.10: **3BRANCH-VIT architecture**. Each branch splits the spectrogram into patches of varying shapes: square, frequency-rectangular, and time-rectangular.

Concatenation is possible because the number of patches varies, but the embedding dimension remains the same. As a result, the attention layer operates on an input  $x$  of shape  $(3N \times D)$ , where  $N$  is the number of patches per branch and 3 is the number of branches in this implementation. The attention mechanism then computes the relevance among embeddings extracted from patches of different sizes, allowing interactions between embeddings that focus more on the temporal or frequency domain. Finally, either the  $[CLS]$  token or the mean of the outputs is extracted and passed to the classification head.

### 3.3.4 LAMBDA-ViT: a CNN–ViT hybrid architecture

The architecture of this model is the same as the previous one, however, an additional branch is introduced that allows for direct classification of the input  $x$  of shape  $(3N \times D)$  right after the patch embedding layer. The output of this classification is used to build a composite loss:

- $L_1$  is the standard cross-entropy loss for the classification of the three branches;
- $L_2$  is the cross-entropy loss of the branch that classifies the concatenated input directly;

and the final loss is defined as

$$L_c = (1 - \lambda) L_1 + \lambda L_2.$$

This implementation was developed to preserve a component that more closely reflects the original data, classifying it in the same way as an image divided into patches. Moreover, this classification is important because this model employs an adaptive patch embedding layer, constructed using a single convolutional layer followed by a flatten and transpose operation.

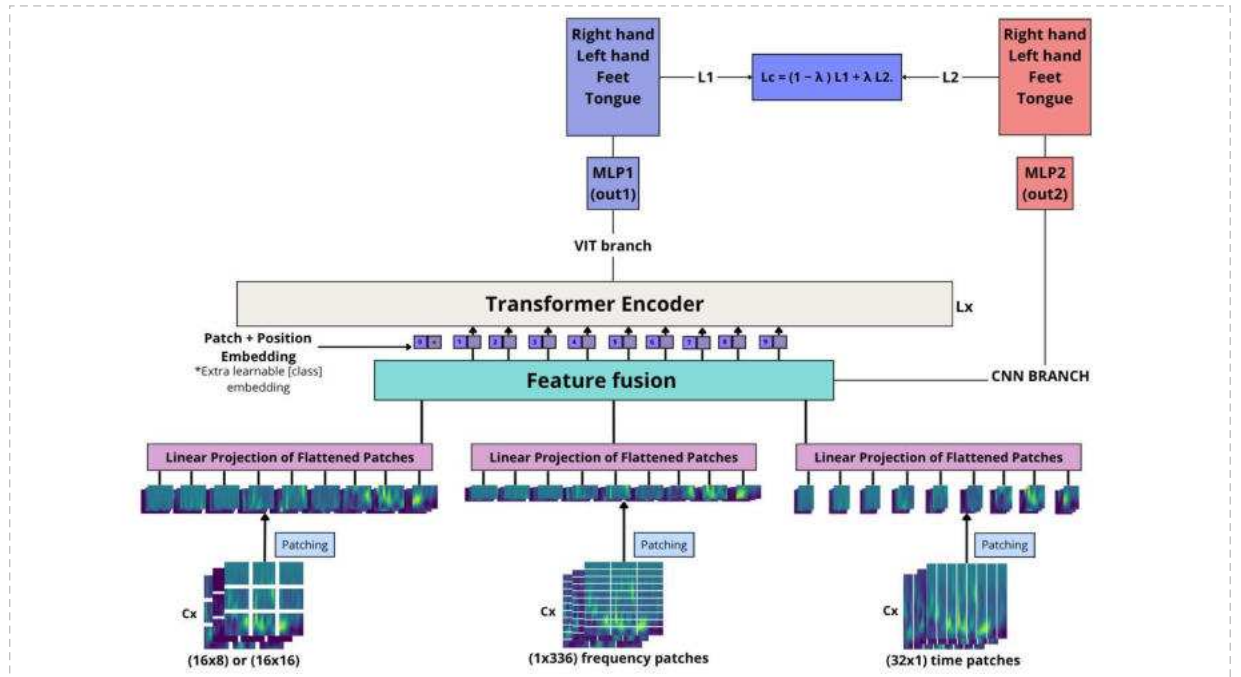


Figure 3.11: **LAMBDA-ViT architecture.** Each branch divides the spectrogram into patches of different sizes: square, frequency-rectangular, and time-rectangular, as in the 3BRANCH-ViT. However, it includes an additional branch that does not pass through the Transformer encoder, in order to give more weight to the effect of the adaptive patch embedding built through a CNN layer.

This design is expected to improve performance by introducing an effective inductive bias, particularly suited to the limited amount of available data. By tuning the  $\lambda$  parameter, it is possible to balance the contribution of the Transformer branch and the single CNN branch, thus analyzing the performance impact of using just a single CNN layer and quantifying the role of the inductive bias in data-scarce scenarios.

### 3.4 Methodological innovations

The inductive bias enables convolutional networks to perform well even with relatively small datasets. Nevertheless, when enough data is available, this bias can limit the model’s performance potential, causing it to focus excessively on patterns in the training data and hindering its generalization capability. Transformer-based models, including Vision Transformers, rely mainly on the attention mechanism and consequently lack inductive bias. The absence of inductive bias represents a clear drawback when only limited data are available, as seen in audio-related tasks or in medical applications like motor imagery. On the other hand, the absence of inductive bias can also be considered an advantage, since it eliminates the performance ceiling inherent to CNNs: if a CNN and a Transformer-based model with an equal number of parameters are trained on a sufficiently large dataset, the Transformer-based model outperforms the CNN [3]. The training of models for motor imagery using EEG signals is divided, as previously mentioned, into intra-subject and cross-subject. The reason for this division lies in the distribution of EEG signals of each individual, which is considered to differ from the distribution of other subjects. This difference explains why cross-subject performance is significantly lower than intra-subject performance, and consequently represents the main challenge that prevents BCIs from being generalizable. This issue also creates the need to build subject-specific BCIs that are calibrated for each individual. In existing literature, research primarily emphasizes subject-specific performance, while approaches that involve pre-training on data to develop a general model are practically absent. For this reason, a self supervised pre-training stage was implemented. If this approach leads to improved performance, it could open the door for the development of general BCIs that are not related to a specific patient, as it would indicate that a sufficiently large model, trained on a large amount of data, can identify similarities between EEG signal distributions to enhance task performance. Choosing how to pre-train the model specifically on EEG signals is crucial: one strategy could involve using a model that inherits weights from another model pre-trained on images. However, as already discussed in **section 2.6**, although this approach provides a slight performance gain, a comparison with a model pre-trained directly on the target signals reveals clear differences in performance.

The core of this work will be presented in the following section. A self supervised pre-training is carried out on the EEG-MVIT after analyzing the motivations behind the performance of other ViT architectures, as it is considered the most general model for this task, given its well-defined channel division and the suitable number of parameters for performing pre-training. This model is pre-trained on a dataset larger than BCI Competition IV 2a and 2b datasets. To build this dataset, which will be referred to as extended 2b, the entire signal trace from the BCI Competition IV 2b dataset was used. Four-second windows were extracted with a stride of 1 second, resulting in an overlap of 3 seconds. Each 4-second window is treated as a valid sample for pre-training, increasing the number of available samples from around 400 to roughly 6000 per subject.

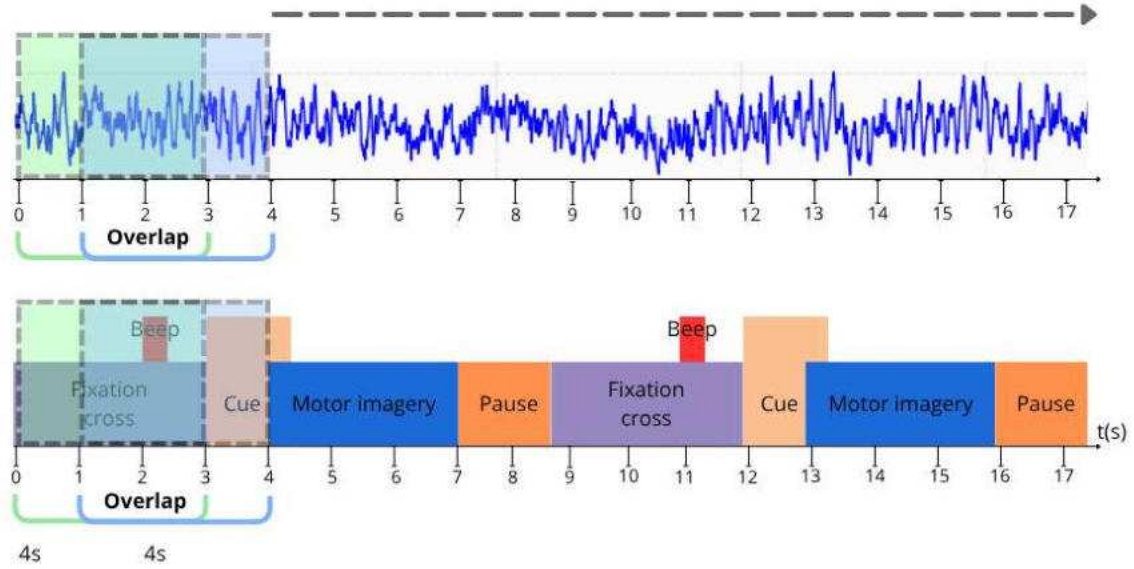


Figure 3.12: Extended 2b dataset. Windowing on the entire signal trace from the BCI Competition IV 2b dataset, every 4 seconds (with 1s of overlap) a new signal for pretraining is obtained. From the image, it can be seen that the signals obtained do not only belong to the MI task, but also to the period before the recording and during the pause before starting the next trial.

Each of these samples, as shown in figure **figure 3.12**, does not necessarily correspond to the actual motor imagery task. Some segments may belong to the fixation cross phase or the resting period. These samples are still valuable, as they provide the model with information about the general structure of EEG signals. Using signals not strictly related to the motor imagery task and still achieving performance improvements would pave the way for pre-training on an even larger set of EEG data collected from different tasks, such as epileptic seizure recordings. All samples obtained are subjected to the preprocessing detailed in **section 3.1**, and then converted into spectrograms. Here, the decision to use the spectrogram as the input transform is even clearer and crucial: it enables the creation of the input without additional convolutional layers, while ensuring that the patches extracted during pre-training carry meaningful semantic information.

### 3.4.1 Supervised pre-training on limited data

This work also implements an additional pre-training strategy tested before the self supervised pre-training, to evaluate the model’s generalization capability, considering distributions from other patients that are expected to differ from the patient under train. In the BCI motor imagery domain, the main training strategies used so far are mainly two, as already discussed:

- intra-subject, in which the model is trained using patient-specific data and its performance is evaluated by averaging the results over all patients;
- cross-subject (leave one subject out), in which the model is trained on the data of all patients and tested on the patient whose data was excluded from the training set;

A supervised pre-training strategy is introduced, combining both approaches: when data from multiple patients is available, one can first train a model on the training data of all patients and subsequently fine-tune it on the specific patient. In this scenario, a supervised pre-training can be carried out by leveraging the labeled data from the 2a and 2b datasets to initialize the model weights, thus using a smaller amount of data compared to the self supervised approach to test the model’s ability to learn from multiple subjects. In **chapter 4**, this strategy will be investigated to analyze whether the distribution of a single patient can benefit from the EEG signals (specifically from the motor imagery task) of other patients.

### **3.4.2 A method to overcome the lack of inductive bias: self supervised curriculum masked pre-training**

The framework used for pre-training in this work integrates curriculum learning, self supervised and generative masked spectrogram patch modeling (MSPM) pre-training from SSAST [24]. SSAST assumes that AST achieves excellent performance but only thanks to pre-training on a very large labeled dataset, which is difficult to obtain. This is the reason for adopting a self-supervised learning strategy. Such difficulty is also present in the motor imagery domain. Until now, the models only masked temporal frames of the spectrogram, meaning the network learned only temporal structure. MSPM aims to predict a specific frequency band within a given time range, referred to as a patch. This patch concept further supports the choice of a Vision Transformer as the base architecture of the neural network. ViT relies on a patch embedding layer that divides the signal into patches before passing it to a Transformer Encoder; predicting a patch therefore involves predicting elements essential for the construction of the final embedding output by the encoder fully leveraging the time-frequency representation provided by the wavelet transform input.

Denote  $O$  as the output of the Transformer encoder. According to the SSAST procedure, a cluster factor  $C$  is selected, then a random patch is picked and the square of patches centered on it is masked. For instance, with  $C=3$ , a cluster of 9 patches is created, so if each patch measures  $16 \times 16$ , the resulting square will be  $48 \times 48$ , with 3 patches forming one side of the square. The larger the value of  $C$ , the more the model learns a global structure, while smaller values of  $C$  focus on local structures.  $C$  is randomly chosen in the range  $[3,5]$  during pre-training to allow the model to learn both local and global components.  $C$  replaces the usual masking ratio, which is typically used; this choice is more suitable since the patches must be identified in both time and frequency, and not merely as a portion of the spectrogram to be hidden. The choice of using the spectrogram is crucial: by masking regions in both time and frequency, the model can better learn the structure of the data. A set  $I$  of  $N$  masked patch position indexes is generated. For each patch to be masked, the patch embedding is replaced with a learnable mask embedding or, as in this work, with a zero embedding (all zeros). The positional embedding is applied, and the masked input is fed to the Transformer encoder. The encoder outputs the embedding of

the input, so for each masked patch  $x_i$ , the corresponding encoder output  $O_i$  is obtained. Subsequently,  $O_i$  is fed into a classification head and a reconstruction head to obtain outputs  $ch_i$  and  $rh_i$ . These two heads map  $O_i$  (dimension 768, for example) to the dimension of  $x_i$  (256 if the patches are  $16 \times 16$  or 64 if the patches are  $8 \times 8$ ). The reconstruction  $rh_i$  is expected to be close to the original removed patch  $x_i$ , and the model can consider  $(x_i, ch_i)$  pairs as similar. The InfoNCE loss  $L_d$  is used for the discriminative objective, and the MSE loss  $L_g$  is used for the generative objective.  $N$  is the number of masked patches. For the discriminative loss  $L_d$ , negative samples are taken from the same spectrogram, making the task more challenging and saving memory.

$$L_d = -\frac{1}{N} \sum_{i=1}^N \log \frac{\exp(ch_i^T x_i)}{\sum_{j=1}^N \exp(ch_i^T x_j)}$$

The goal of the InfoNCE loss  $L_d$  is to minimize the distance between representations of positive pairs, here the reconstructed patch and the removed patch from the same spectrogram, while maximizing the distance between representations of negative pairs, in this context, the reconstructed patch and a patch from another spectrogram. InfoNCE evaluates the model's ability to separate similar and dissimilar examples in the embedding space. In other words, InfoNCE measures how well a model can distinguish between similar and dissimilar examples in an embedding space. The numerator computes the similarity between  $ch_i^T$  and  $x_i$ , i.e., the positive patch to be reconstructed from the same spectrogram. The denominator computes the similarity between  $ch_i^T$  and  $x_j$ , i.e., all patches in the batch, including negative ones. This ratio measures how much  $ch_i$  is more similar to  $x_i$  than to other  $x_j$ , and it is a log-likelihood averaged over all masked patches in the batch. Minimizing  $L_d$  ensures that  $c_i^T x_i$  is high, while  $c_i^T x_j$  is low for patches not belonging to the same spectrogram.

$$L_g = \frac{1}{N} \sum_{i=1}^N (rh_i - x_i)^2$$

Each reconstructed patch is compared with its original. The MSE  $L_g$  measures how much the reconstructed patch deviates from the original and minimizing  $L_g$  means accurately predicting the masked patches.

The final loss is a composite loss:

$$L = L_d + \lambda * L_g$$

where the value of  $\lambda$  is 10. This weighting ensures that the MSE contribution is predominant over InfoNCE, since the goal is for the masked patches to be reconstructed closely to the originals. If  $\lambda$  were applied to  $L_d$  instead, the model would focus on identifying as many masked patches as possible but not on reconstructing them correctly. While  $L_g$  improves reconstruction quality,  $L_d$  enhances predictive capability.

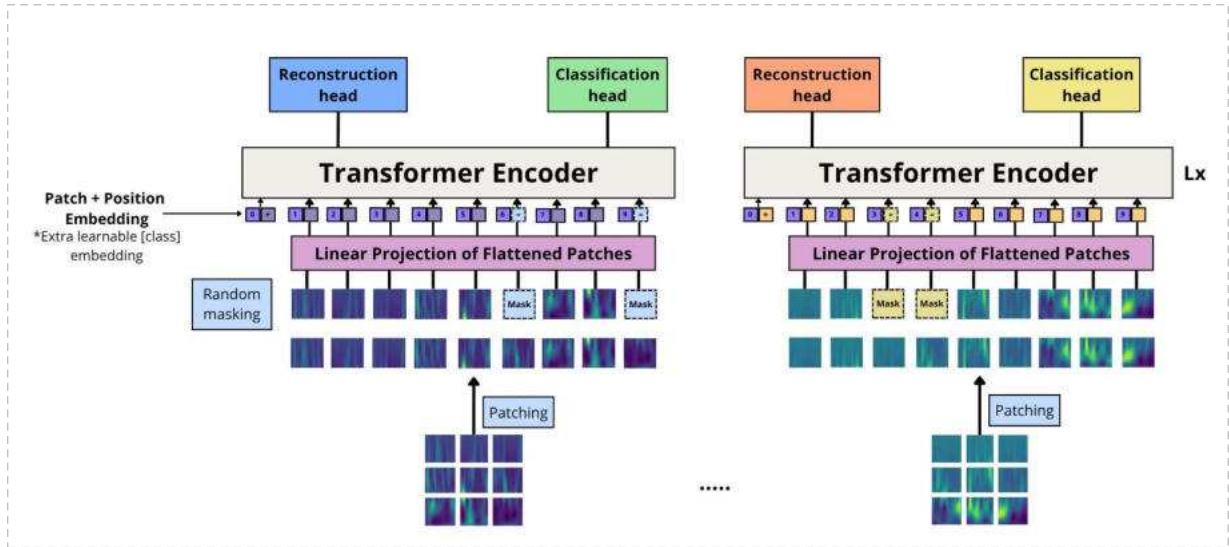


Figure 3.13: **EEG-MVIT architecture during the pretrain phase.** Each branch for each channel is subject to the patch masking. For the first channel, the sixth and ninth patches were masked, while for the  $i$ -th channel, the third and fourth patches were masked. Instead of the classic MLP, there are the reconstruction and classification heads, which will be removed during the fine-tuning phase.

The procedure just described corresponds to the pre-training stage. This is combined with curriculum learning to make the pre-training more effective. Curriculum learning consists of gradually teaching the network to predict an increasing number of masked patches, starting from a small amount. Moreover, as will be discussed and demonstrated in the experiments of **chapter 4**, the strategy is to begin with a simpler task but with a relatively high number of masked patches: during the pre-training presented earlier, the masked patches are randomly distributed in both time and frequency, because the indexes are randomly selected. To simplify the task at the beginning, the indexes are chosen to form a continuous block, so that a contiguous region of the spectrogram is masked. Although the masked region can vary along the frequency axis, the patches remain connected. This allows the model to first learn to detect the “missing” part of the spectrogram, facilitating the complete reconstruction task later on.

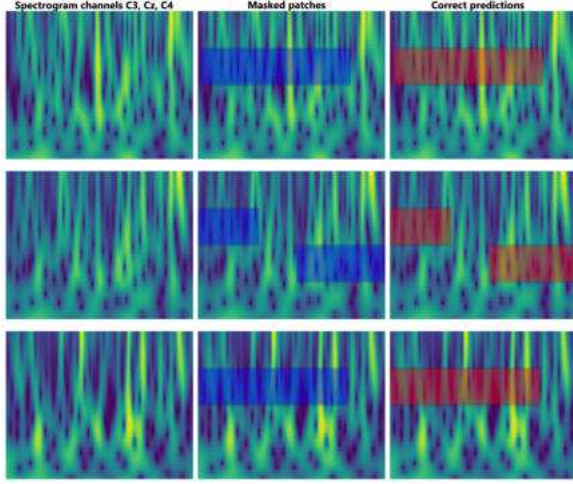


Figure 3.14: First step of the curriculum learning: 100 consecutive patches masked in time and frequency on channels C3, Cz, and C4. The masked patches are shown in red.

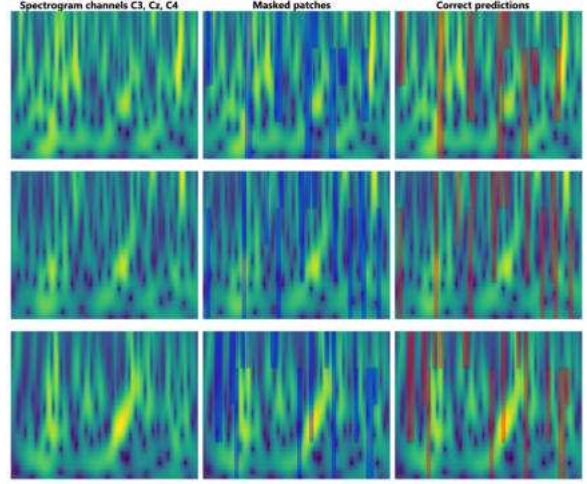


Figure 3.15: Second step of the curriculum learning: 100 random patches masked in time and frequency on channels C3, Cz, and C4.

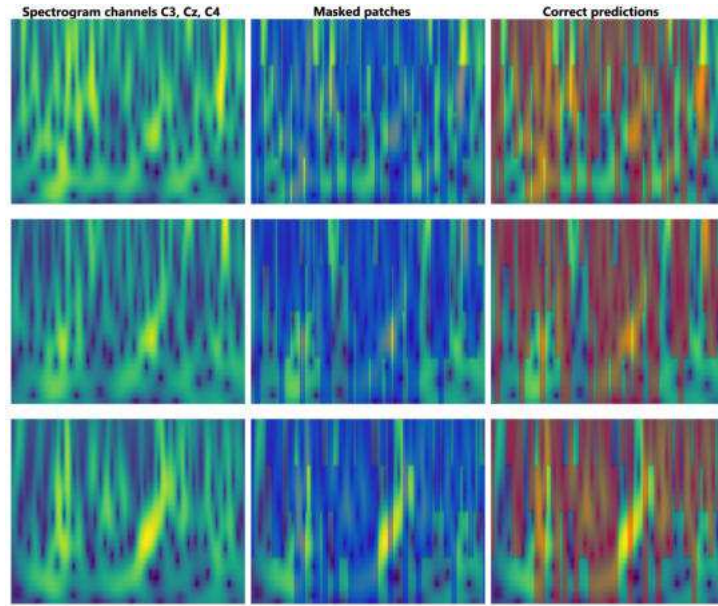


Figure 3.16: Third and last step of the curriculum learning: 300 random patches masked in time and frequency on channels C3, Cz, and C4.

Once the model learns the simple task, the previously described MSPM masking is applied, starting with 100 masked patches, and increasing to 300. The transition from one number of masked patches to another can be made either by visually inspecting the reconstruction or by monitoring the model's loss.

### 3.4.3 Fine-tuning on the target task

Once the model has been pre-trained, a fine-tuning phase is required for the specific task. In this work, fine-tuning is performed by first training a classification head for the downstream task,

which in this case involves classifying 2 or 4 classes, corresponding to BCI Competition IV 2b and BCI Competition IV 2a, while freezing all other model weights. After this initial step, the weights are unfrozen, and the remaining parts of the model are also trained. A higher learning rate is used for the classification head and a slightly lower one for the encoder. This type of fine-tuning is designed to better preserve the feature learned by the model during pre-training and adapt it to the downstream task only in a later stage. In this way, the goal is to ensure that the model remains both effective for the specific task and sufficiently general. In this context, “general” means that the model performs well across all patients.

As a result, the pre-training pipeline consists of the following steps:

- Apply continuous masking to 100 patches
- Apply MSPM masking with 100, and 300 patches
- Fine-tune the model by freezing all weights except the classification head
- Unfreeze the entire model and train it, setting a higher learning rate for the classification head and a lower one for the encoder

The pre-training fully exploits the structure of the Vision Transformer and the EEG signal spectrogram as input. The goal is to use a large amount of EEG data to learn general characteristics of the signal by masking components in both time and frequency, and then perform fine-tuning to benefit the classification task and improve accuracy across subjects.

### **3.5 Tools, technologies, and models employed for the development**

The MVIT model and its variants were trained and tested on the university server equipped with Tesla V100-DGXS-32GB GPUs. The EEG-MVIT model was pre-trained on a NVIDIA GeForce RTX 5090. The BCI Competition IV 2a and BCI Competition IV 2b datasets [8] were used, and their data were pre-processed following the procedure described in **section 3.1**, using the MEG + EEG Analysis & Visualization (MNE) library, with Python 3.12.3 and PyTorch 2.8.0. To conduct a feature interpretability analysis, the t-distributed stochastic neighbor embedding (t-SNE) distribution was visualized using the Scikit-learn library.

## 4 Experimental evaluation and application perspectives

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This chapter presents the performance of the models introduced in **section 3.2** and **section 3.3**, which were trained on the BCI Competition IV 2a and 2b datasets. The evaluation metrics and the verification protocol will be described, and the results of the models will be compared to understand the causes of performance variations. After analyzing the results, the focus will shift to the EEG-MVIT model to show the main conclusion of this work, that is, the performance differences following pre-training. Insights on the inductive bias and the possibility that some EEG signals share a common distribution will be provided. Finally, a dataset for future general pre-training will be presented.

### 4.1 Description of the evaluation metrics

The evaluation mainly relied on accuracy and Cohen’s kappa-score, which are described in the following. In addition, the model loss was closely monitored to verify the convergence of the model. During pre-training, the loss played a key role alongside the visual feedback of the reconstructed patches, which provided insights into whether the model was effectively learning the reconstruction task.

#### 4.1.1 Accuracy

Accuracy measures the proportion of correctly classified samples with respect to the total number of samples.

$$\text{Accuracy} = \frac{1}{N} \sum_{i=1}^N \mathbf{1}(\hat{y}_i = y_i)$$

- $N$  is the total number of samples;
- $y_i$  is the true label of sample  $i$ ;

- $\hat{y}_i$  is the predicted label by the model;
- $\mathbf{1}(\cdot)$  is the indicator function, which equals 1 if the condition is true and 0 otherwise;

### 4.1.2 Cohen's kappa score

The kappa score is a metric that measures the agreement between two classifications while taking into account the possibility of random agreement. The model predictions are compared with the true labels, and the analysis evaluates whether such predictions occurred by chance.

$$\kappa = \frac{p_0 - p_e}{1 - p_e}$$

where  $p_0$  is the proportion of observed agreement, and  $p_e$  is the proportion of agreement expected by chance, which is computed based on the class frequencies in the predictions and in the true labels.

- If  $\kappa = 1$ , the agreement is perfect, meaning the model is able to fully understand the data;
- if  $\kappa = 0$ , the model behaves almost randomly;
- if  $\kappa < 0$ , there is no agreement, indicating that the model likely fails to interpret the data or has not learned the task;

In practice, the kappa value often falls within intermediate ranges. Unlike accuracy, the kappa-score takes into account the fact that a classifier can achieve high accuracy even by making trivial choices, for example by following a prediction pattern. This behavior becomes even more evident in the presence of unbalanced classes, but it can also occur in cases of skewed distributions, such as those that may be found in EEG datasets.

### 4.1.3 Confusion matrix

The confusion matrix shows how the model "confuses" classes with each other during classification. For a classification problem with  $C$  classes, the confusion matrix is a  $C \times C$  matrix where:

- the rows represent the true labels;
- the columns represent the model's predictions;
- if we denote the rows by  $i$  and the columns by  $j$ , then  $M_{i,j}$  indicates the number of samples of true class  $i$  classified as class  $j$ .

The confusion matrix is useful because it can help identify potential biases of the model toward certain classes.

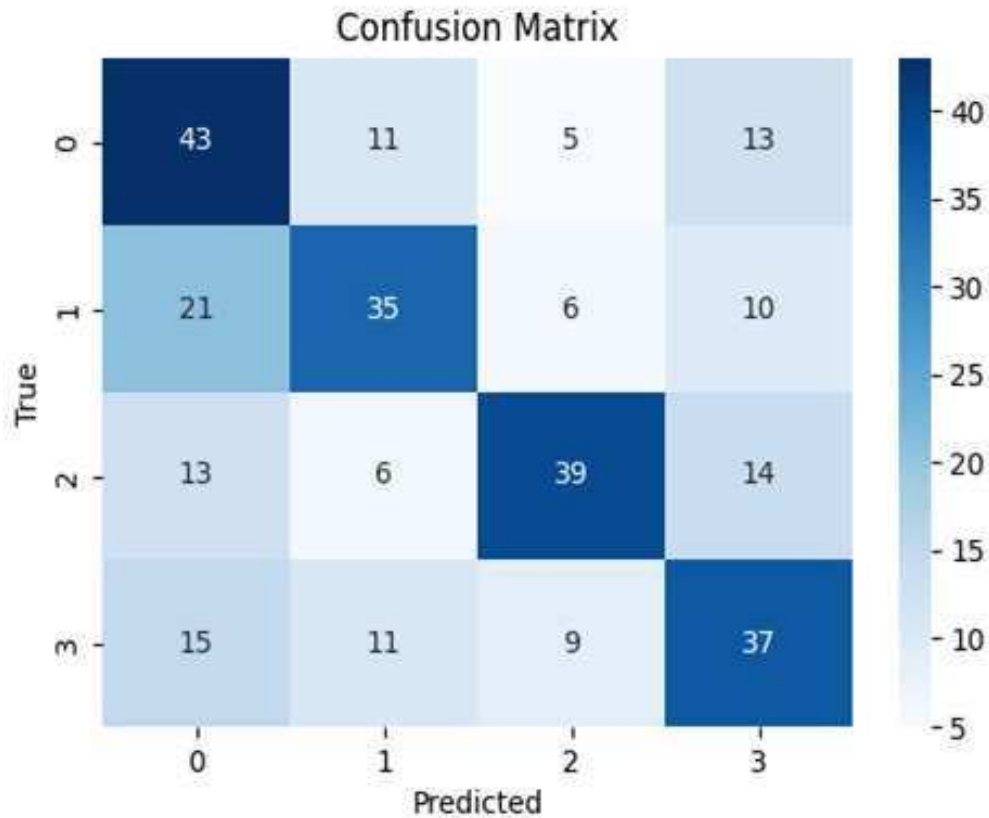


Figure 4.1: Confusion matrix of subject 6 from the BCI Competition IV 2a dataset. The model's classification is quite accurate; however, it can be seen that the model confuses class 2, which in this case corresponds to right hand movement, with left hand movement.

#### 4.1.4 P-value

The p-value is used to assess whether the difference between two results is statistically significant, and it will therefore be used to validate the differences between the final results obtained and the initial ones. In this case, the p-value derived from the paired Wilcoxon Signed-Rank Test is used:

- When  $p < 0.05$  (\*) or  $p < 0.01$  (\*\*), the observed outcomes are highly unlikely under the null hypothesis of randomness; therefore, it can be rejected, indicating that the results are statistically relevant.
- When  $p > 0.05$ , there is not enough evidence to reject the null hypothesis, so the observed results might be caused by chance.

It is important to point out that it is necessary to analyze the individual accuracies, the average accuracy, and the kappa together with the p-value to get an overall idea of the model's performance.

## 4.2 Experimental protocol definition

The models were trained on the BCI Competition IV 2a and BCI Competition IV 2b datasets, following the canonical split already specified by the provided train and test data files. For intra-subject training, the samples were divided into 80% for training and 20% for validation, while the test dataset remained unchanged. Intra-subject training involves training one model for each subject using their own training data. Each model is then tested individually, and finally, the average accuracy across all subjects is computed. For cross-subject training, the leave one subject out (LOSO) procedure is followed where the model must be trained on the training and test data of all subjects except the one on which it will be tested. For this reason, an 80% and 20% split is applied to the entire set of data from the training subjects to obtain a training and a validation set. In the end, the average accuracy across all tested subjects is computed, which serves as an indicator of the model’s generalization capability. For supervised pre-train, the model is pre-trained on the training data of all subjects and then proceeds with intra-subject training, the data is split into 80% for training and 20% for validation.

For curriculum masked pre-training, the dataset was generated using the method described in **section 3.4**. In this case as well, 80% of the data was used for training and the remaining 20% for validation. The model selected at each step of curriculum learning is the one with the lowest validation loss (MSE+InfoNCE). This model was then used to perform intra-subject or cross-subject training. Validation was performed every 2 training epochs for all training procedures. Training was stopped using the early stopping technique after 20 validation epochs during which the validation loss did not improve compared to the best loss achieved so far. The selected model is therefore the one with the lowest validation loss. All training runs use the Adam optimizer with a weight decay of 0.01, a cosine scheduler with 5% warmup epochs, and a gradient max norm of 2. During the experiments, it was observed that the optimal number of parameters for achieving better performance in the considered ViT models was 2 encoder layers (depth) and 6 heads for intra-subject training, while 6 depth and 6 heads for pre-training. The data files used are already divided into train and test sets and are publicly available.

## 4.3 EEG-MVIT: architectures comparison results

In this section, the results obtained using the models and training methods presented in the previous chapters are shown. The analysis begins with a comparison between the EEG-MVIT model and the TSFF model to understand the performance that can be achieved by the model. Subsequently, the architectures presented in **section 3.3** are analyzed to investigate the reasons behind the obtained performance.

### 4.3.1 Comparison between TSFF-img and EEG-MVIT

To determine whether the EEG-MVIT model, which is primarily a Vision Transformer, is comparable in performance to TSFF-img, a CNN that is not based on Transformers or attention modules, a comparison is carried out by evaluating the intra-subject performance of the two models. The EEG-MVIT model was trained with a learning rate of  $2e-4$  on the 2a dataset, a batch size of 32, depth = 2, heads = 6, and a patch size of  $16 \times 16$ ; on the 2b dataset, the patch size was set to  $8 \times 8$ . The TSFF model was also trained with a learning rate of  $2e-4$  and a batch size of 32.

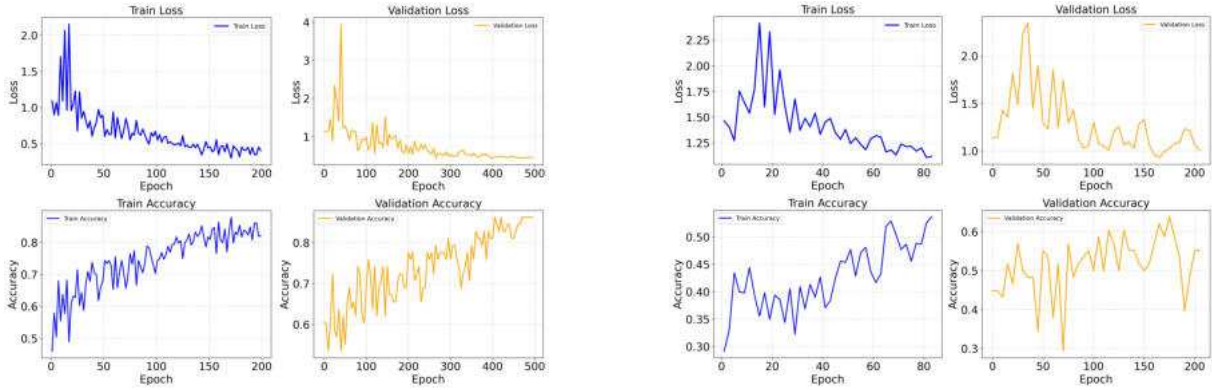


Figure 4.2: Loss and accuracy on the training and validation sets per epoch of subjects 1(left) and 4(right) from BCI Competition IV 2a using EEG-MVIT.

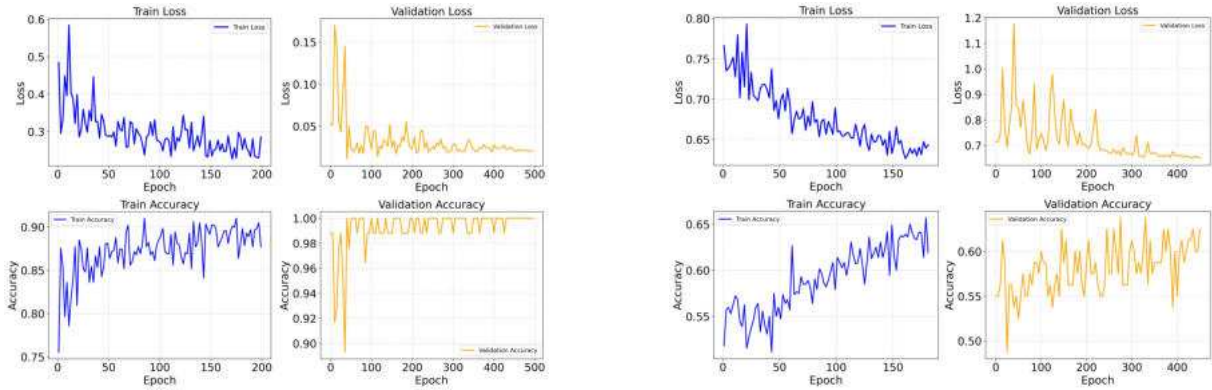


Figure 4.3: Loss and accuracy on the training and validation sets per epoch of subjects 4(left) and 2(right) from BCI Competition IV 2b using EEG-MVIT.

The 2a graphs indicate that the EEG-MVIT model converges relatively fast, probably because of the amount of input data. While training accuracy rises consistently, validation accuracy shows more variability, revealing the model's challenges in generalizing. The same pattern is observed on the 2b dataset, where the model begins with a high loss that quickly drops in the initial epochs before early stopping is triggered.

| Subject | Accuracy | Kappa |
|---------|----------|-------|
| 1       | 72.57    | 0.63  |
| 2       | 46.88    | 0.29  |
| 3       | 75.35    | 0.67  |
| 4       | 48.61    | 0.31  |
| 5       | 40.97    | 0.21  |
| 6       | 51.39    | 0.35  |
| 7       | 71.18    | 0.61  |
| 8       | 69.79    | 0.59  |
| 9       | 71.88    | 0.62  |

Table 4.1: Accuracy and kappa score of all subjects on the test set of BCI Competition IV 2a using EEG-MVIT.

| Subject | Accuracy | Kappa |
|---------|----------|-------|
| 1       | 76.56    | 0.53  |
| 2       | 57.14    | 0.14  |
| 3       | 55.94    | 0.11  |
| 4       | 93.75    | 0.87  |
| 5       | 87.50    | 0.75  |
| 6       | 87.81    | 0.75  |
| 7       | 75.31    | 0.50  |
| 8       | 85.94    | 0.71  |
| 9       | 83.44    | 0.66  |

Table 4.2: Accuracy and kappa score of all subjects on the test set of BCI Competition IV 2b using EEG-MVIT.

| Model    | Dataset | Sub1  | Sub2  | Sub3  | Sub4  | Sub5  | Sub6  | Sub7  | Sub8  | Sub9  | Mean Acc | Mean Kappa |
|----------|---------|-------|-------|-------|-------|-------|-------|-------|-------|-------|----------|------------|
| TSFF     | 2a      | 66.32 | 36.81 | 63.89 | 47.22 | 34.38 | 35.42 | 73.26 | 72.57 | 73.26 | 55.90    | 0.41       |
| EEG-MVIT | 2a      | 72.57 | 46.88 | 75.35 | 48.61 | 40.97 | 51.39 | 71.18 | 69.79 | 71.88 | 61.00    | 0.48       |
| TSFF     | 2b      | 69.69 | 60.36 | 52.19 | 96.88 | 91.56 | 76.88 | 80.94 | 92.19 | 80.94 | 77.96    | 0.56       |
| EEG-MVIT | 2b      | 76.56 | 57.14 | 55.94 | 93.75 | 87.50 | 87.81 | 75.31 | 85.94 | 83.44 | 78.15    | 0.56       |

Table 4.3: Comparison between the accuracy of all subjects on the test sets of BCI Competition IV 2a and 2b for the EEG-MVIT and TSFF models. The last columns report the average accuracy and average kappa score of both models, serving as indicators of overall performance.

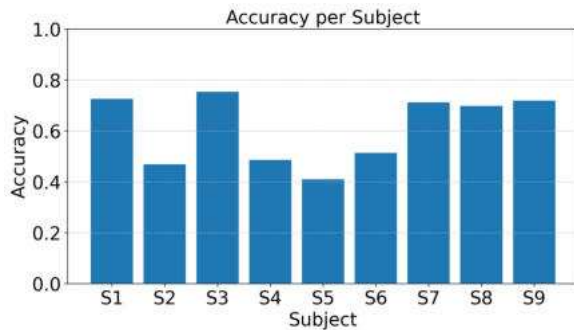


Figure 4.4: Accuracy bar chart for all subjects on the BCI Competition IV 2a test set using EEG-MVIT.

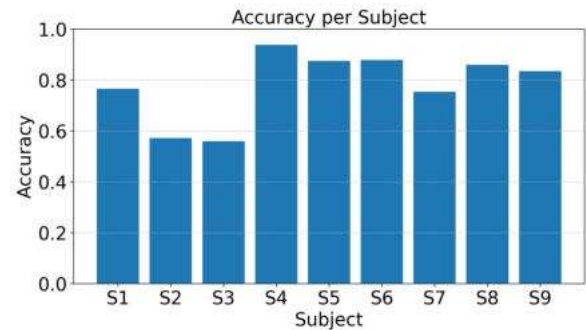


Figure 4.5: Accuracy bar chart for all subjects on the BCI Competition IV 2b test set using EEG-MVIT.

Analyzing the individual subject accuracies for the EEG-MVIT model reveals considerable performance variability. Subjects 1, 3, 7, 8, and 9 surpass 70% accuracy on dataset 2a, whereas subjects like 2 and 4 struggle to exceed 50%. This indicates substantial differences in EEG signal distributions between subjects, demonstrating the model's difficulty in generalizing with this amount of data. On dataset 2b, most subjects achieve high performance, such as subject 4, who reaches 93% accuracy; nevertheless, and as will be repeatedly observed, subjects 2 and 3 obtain accuracies well below the average of the other participants. The TSFF analysis supports this result, showing that the EEG signals of subjects 2 and 3 from dataset 2b might not carry the same information as those of other subjects, or that the networks fail to extract it properly. This

could imply that these two subjects need a specialized preprocessing approach or may benefit more from using raw data instead of the input scalogram. Nevertheless, this would lead to a subject-specific BCI rather than a model capable of generalization.

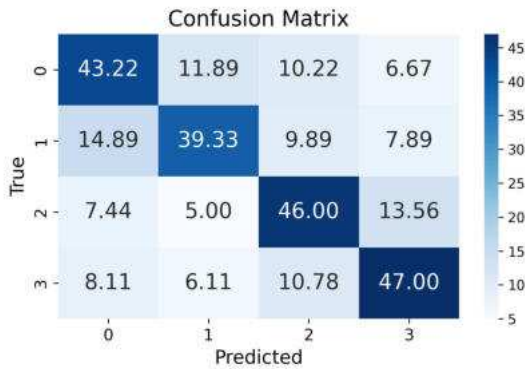


Figure 4.6: Average confusion matrix on the BCI Competition IV 2a test set using EEG-MVIT.

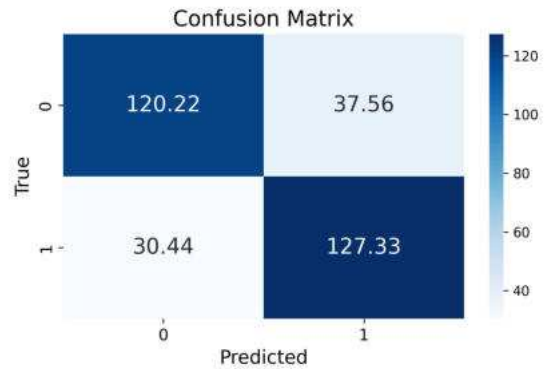


Figure 4.7: Average confusion matrix on the BCI Competition IV 2b test set using EEG-MVIT.

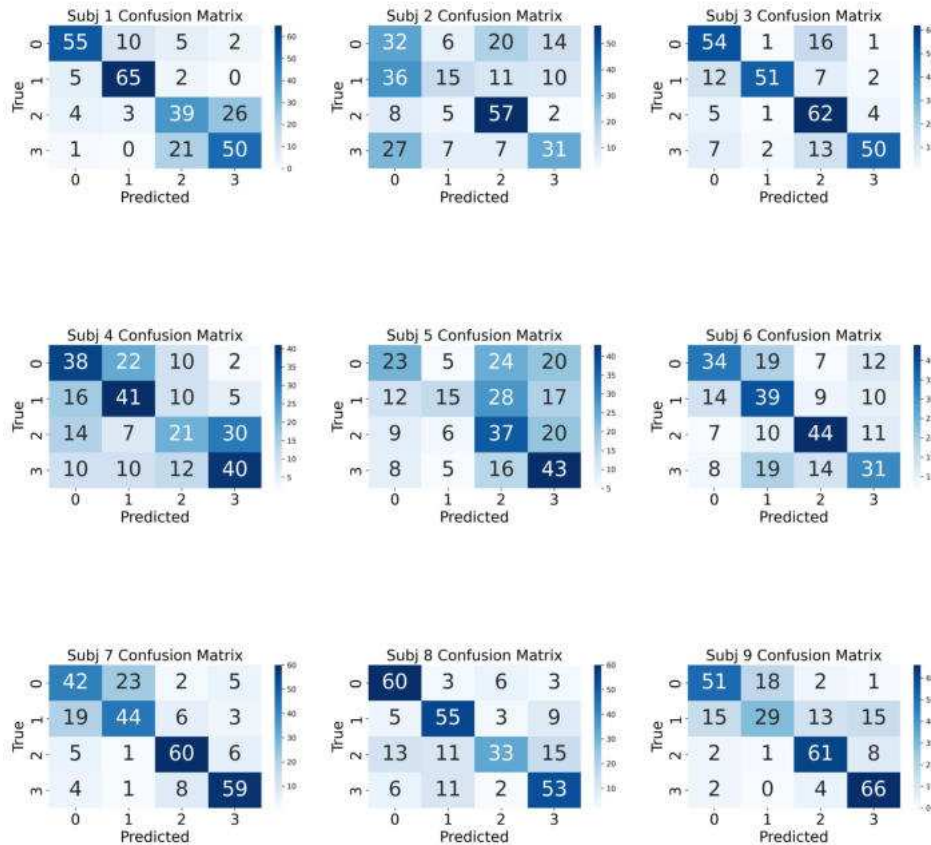


Figure 4.8: Confusion matrix of all subjects on the BCI Competition IV 2a test set using EEG-MVIT. Class 1 corresponds to left-hand movement, class 2 to right-hand movement, class 3 to foot movement, and class 4 to tongue movement.

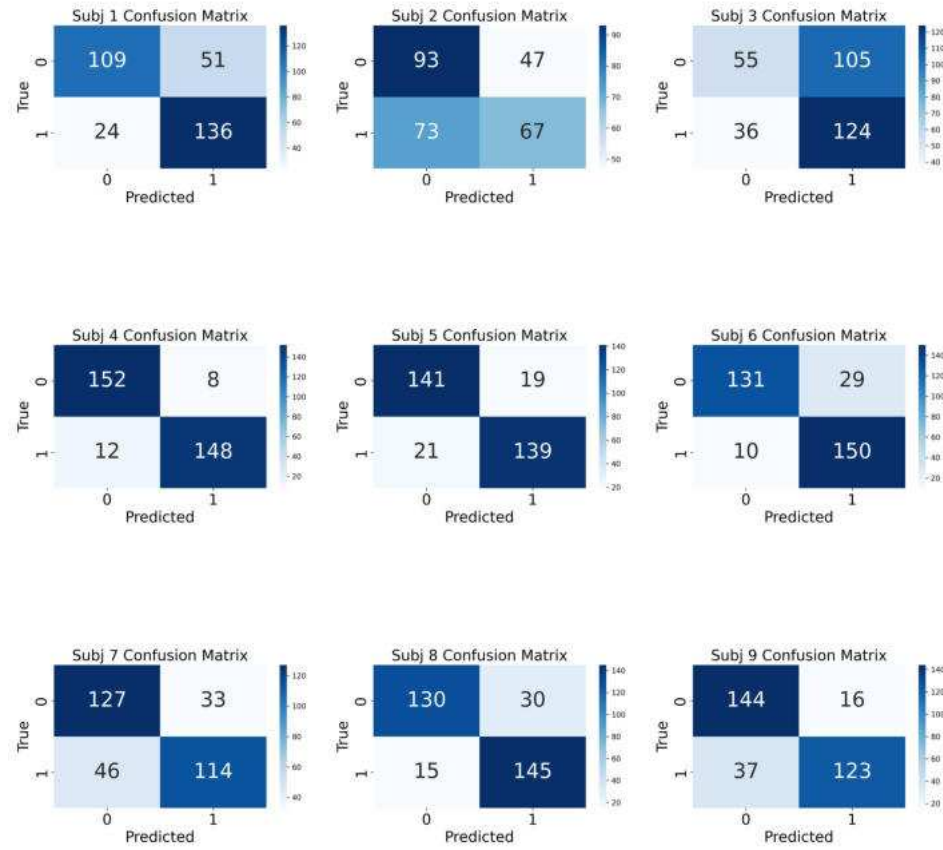


Figure 4.9: Confusion matrix of all subjects on the BCI Competition IV 2b test set using EEG-MVIT. Class 1 corresponds to left-hand movement and class 2 to right-hand movement.

Analyzing the confusion matrices of subjects on dataset 2b in **figure 4.9** is revealing. While most patients achieve correct classification, subjects 2 and 3 show failures for different reasons: the model for subject 2 successfully classifies the first class but struggles with the second, accurately detecting left-hand movements but not right-hand ones. In contrast, subject 3 correctly identifies the second class but not the first, showing better detection of right-hand movements over left-hand movements.

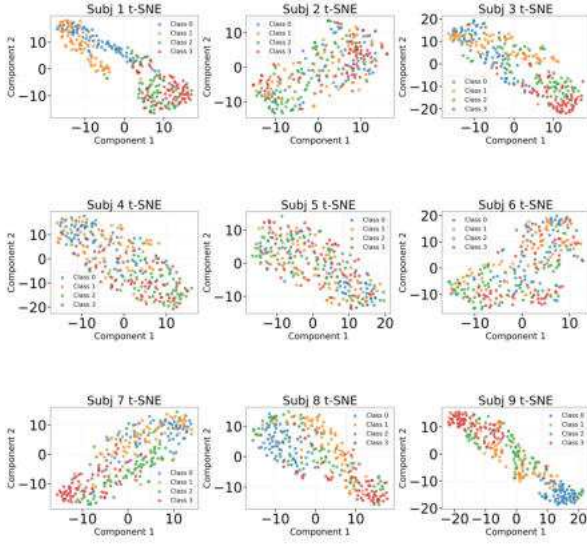


Figure 4.10: T-SNE of all subjects on the BCI Competition IV 2a test set using EEG-MVIT: blue points represent left-hand features, orange represent right-hand features, green represent feet, and red represent tongue features before classification.

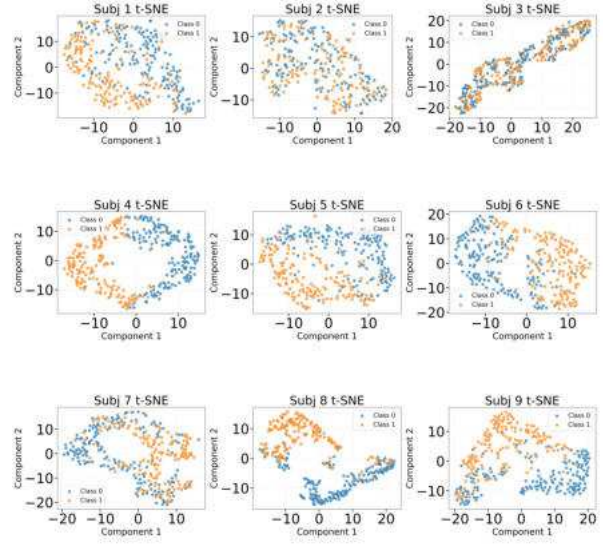


Figure 4.11: T-SNE of all subjects on the BCI Competition IV 2b test set using EEG-MVIT: blue points represent left-hand features and orange points represent right-hand features before classification.

The t-SNE plots for both datasets reveal that features from some subjects are clearly more discriminative than others. On dataset 2b, the separation is particularly clear due to the binary nature of the classification. These plots also highlight that the features generated by the model for subjects 2 and 3 do not make classification straightforward. EEG-MVIT achieves a 5% higher accuracy than TSFF on the 2a dataset, indicating that it can outperform a CNN without attention modules or Transformers. Nonetheless, it remains far from the performance levels seen in state-of-the-art models, which combine CNNs and Transformers in their architectures. We therefore proceed with the EEG-MVIT approach, which still surpasses TSFF, to explore the reasons behind its performance.

### 4.3.2 ENC-MVIT architecture analysis: comparison between channel feature fusion before or after the encoder

From the analysis carried out in **subsection 4.3.1**, it can be observed that a Vision Transformer, in this case the EEG-MVIT variant, does not reach the performance of the state-of-the-art models. However, it is able to achieve performance similar to that of the TSFF-img CNN model. To understand why, in this and the following sections, all the architectures presented in **section 3.3** will be implemented, tested, and compared, namely: ENC-MVIT, DEPTH-VIT, 3BRANCH-VIT, and LAMBDA-VIT, in order to analyze how performance varies depending on the architecture. For this reason, as previously stated, we analyze architectures that modify the main components of a ViT, namely the fusion at the encoder level, the structure of the input channels, and the patch embedding stage.

The first architecture to be implemented is ENC-MVIT, described in **subsection 3.3.1**, to verify the effect of fusing the features at the output of the patch embedding layer, for this reason there will no longer be multiple encoders per channel, but a single encoder. The ENC-MVIT model was trained with a learning rate of  $2e-4$  on the 2a dataset, a batch size of 32, depth = 2, heads = 6, and a patch size of  $16 \times 16$ ; on the 2b dataset, the patch size was set to  $8 \times 8$ .

| Subject | Accuracy | Kappa |
|---------|----------|-------|
| 1       | 74.31    | 0.65  |
| 2       | 44.79    | 0.26  |
| 3       | 78.12    | 0.70  |
| 4       | 48.26    | 0.31  |
| 5       | 40.97    | 0.21  |
| 6       | 50.00    | 0.33  |
| 7       | 64.93    | 0.53  |
| 8       | 67.36    | 0.56  |
| 9       | 70.49    | 0.60  |

Table 4.4: Accuracy and kappa score of all subjects on the test set of BCI Competition IV 2a using ENC-MVIT.

| Subject | Accuracy | Kappa |
|---------|----------|-------|
| 1       | 79.37    | 0.58  |
| 2       | 56.79    | 0.13  |
| 3       | 54.37    | 0.08  |
| 4       | 93.75    | 0.87  |
| 5       | 90.00    | 0.80  |
| 6       | 87.19    | 0.74  |
| 7       | 76.25    | 0.52  |
| 8       | 89.06    | 0.78  |
| 9       | 85.31    | 0.70  |

Table 4.5: Accuracy and kappa score of all subjects on the test set of BCI Competition IV 2b using ENC-MVIT.

| Model    | Dataset | Sub1  | Sub2  | Sub3  | Sub4  | Sub5  | Sub6  | Sub7  | Sub8  | Sub9  | Mean Acc | Mean Kappa |
|----------|---------|-------|-------|-------|-------|-------|-------|-------|-------|-------|----------|------------|
| ENC-MVIT | 2a      | 74.31 | 44.79 | 78.12 | 48.26 | 40.97 | 50.00 | 64.93 | 67.36 | 70.49 | 59.91    | 0.46       |
| EEG-MVIT | 2a      | 72.57 | 46.88 | 75.35 | 48.61 | 40.97 | 51.39 | 71.18 | 69.79 | 71.88 | 61.00    | 0.48       |
| ENC-MVIT | 2b      | 79.37 | 56.79 | 54.37 | 93.75 | 90.00 | 87.19 | 76.25 | 89.06 | 85.31 | 79.12    | 0.58       |
| EEG-MVIT | 2b      | 76.56 | 57.14 | 55.94 | 93.75 | 87.50 | 87.81 | 75.31 | 85.94 | 83.44 | 78.15    | 0.56       |

Table 4.6: Comparison between the accuracy of all subjects on the test sets of BCI Competition IV 2a and 2b for the EEG-MVIT and ENC-MVIT models. The last columns report the average accuracy and average kappa-score of both models, serving as indicators of overall performance.

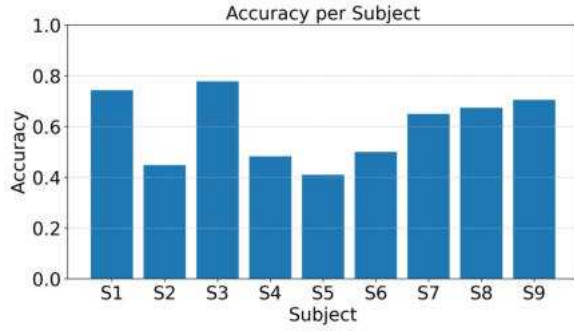


Figure 4.12: Accuracy bar chart for all subjects on the BCI Competition IV 2a test set using ENC-MVIT.

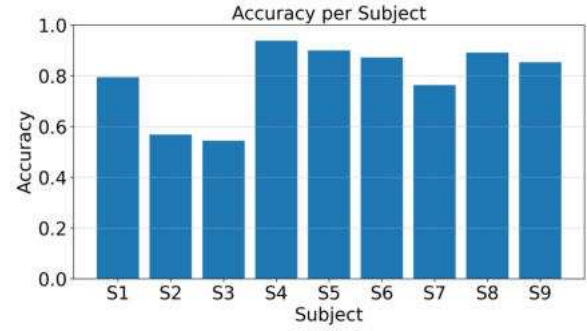


Figure 4.13: Accuracy bar chart for all subjects on the BCI Competition IV 2b test set using ENC-MVIT.

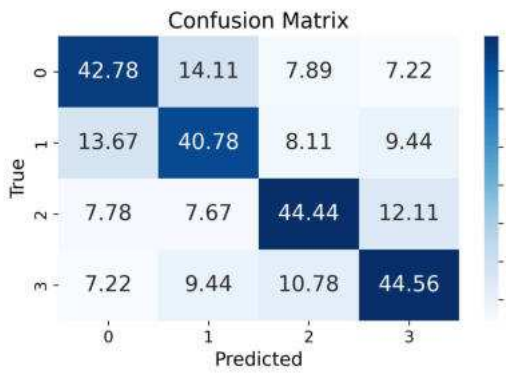


Figure 4.14: Average confusion matrix on the BCI Competition IV 2a test set using ENC-MVIT. The main error occurs in hand classification.

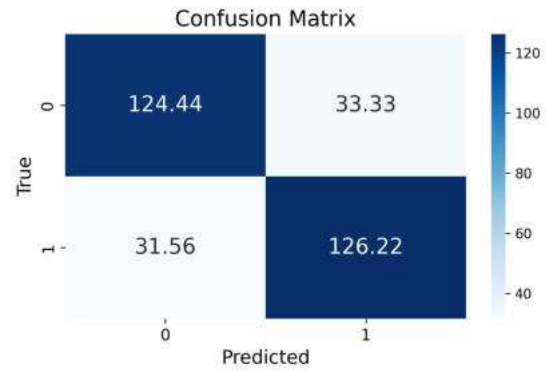


Figure 4.15: Average confusion matrix on the BCI Competition IV 2b test set using ENC-MVIT. The classification appears balanced, and the error in the hands is lower than in 2a.

The results suggest that this architecture does not lead to meaningful performance gains, in fact, even if it gains 1% on 2b, it loses 1% accuracy on 2a. In the plots for dataset 2b, even for subjects 1, 2, and 3, the behavior is almost identical to that of the EEG-MVIT. An inspection of the confusion matrices and bar plots reveals no major differences compared to ENC-MVIT. Thus, while employing a single encoder can be beneficial in terms of memory efficiency, it comes at the cost of losing the initial channel-wise independence and 1% of accuracy on 2a.

### 4.3.3 DEPTH-VIT architecture analysis: comparison of input channel structure

The next architecture implemented is DEPTH-VIT, as described in **subsection 3.3.2**, to test an alternative configuration of the input channels. Here, the channels are concatenated along the depth dimension, creating an image-like representation, while still using a single Transformer encoder that processes all channels, similar to the previous architecture. The DEPTH-VIT model was trained with a learning rate of  $2e-6$  on the 2a dataset, a batch size of 32, depth = 2, heads = 6, and a patch size of  $16 \times 16$ ; on the 2b dataset, the patch size was set to  $8 \times 8$  and the learning rate was  $2e-4$ .

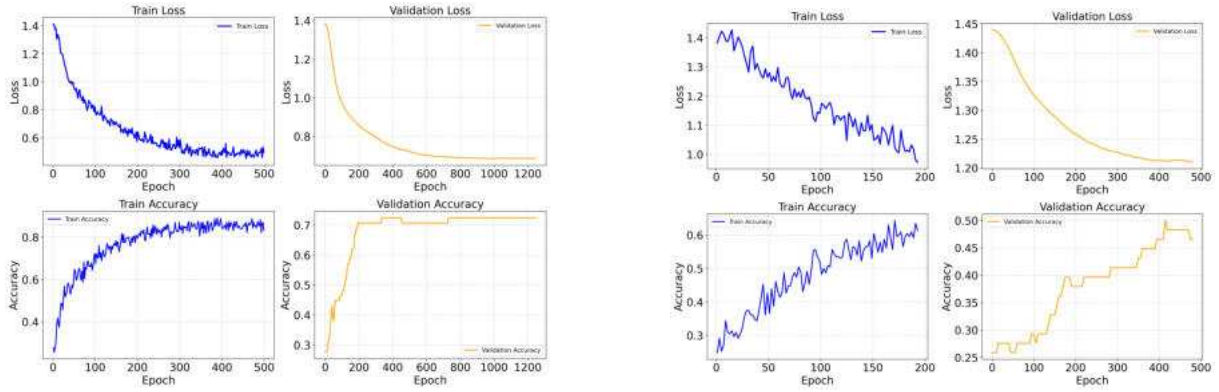


Figure 4.16: Loss and accuracy on the training and validation sets per epoch of subjects 1(left) and 4(right) from BCI Competition IV 2a using DEPTH-VIT.

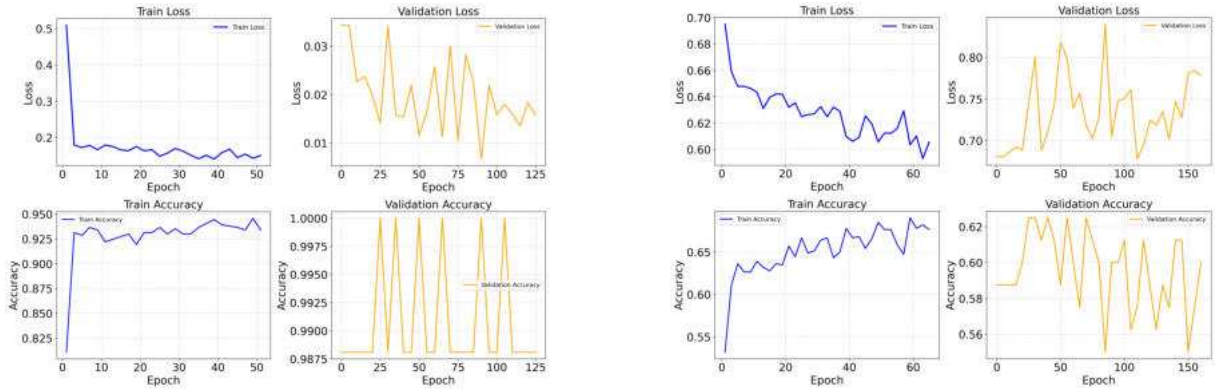


Figure 4.17: Loss and accuracy on the training and validation sets per epoch of subjects 4(left) and 2(right) from BCI Competition IV 2b using DEPTH-VIT.

| Subject | Accuracy | Kappa |
|---------|----------|-------|
| 1       | 76.74    | 0.69  |
| 2       | 53.12    | 0.37  |
| 3       | 81.94    | 0.76  |
| 4       | 57.99    | 0.44  |
| 5       | 45.49    | 0.27  |
| 6       | 55.21    | 0.40  |
| 7       | 77.43    | 0.70  |
| 8       | 69.10    | 0.59  |
| 9       | 75.35    | 0.67  |

Table 4.7: Accuracy and kappa score of all subjects on the test set of BCI Competition IV 2a using DEPTH-VIT.

| Subject | Accuracy | Kappa |
|---------|----------|-------|
| 1       | 76.88    | 0.54  |
| 2       | 58.93    | 0.18  |
| 3       | 55.94    | 0.12  |
| 4       | 94.06    | 0.88  |
| 5       | 88.44    | 0.77  |
| 6       | 87.19    | 0.74  |
| 7       | 72.19    | 0.44  |
| 8       | 88.12    | 0.76  |
| 9       | 85.94    | 0.71  |

Table 4.8: Accuracy and kappa score of all subjects on the test set of BCI Competition IV 2b using DEPTH-VIT.

| Model     | Dataset | Sub1  | Sub2  | Sub3  | Sub4  | Sub5  | Sub6  | Sub7  | Sub8  | Sub9  | Mean Acc | Mean Kappa |
|-----------|---------|-------|-------|-------|-------|-------|-------|-------|-------|-------|----------|------------|
| DEPTH-VIT | 2a      | 76.74 | 53.12 | 81.94 | 57.99 | 45.49 | 55.21 | 77.43 | 69.10 | 75.35 | 65.90    | 0.54       |
| EEG-MVIT  | 2a      | 72.57 | 46.88 | 75.35 | 48.61 | 40.97 | 51.39 | 71.18 | 69.79 | 71.88 | 61.00    | 0.48       |
| DEPTH-VIT | 2b      | 76.88 | 58.93 | 55.94 | 94.06 | 88.44 | 87.19 | 72.19 | 88.12 | 85.94 | 78.63    | 0.57       |
| EEG-MVIT  | 2b      | 76.56 | 57.14 | 55.94 | 93.75 | 87.50 | 87.81 | 75.31 | 85.94 | 83.44 | 78.15    | 0.56       |

Table 4.9: Comparison between the accuracy of all subjects on the test sets of BCI Competition IV 2a and 2b for the EEG-MVIT and DEPTH-VIT models. The last columns report the average accuracy and average kappa-score of both models, serving as indicators of overall performance.

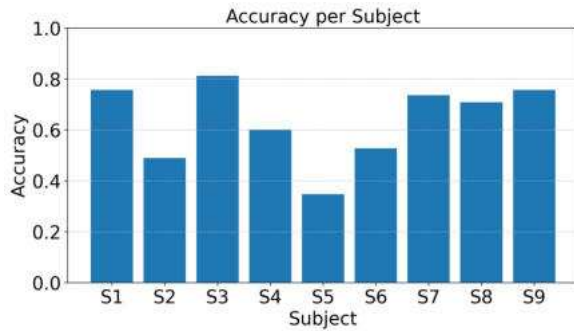


Figure 4.18: Accuracy bar chart for all subjects on the BCI Competition IV 2a test set using DEPTH-VIT.

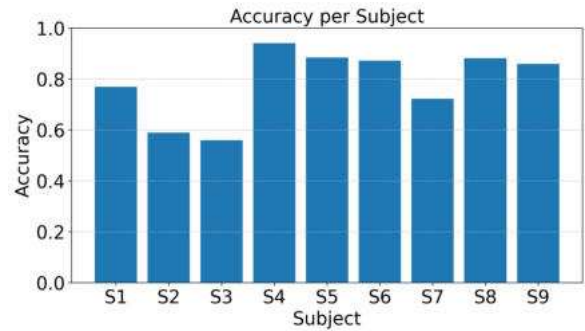


Figure 4.19: Accuracy bar chart for all subjects on the BCI Competition IV 2b test set using DEPTH-VIT.

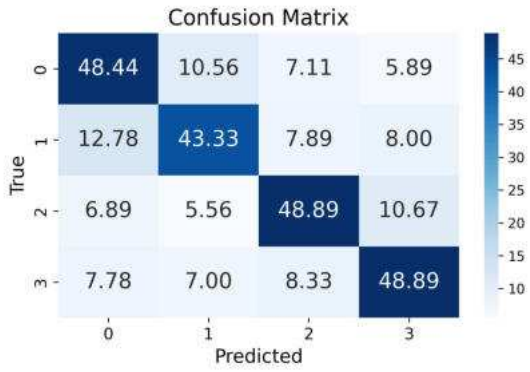


Figure 4.20: Average confusion matrix on the BCI Competition IV 2a test set using DEPTH-VIT. Compared to the previous models, the error mainly occurs on the right hand rather than on both.

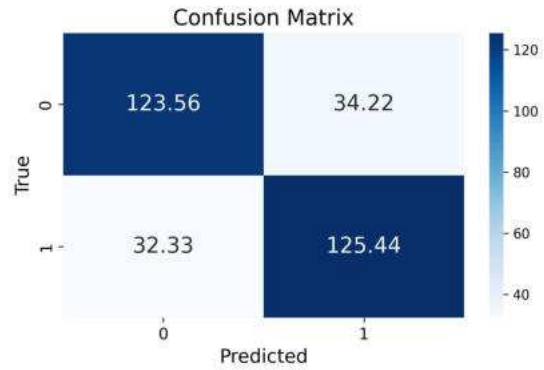


Figure 4.21: Average confusion matrix on the BCI Competition IV 2b test set using DEPTH-VIT. The classification does not differ much from the previous one.

Depth-VIT reaches 66% accuracy on 2a, with subjects 2, 3, and 4 showing an accuracy boost of 7 to 10%. For dataset 2b, the performance gain is limited to just 1%. Such a difference can also be noticed in the loss pattern of dataset 2b, which remains quite similar to that of the EEG-MVIT, while the loss on dataset 2a exhibits a smoother, less oscillatory behavior. This difference in performance gains between the two datasets explains these results: dataset 2a has fewer samples than dataset 2b, therefore a network with fewer parameters than those previously considered and a single branch for the channels performs better. Moreover, the most important factor to consider is that with a network having fewer parameters, the patch embedding layer becomes increasingly important. This hypothesis will serve as the starting point to analyze the last two architectures.

#### 4.3.4 3BRANCH-VIT architecture analysis: effect of varying patch sizes

The next architecture implemented is 3BRANCH-VIT, as outlined in **subsection 3.3.3**, to conduct a deeper analysis of the patch embedding stage. The patch embedding layer, as previously mentioned, also includes a CNN layer responsible for dividing the input into patches. In this architecture, patches of different sizes are implemented, aiming to emphasize either the temporal or frequency components. The 3BRANCH-VIT model was trained with a learning rate of  $2e-4$  on the 2a dataset, a batch size of 32, depth = 2, heads = 6, and a patch size of  $1 \times 336$ ,  $32 \times 1$ ,  $16 \times 8$ ; on the 2b dataset, the patch size was set to  $1 \times 125$ ,  $32 \times 1$ ,  $16 \times 8$  and the learning rate was  $2e-4$ .

| Subject | Accuracy | Kappa score |
|---------|----------|-------------|
| 1       | 72.92    | 0.64        |
| 2       | 51.39    | 0.35        |
| 3       | 78.12    | 0.70        |
| 4       | 60.42    | 0.47        |
| 5       | 37.85    | 0.17        |
| 6       | 54.83    | 0.28        |
| 7       | 75.35    | 0.67        |
| 8       | 70.83    | 0.61        |
| 9       | 77.78    | 0.70        |

Table 4.10: Accuracy and kappa score of all subjects on the test set of BCI Competition IV 2a using 3BRANCH-VIT.

| Subject | Accuracy | Kappa score |
|---------|----------|-------------|
| 1       | 78.12    | 0.56        |
| 2       | 56.79    | 0.13        |
| 3       | 58.13    | 0.16        |
| 4       | 95.94    | 0.92        |
| 5       | 90.94    | 0.82        |
| 6       | 90.31    | 0.80        |
| 7       | 72.50    | 0.45        |
| 8       | 91.87    | 0.83        |
| 9       | 86.88    | 0.74        |

Table 4.11: Accuracy and kappa score of all subjects on the test set of BCI Competition IV 2b using 3BRANCH-VIT.

| Model                   | Dataset | Sub1  | Sub2  | Sub3  | Sub4  | Sub5  | Sub6  | Sub7  | Sub8  | Sub9  | Mean Acc | Mean Kappa |
|-------------------------|---------|-------|-------|-------|-------|-------|-------|-------|-------|-------|----------|------------|
| 3BRANCH-VIT<br>EEG-MVIT | 2a      | 72.92 | 51.39 | 78.12 | 60.42 | 37.85 | 45.83 | 75.35 | 70.83 | 77.78 | 63.39    | 0.51       |
|                         |         | 72.57 | 46.88 | 75.35 | 48.61 | 40.97 | 51.39 | 71.18 | 69.79 | 71.88 | 61.00    | 0.48       |
| 3BRANCH-VIT<br>EEG-MVIT | 2b      | 78.12 | 56.79 | 58.13 | 95.94 | 90.94 | 90.31 | 72.50 | 91.87 | 86.88 | 80.00    | 0.60       |
|                         |         | 76.56 | 57.14 | 55.94 | 93.75 | 87.50 | 87.81 | 75.31 | 85.94 | 83.44 | 78.15    | 0.56       |

Table 4.12: Comparison between the accuracy of all subjects on the test sets of BCI Competition IV 2a and 2b for the EEG-MVIT and 3BRANCH-VIT models. The last columns report the average accuracy and average kappa-score of both models, serving as indicators of overall performance.

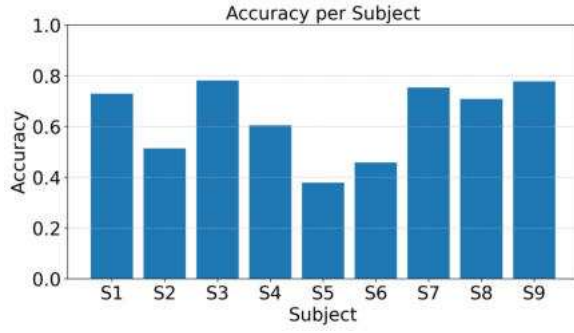


Figure 4.22: Accuracy bar chart for all subjects on the BCI Competition IV 2a test set using 3BRANCH-VIT.

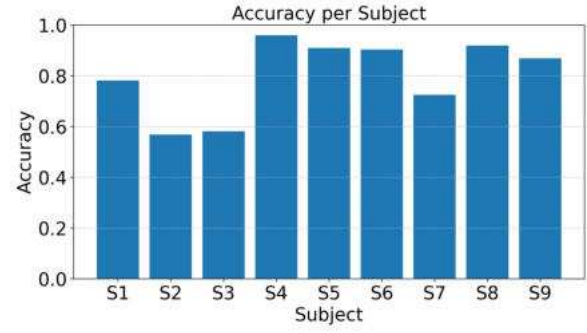


Figure 4.23: Accuracy bar chart for all subjects on the BCI Competition IV 2b test set using 3BRANCH-VIT.

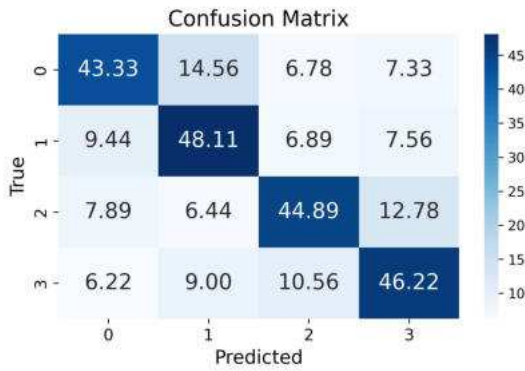


Figure 4.24: Average confusion matrix on the BCI Competition IV 2a test set using 3BRANCH-VIT.

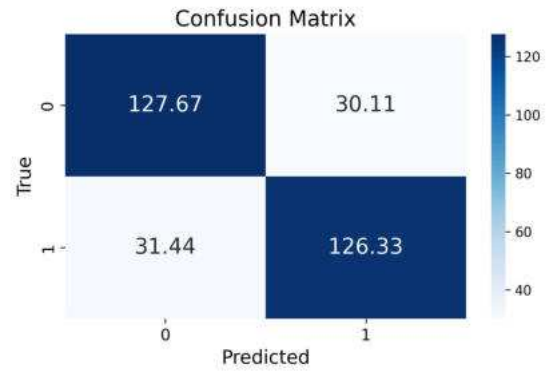


Figure 4.25: Average confusion matrix on the BCI Competition IV 2b test set using 3BRANCH-VIT.

The 3BRANCH-VIT achieves 63% accuracy on dataset 2a and reaches 80% accuracy on dataset 2b. The effect of using patches with different sizes is positive; as shown by other works, the use of rectangular patches leads to performance improvements. Nevertheless, the performance improvement on 2b is still less evident compared to 2a, highlighting the larger amount of available data that helps stabilize training. The final experiment follows this direction, further investigating the patch embedding layer.

### 4.3.5 LAMBDA-VIT architecture analysis: effect of inductive bias on performance

The last architecture to be implemented is LAMBDA-VIT, described in **subsection 3.3.4**. This architecture has a direct connection with the single convolutional layer, and it is the closest to state-of-the-art models, which are often CNN-Transformer hybrids. Through this architecture, it is possible to investigate the reasons behind the use of hybrid architectures rather than standalone Vision Transformers. The LAMBDA-VIT model was trained with a learning rate of  $2e-4$  on the 2a dataset, a batch size of 32, depth = 2, heads = 6, and a patch size of  $1 \times 336$ ,  $32 \times 1$ ,  $16 \times 8$ ; on the 2b dataset, the patch size was set to  $1 \times 125$ ,  $32 \times 1$ ,  $16 \times 8$  and the learning rate was  $2e-4$ . The final value of lambda chosen is 0.7.

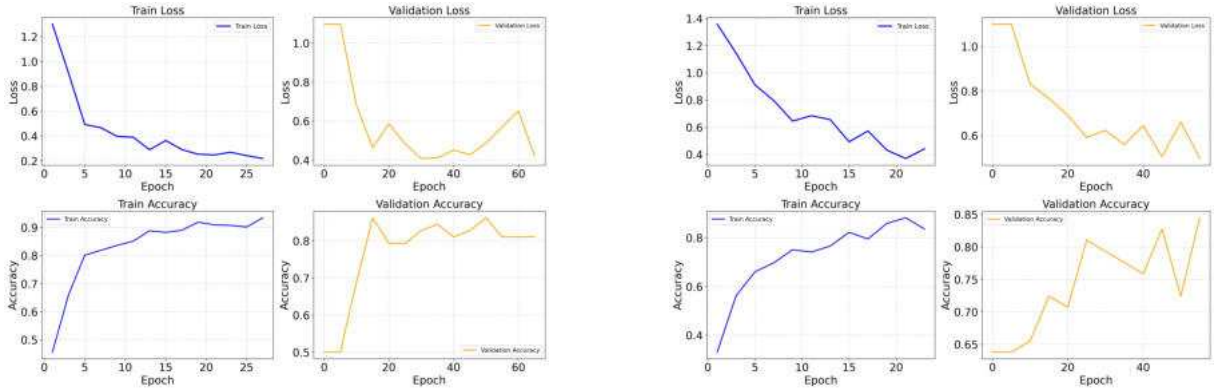


Figure 4.26: Loss and accuracy on the training and validation sets per epoch of subjects 1(left) and 4(right) from BCI Competition IV 2a using LAMBDA-VIT.

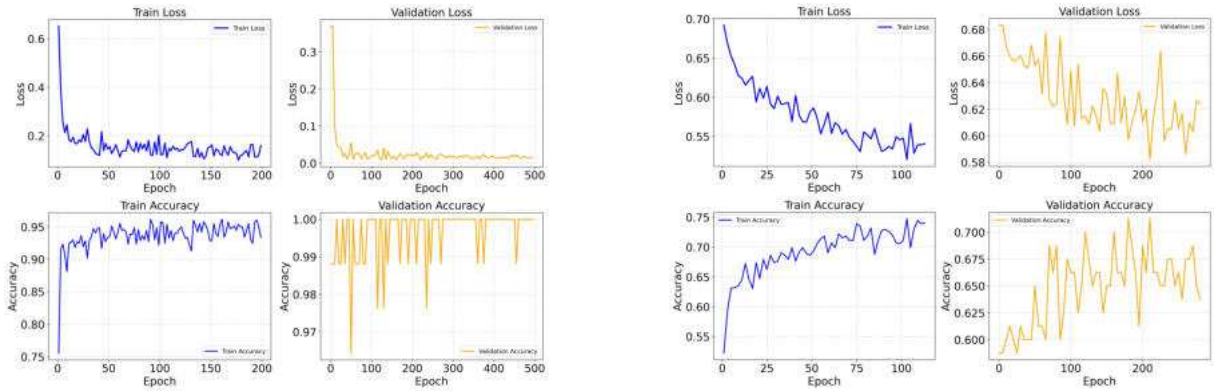


Figure 4.27: Loss and accuracy on the training and validation sets per epoch of subjects 4(left) and 2(right) from BCI Competition IV 2b using LAMBDA-VIT.

| Model      | $\lambda$  | Sub1  | Sub2  | Sub3  | Sub4  | Sub5  | Sub6  | Sub7  | Sub8  | Sub9  | Mean Acc |
|------------|------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|----------|
| LAMBDA-VIT | 0.4        | 76.73 | 49.30 | 84.02 | 60.41 | 42.01 | 51.73 | 79.16 | 75.00 | 71.18 | 65.50    |
|            | 0.5        | 80.90 | 53.47 | 83.33 | 61.11 | 43.75 | 46.18 | 81.94 | 75.34 | 71.52 | 66.39    |
|            | <b>0.7</b> | 79.86 | 51.39 | 85.07 | 58.68 | 45.14 | 51.74 | 81.25 | 75.00 | 74.65 | 66.98    |
|            | 0.9        | 78.81 | 52.08 | 86.80 | 60.76 | 46.87 | 50.00 | 81.25 | 75.69 | 71.18 | 67.00    |

Table 4.13: Comparison between the accuracy of all subjects on the test sets of BCI Competition IV 2a varying lambda. Increasing  $\lambda$  enhances the contribution of the CNN layer, leading to better performance until  $\lambda = 0.9$ , highlighting the significant impact that even a single CNN layer can have on the model's performance when working with a dataset containing such a limited number of samples.

| Subject | Accuracy | Kappa |
|---------|----------|-------|
| 1       | 79.86    | 0.73  |
| 2       | 51.39    | 0.35  |
| 3       | 85.07    | 0.80  |
| 4       | 58.68    | 0.45  |
| 5       | 45.14    | 0.27  |
| 6       | 51.74    | 0.36  |
| 7       | 81.25    | 0.75  |
| 8       | 75.00    | 0.66  |
| 9       | 74.65    | 0.66  |

Table 4.14: Accuracy and kappa score of all subjects on the test set of BCI Competition IV 2a using LAMBDA-VIT.

| Subject | Accuracy | Kappa |
|---------|----------|-------|
| 1       | 78.75    | 0.58  |
| 2       | 57.86    | 0.16  |
| 3       | 53.75    | 0.07  |
| 4       | 96.25    | 0.92  |
| 5       | 90.00    | 0.80  |
| 6       | 89.69    | 0.79  |
| 7       | 74.06    | 0.48  |
| 8       | 91.56    | 0.83  |
| 9       | 88.12    | 0.76  |

Table 4.15: Accuracy and kappa score of all subjects on the test set of BCI Competition IV 2b using LAMBDA-VIT.

| Model      | Dataset | Sub1  | Sub2  | Sub3  | Sub4  | Sub5  | Sub6  | Sub7  | Sub8  | Sub9  | Mean Acc | Mean Kappa |
|------------|---------|-------|-------|-------|-------|-------|-------|-------|-------|-------|----------|------------|
| LAMBDA-VIT | 2a      | 79.86 | 51.39 | 85.07 | 58.68 | 45.14 | 51.74 | 81.25 | 75.00 | 74.65 | 66.98    | 0.56       |
| EEG-MVIT   |         | 72.57 | 46.88 | 75.35 | 48.61 | 40.97 | 51.39 | 71.18 | 69.79 | 71.88 | 61.00    | 0.48       |
| LAMBDA-VIT | 2b      | 78.75 | 57.86 | 53.75 | 96.25 | 90.00 | 89.69 | 74.06 | 91.56 | 88.12 | 80.00    | 0.60       |
| EEG-MVIT   |         | 76.56 | 57.14 | 55.94 | 93.75 | 87.50 | 87.81 | 75.31 | 85.94 | 83.44 | 78.15    | 0.56       |

Table 4.16: Comparison between the accuracy of all subjects on the test sets of BCI Competition IV 2a and 2b for the EEG-MVIT and LAMBDA-VIT models. The last columns report the average accuracy and average kappa-score of both models, serving as indicators of overall performance.

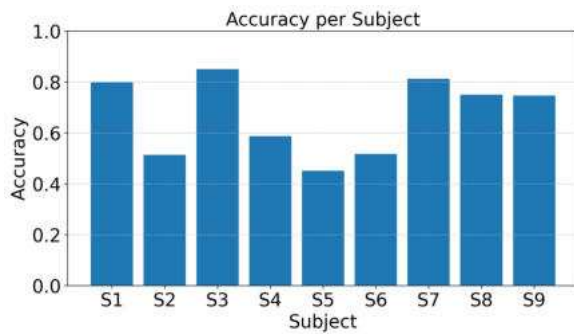


Figure 4.28: Accuracy bar chart for all subjects on the BCI Competition IV 2a test set using LAMBDA-VIT.

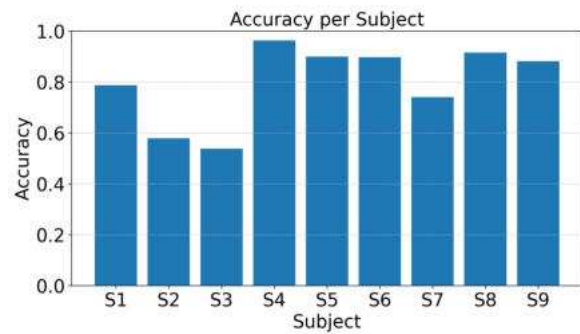


Figure 4.29: Accuracy bar chart for all subjects on the BCI Competition IV 2b test set using LAMBDA-VIT.

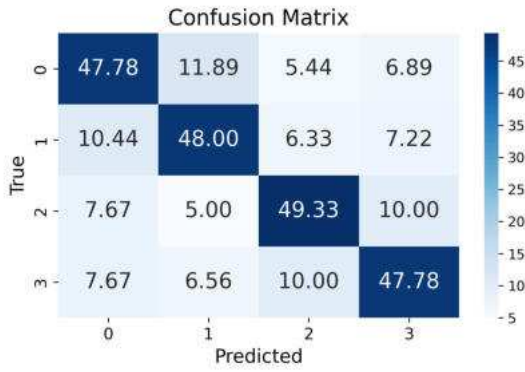


Figure 4.30: Average confusion matrix on the BCI Competition IV 2a test set using LAMBDA-VIT.

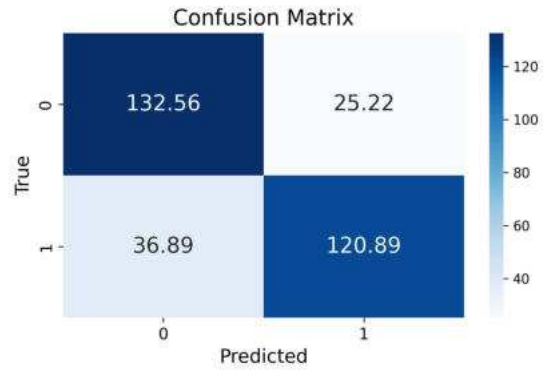


Figure 4.31: Average confusion matrix on the BCI Competition IV 2b test set using LAMBDA-VIT.

The LAMBDA-VIT reaches 67% accuracy on 2a and 80% accuracy on 2b, making it the best performing architecture. The performance increase is particularly evident on the 2a dataset, where there is a 6% improvement compared to EEG-MVIT. From the confusion matrix of 2a, it can be observed that the accuracy of each class has increased; in particular, the gap between the classification of the hand-related classes has been reduced. An improvement in the performance of certain subjects is noticeable: both Subject 3 and Subject 7 show a 10% increase in performance relative to the EEG-MVIT model. This clearly highlights the direct effect of the CNN, which, on small datasets like 2a, performs better than a Vision Transformer model thanks to the presence of inductive bias. It is worth noting that, on the BCI Competition IV 2b dataset, the performance improvement is significantly smaller than that observed on 2a, performance gains are observed, although they are much smaller compared to those on the 2a dataset. This result provides evidence that the inductive bias has a stronger effect in data-scarce scenarios.

### 4.3.6 Architectures comparison conclusions and supervised pre-train: larger-scale pretraining could improve performance

By examining the results reported in **table 4.17**, it can be observed that the performance ranges between 60% and 67%, reaching its peak when using LAMBDA-VIT, which emphasizes the single convolutional layer present in the patch embedding layer. On the 2a dataset, there is a performance increase of 6%; however, on the 2b dataset, the performance is already much more stable, thanks to the larger amount of data available.

| Model       | Dataset | Sub1  | Sub2  | Sub3  | Sub4  | Sub5  | Sub6  | Sub7  | Sub8  | Sub9  | Mean Acc $\pm$ std | Mean Kappa $\pm$ std |
|-------------|---------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--------------------|----------------------|
| EEG-MVIT    | 2a      | 72.57 | 46.88 | 75.35 | 48.61 | 40.97 | 51.39 | 71.18 | 69.79 | 71.88 | 61.00 $\pm$ 13.63  | 0.48 $\pm$ 0.18      |
| ENC-MVIT    |         | 74.31 | 44.79 | 78.12 | 48.26 | 40.97 | 50.00 | 64.93 | 67.36 | 70.49 | 59.91 $\pm$ 13.93  | 0.46 $\pm$ 0.18      |
| 3BRANCH-VIT |         | 72.92 | 51.39 | 78.12 | 60.42 | 37.85 | 45.83 | 75.35 | 70.83 | 77.78 | 63.39 $\pm$ 15.11  | 0.51 $\pm$ 0.20      |
| DEPTH-VIT   |         | 76.74 | 53.12 | 81.94 | 57.99 | 45.49 | 55.21 | 77.43 | 69.10 | 75.35 | 65.90 $\pm$ 13.06  | 0.54 $\pm$ 17        |
| LAMBDA-VIT  |         | 79.86 | 51.39 | 85.07 | 58.68 | 45.14 | 51.74 | 81.25 | 75.00 | 74.65 | 66.98 $\pm$ 15.17  | 0.56 $\pm$ 0.20      |

Table 4.17: Comparison between the accuracy of all subjects on the BCI Competition IV 2a test set across all EEG-MVIT variations. Accuracy increases as the number of model parameters decreases, reaching its maximum with the LAMBDA-VIT model, which gives more importance to the convolutional layer by leveraging the inductive bias.

| Model       | Dataset | Sub1  | Sub2  | Sub3  | Sub4  | Sub5  | Sub6  | Sub7  | Sub8  | Sub9  | Mean Acc $\pm$ std | Mean Kappa $\pm$ std |
|-------------|---------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--------------------|----------------------|
| EEG-MVIT    | 2b      | 76.56 | 57.14 | 55.94 | 93.75 | 87.50 | 87.81 | 75.31 | 85.94 | 83.44 | 78.15 $\pm$ 13.50  | 0.56 $\pm$ 0.27      |
| DEPTH-VIT   |         | 76.88 | 58.93 | 55.94 | 94.06 | 88.44 | 87.19 | 72.19 | 88.12 | 85.94 | 78.63 $\pm$ 13.69  | 0.57 $\pm$ 0.27      |
| ENC-MVIT    |         | 79.37 | 56.79 | 54.37 | 93.75 | 90.00 | 87.19 | 76.25 | 89.06 | 85.31 | 79.12 $\pm$ 14.38  | 0.58 $\pm$ 0.28      |
| 3BRANCH-VIT |         | 78.12 | 56.79 | 58.13 | 95.94 | 90.94 | 90.31 | 72.50 | 91.87 | 86.88 | 80.00 $\pm$ 14.77  | 0.60 $\pm$ 0.29      |
| LAMBDA-VIT  |         | 78.75 | 57.86 | 53.75 | 96.25 | 90.00 | 89.69 | 74.06 | 91.56 | 88.12 | 80.00 $\pm$ 15.30  | 0.60 $\pm$ 0.30      |

Table 4.18: Comparison between the accuracy of all subjects on the BCI Competition IV 2b test set across all EEG-MVIT variations. Accuracy does not vary significantly compared to 2a, mainly due to the larger amount of data. It is worth noting that subjects 2 and 3 show accuracy values that differ substantially from the others, which significantly lowers the overall performance.

However, during the experiments, it became clear that the significant performance gain is not only a result of using a dedicated branch to emphasize the CNN, but also of applying the S&R augmentation described in **subsection 3.1.4**.

| Model                         | Dataset | Sub1  | Sub2  | Sub3  | Sub4  | Sub5  | Sub6  | Sub7  | Sub8  | Sub9  | Mean Acc $\pm$ std | Mean Kappa $\pm$ std |
|-------------------------------|---------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--------------------|----------------------|
| EEG-MVIT (no augmentation)    | 2a      | 68.06 | 46.53 | 79.86 | 49.65 | 36.81 | 48.26 | 63.54 | 69.44 | 72.92 | 59.45 $\pm$ 14.53  | 0.46 $\pm$ 0.19      |
| ENC-MVIT (no augmentation)    |         | 74.65 | 44.10 | 80.90 | 47.92 | 35.42 | 50.35 | 68.40 | 68.06 | 70.14 | 60.00 $\pm$ 15.76  | 0.46 $\pm$ 0.21      |
| 3BRANCH-VIT (no augmentation) |         | 69.10 | 52.08 | 81.25 | 53.47 | 35.76 | 48.96 | 77.43 | 68.75 | 64.24 | 61.23 $\pm$ 14.72  | 0.48 $\pm$ 0.19      |
| DEPTH-VIT (no augmentation)   |         | 75.69 | 48.96 | 81.25 | 60.07 | 34.72 | 52.78 | 73.61 | 70.83 | 75.69 | 63.73 $\pm$ 15.55  | 0.52 $\pm$ 0.20      |
| LAMBDA-VIT (no augmentation)  |         | 77.08 | 54.86 | 80.55 | 55.90 | 34.72 | 49.65 | 72.22 | 74.31 | 73.26 | 63.62 $\pm$ 15.48  | 0.51 $\pm$ 0.20      |

Table 4.19: Comparison between the accuracy of all subjects on the BCI Competition IV 2a test set across all EEG-MVIT variations without augmentation. The performance is fairly consistent and shows about 2% lower accuracy compared to the augmented models.

| Model                         | Dataset | Sub1  | Sub2  | Sub3  | Sub4  | Sub5  | Sub6  | Sub7  | Sub8  | Sub9  | Mean Acc $\pm$ std | Mean Kappa $\pm$ std |
|-------------------------------|---------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--------------------|----------------------|
| EEG-MVIT (no augmentation)    | 2b      | 78.44 | 57.14 | 51.56 | 95.00 | 90.31 | 85.00 | 68.13 | 84.06 | 84.06 | 77.08 $\pm$ 14.95  | 0.54 $\pm$ 0.29      |
| DEPTH-VIT (no augmentation)   |         | 78.44 | 56.79 | 54.37 | 93.44 | 86.25 | 88.44 | 73.75 | 87.50 | 85.31 | 78.25 $\pm$ 14.08  | 0.56 $\pm$ 0.28      |
| ENC-MVIT (no augmentation)    |         | 78.44 | 50.00 | 55.31 | 92.19 | 90.00 | 79.69 | 74.69 | 84.06 | 85.94 | 76.70 $\pm$ 15.76  | 14.76 $\pm$ 0.29     |
| 3BRANCH-VIT (no augmentation) |         | 73.75 | 57.86 | 58.75 | 95.94 | 87.81 | 83.75 | 75.31 | 90.00 | 87.81 | 79.00 $\pm$ 13.62  | 0.58 $\pm$ 0.27      |
| LAMBDA-VIT (no augmentation)  |         | 75.00 | 55.00 | 53.12 | 96.88 | 89.06 | 90.00 | 72.50 | 89.06 | 85.94 | 78.51 $\pm$ 15.78  | 0.57 $\pm$ 0.31      |

Table 4.20: Comparison between the accuracy of all subjects on the BCI Competition IV 2b test set across all EEG-MVIT variations without augmentation. Even on dataset 2b, the performance is quite consistent, showing about 2% lower accuracy compared to the augmented models.

As can be seen from the tables, the performance of EEG-MViT and its variations increases by about 2% on 2a thanks to S&R. On 2b, the increase is slightly smaller, because the initial dataset contains more data and the baseline evaluation is already more stable. From these results, it can be concluded that S&R is an effective method for the motor imagery task. The most important finding, however, is that the architecture responds positively to an increasing amount of data, being able to better interpret input signals, even from signals reconstructed from the original ones, demonstrating its ability to generalize as the amount of data grows. To explore more deeply the link between dataset size and performance in a Vision Transformer, an alternative to intra-subject training was adopted, leveraging data from all patients, as described in **sub-section 3.4.1**. The supervised pre-train is a procedure designed to enhance the generalization of the network; intra-subject training, on the other hand, concentrates solely on the data of a single subject, based on the assumption that the EEG signals of that subject differ from those of others. With a Vision Transformer, however, the network might generalize and achieve better performance. Indeed, as noted earlier, inductive bias helps when data is scarce, but with larger datasets, it creates an upper limit on performance, hindering generalization.

| Model                      | Dataset | Sub1  | Sub2  | Sub3  | Sub4  | Sub5  | Sub6  | Sub7  | Sub8  | Sub9  | Mean Acc $\pm$ std | Mean Kappa $\pm$ std |
|----------------------------|---------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--------------------|----------------------|
| EEG-MVIT (no augmentation) | 2a      | 68.06 | 46.53 | 79.86 | 49.65 | 36.81 | 48.26 | 63.54 | 69.44 | 72.92 | 59.45 $\pm$ 14.53  | 0.46 $\pm$ 0.19      |
| EEG-MVIT                   |         | 72.57 | 46.88 | 75.35 | 48.61 | 40.97 | 51.39 | 71.18 | 69.79 | 71.88 | 61.00 $\pm$ 13.63  | 0.48 $\pm$ 0.18      |
| EEG-MVIT(supervised)       |         | 72.57 | 52.08 | 83.33 | 52.43 | 34.38 | 53.47 | 71.88 | 73.96 | 73.26 | 63.04 $\pm$ 15.61  | 0.51 $\pm$ 0.20      |
| EEG-MVIT (no augmentation) | 2b      | 78.44 | 57.14 | 51.56 | 95.00 | 90.31 | 85.00 | 68.13 | 84.06 | 84.06 | 77.08 $\pm$ 14.95  | 0.54 $\pm$ 0.29      |
| EEG-MVIT                   |         | 76.56 | 57.14 | 55.94 | 93.75 | 87.50 | 87.81 | 75.31 | 85.94 | 83.44 | 78.15 $\pm$ 13.50  | 0.56 $\pm$ 0.27      |
| EEG-MVIT (supervised)      |         | 78.75 | 60.71 | 58.44 | 94.37 | 90.94 | 86.25 | 75.94 | 88.44 | 84.69 | 79.84 $\pm$ 12.81  | 0.60 $\pm$ 0.25      |

Table 4.21: Comparison between the EEG-MVIT model with and without supervised pre-train on BCI Competition IV 2a and BCI Competition IV 2b.

Looking at the results in **table 4.21**, EEG-MVIT shows a 2% improvement on both datasets when using training data from other subjects. In particular, on dataset 2b, subject 2 gains 4% in accuracy, highlighting how the more challenging subjects can benefit from the data of others; the network, therefore, not only benefits from the amount of data provided by S&R, but also from data coming from other patients, leveraging their distributions to improve generalization.

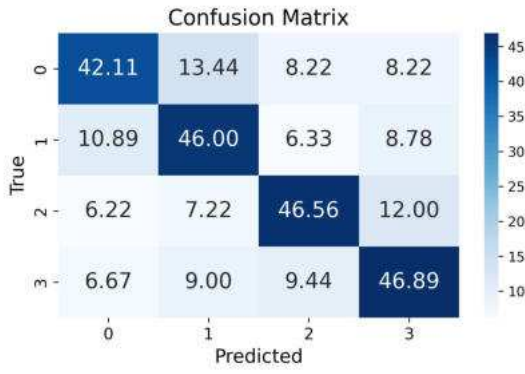


Figure 4.32: Average confusion matrix on the BCI Competition IV 2a test set using EEG-MVIT with supervised pre-train.

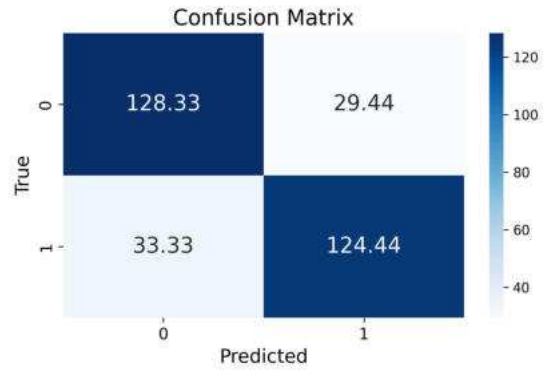


Figure 4.33: Average confusion matrix on the BCI Competition IV 2b test set using LAMBDA-VIT with supervised pre-train.

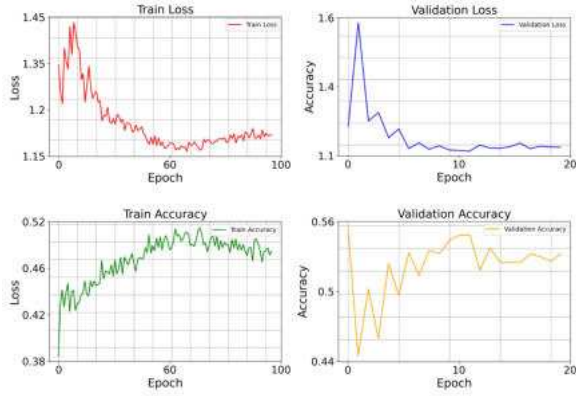


Figure 4.34: EEG-MVIT loss curve using supervised pre-training on the BCI Competition IV 2a dataset.

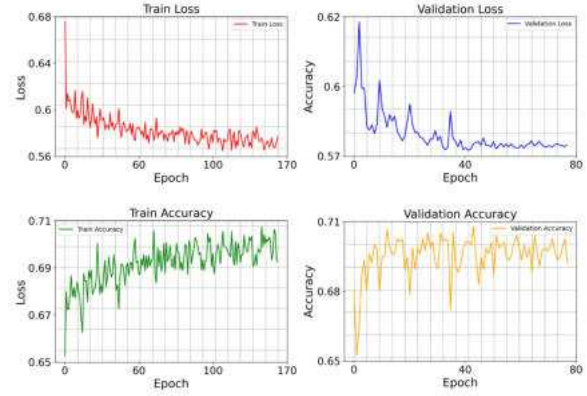


Figure 4.35: EEG-MVIT loss curve using supervised pre-training on the BCI Competition IV 2b dataset.

From the confusion matrices, particularly that of 2a, it can be observed that the overall accuracies have increased for each class. Comparing them with those of the initial EEG-MVIT model in **Figure 4.6**, it is evident that the error reduction mainly occurs in the first two classes. For 2b, instead, the classification of individual classes improves even further compared to the baseline, indicating that the extracted features are more general and that the model benefits from data coming from multiple subjects. This is further supported by the loss behavior across both datasets: the model appears to converge smoothly, effectively leveraging the data from all subjects in each dataset. Based on these considerations, pre-training on a larger set of data is expected to yield further advantages, improving intra-subject performance by finding a shared distribution among EEG signals from different subjects, as well as supporting the development of general BCIs.

## 4.4 EEG-MVIT: curriculum masked spectrogram pre-train results

This section presents the final results achieved through the pre-training procedure performed on the extended 2b dataset described in section **section 3.4**. The outcomes will also be illustrated visually, by showing the spectrogram across the various phases of curriculum pre-training. Observing the spectrogram during the pre-training stage is as crucial as monitoring the loss, since it provides insight into whether the model is truly converging. Lastly, we will run experiments to highlight the improvements in accuracy and kappa-score over the non-pre-trained EEG-MVIT model, with the goal of demonstrating that this pre-training improves the model’s generalization capability. The EEG-MVIT model was chosen for pre-training, as it represents both the central subject of this thesis and the most general architecture among those introduced, being structured with a separate branch for each channel. This structure enables a clear separation of the masked patches, which are applied independently to each channel of the input scalogram. Moreover, in a real-world scenario, the use of only three channels allows for faster inference, thus bringing the model closer to practical deployment. It is enough to consider that computing the spectrogram transform for a signal with 22 channels takes more than twice the time compared to one with only 3 channels. Consequently, in a realistic scenario, the EEG-MVIT stands as a network that maintains a balance between performance and practicality, being sufficiently large for pre-training and clearly defining the concept of a “masked patch”. By employing the extended 2b dataset, three branches are used, corresponding to the C3, Cz, and C4 channels, with a batch size of 24, depth = 6, attention heads = 6, a patch size of 8×8, and a learning rate set to 1e-4.

### 4.4.1 Curriculum Masked Spectrogram pre-training: process overview

The pre-training framework was illustrated in **subsection 3.4.2**, however, this setup was defined after realizing that the model was unable to learn when starting directly from a complex task such as guessing 250 masked patches in both time and frequency, nor from a simple task such as guessing 25 masked patches in time and frequency. This is because the model’s difficulty did not lie in reconstructing the patches as once it understood which patch had been masked, it could reconstruct it without problems, but rather in identifying the masked patches themselves. In fact, even a pre-training procedure starting from 10 masked patches in time and frequency and gradually increasing the number did not succeed.

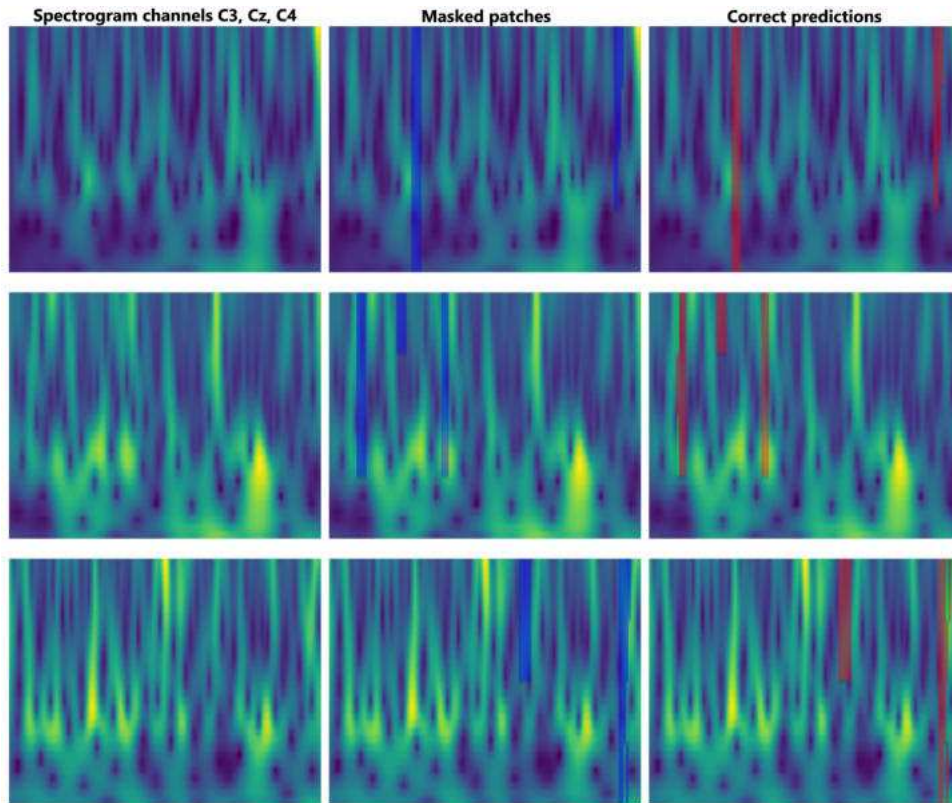


Figure 4.36: Failed attempt of curriculum masked spectrogram pre-training by masking 25 patches in time and frequency: after 80 epochs, the model is able to correctly reconstruct only a very small number of patches for each channel. The red patches are the ones that were removed, while the green patches are those that were correctly reconstructed. The image shows that even after 80 epochs, the red patches outnumber the green ones.

After running these tests, it became clear that, in addition to the number of masked patches, the **position** of the masked patches also played a key role in simplifying the task. This insight led to the definition of the first pre-training step: masking  $P = 100$  consecutive patches in time. Although these patches are consecutive, they can start at different time instants or frequency positions between training steps. In this way, the model learns what is meant by a “masked patch.”

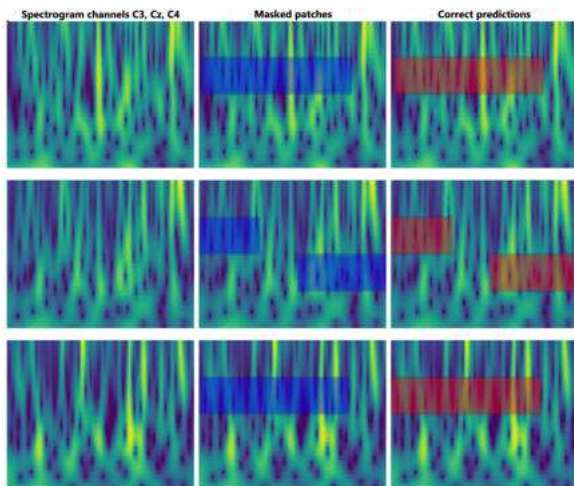


Figure 4.37: Easy step of the masked spectrogram curriculum pre-training: masking 100 consecutive patches in time and frequency. At epoch 5, the model has not yet learned how to reconstruct the patches, as all of them are still marked in red.

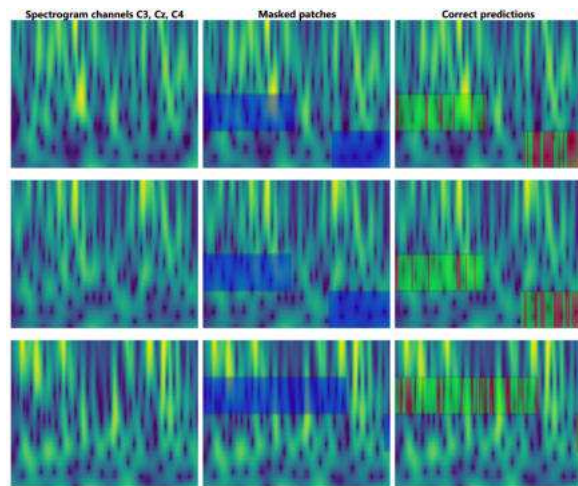


Figure 4.38: Easy step of the masked spectrogram curriculum pre-training: masking 100 consecutive patches in time and frequency. At epoch 10, the model starts to identify which patches need to be reconstructed, and many patches turn green.

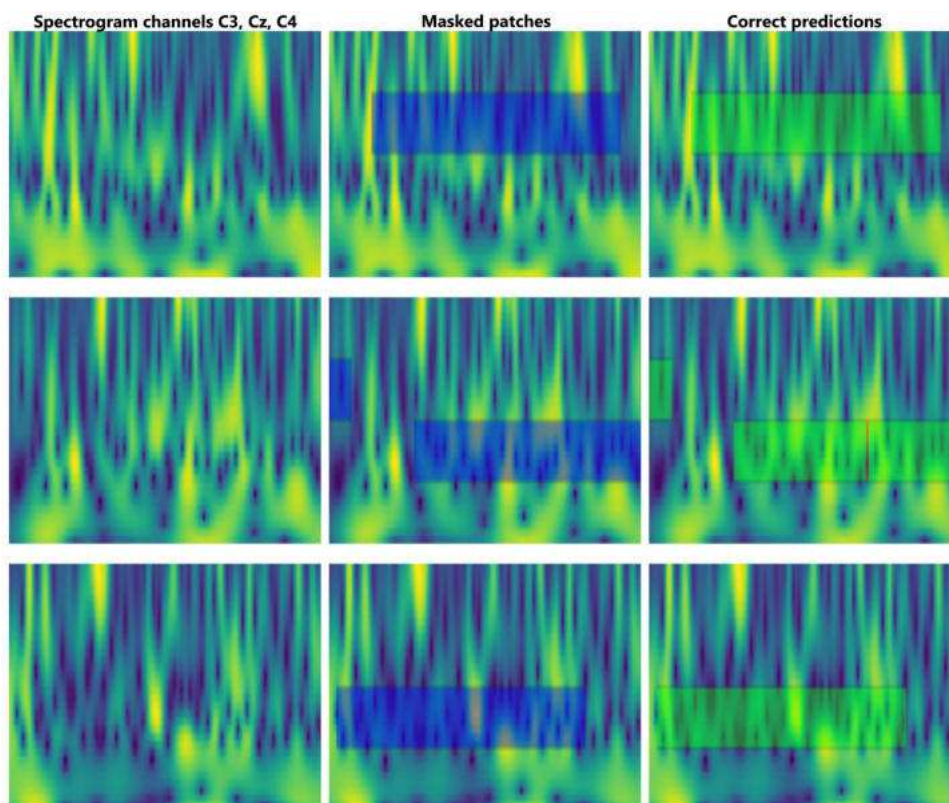


Figure 4.39: Easy step of the masked spectrogram curriculum pre-training: masking 100 consecutive patches in time and frequency. At epoch 20, the model has learned which patches need to be reconstructed, with most of them turning green. This initial step can be considered complete.

From the images, it can be observed that the model gradually learns to reconstruct the patches. The visualized samples belong to the validation set, which the model has never seen during training. Once the “easy” pre-training step is completed, the process moves on to the “medium”

step, where the model must learn to reconstruct the same number of patches ( $P=100$ ), but this time masked in an alternating pattern along time and frequency, with non-consecutive masked patch indices.

This intermediate stage takes the weights from the previous step and continues training them. Experiments with a smaller number of masked patches ( $P=25$ ,  $P=50$ ) showed that the model was able to learn quickly. Therefore, to speed up the following pre-training stages, it was concluded that the “medium” step can start directly from  $P=100$ , which represents the right trade-off, as masking more than 100 patches leads to learning difficulties.

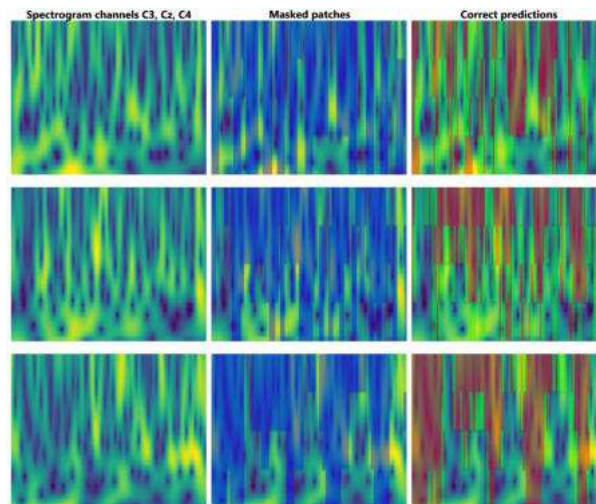


Figure 4.40: Medium step of the curriculum masked spectrogram pre-training, masking 100 random patches in time and frequency. At epoch 1, the model has not yet learned how to reconstruct the patches, and most of them are colored red. However, compared to the first epochs of the “easy” step in **figure 4.37**, the result is significantly better because the model has already acquired prior knowledge.

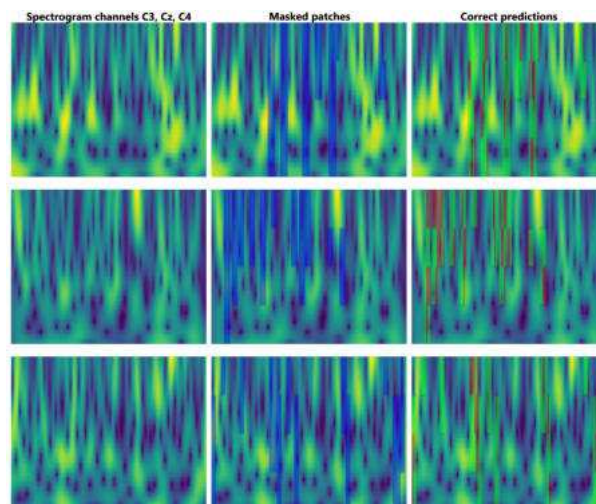


Figure 4.41: Medium step of the curriculum masked spectrogram pre-training, masking 100 random patches in time and frequency. By epoch 5, the model starts to understand how to reconstruct the patches; from epoch 1 to 5, there are clear improvements, indicating that the model is learning quickly.

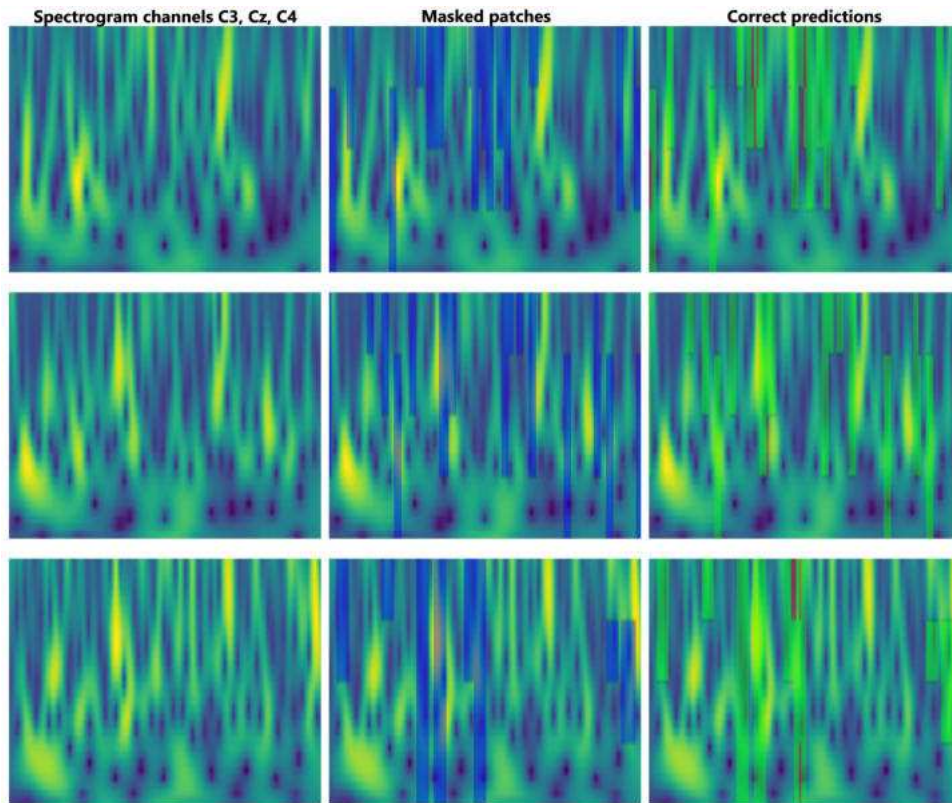


Figure 4.42: Medium step of the curriculum masked spectrogram pre-training, masking 100 random patches in time and frequency. By epoch 20, the model has learned which patches to reconstruct, as most of them are colored green. This medium step can be considered complete.

It is interesting to note the difference between the result obtained at 20 epochs, masking 100 random patches in time and frequency, and the image shown in **figure 4.36**, obtained at 80 epochs by masking only 25 patches without performing the “easy” step. This highlights the importance of the “easy” step, which allowed the model to learn how to identify the masked patches and then reconstruct them in just 20 epochs, something that it could not achieve in 80 epochs with a much smaller number of patches.

After completing the “medium” step of the pre-training, the process moves to the “hard” step, where the model is required to reconstruct  $P=300$  patches, which is about 60% of all spectrogram patches. Even here, the model manages to increase from 100 to 300 patches to reconstruct fairly easily, yet performance declines past 300 patches. This step concludes the curriculum masked spectrogram pre-training, since the model successfully reconstructs a significant percentage of masked patches.

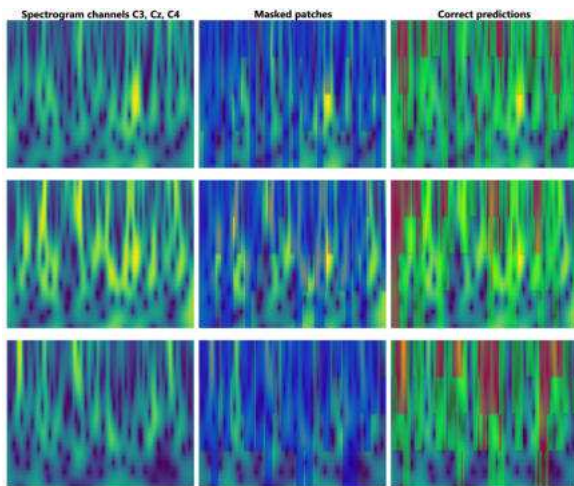


Figure 4.43: Hard step of the curriculum masked spectrogram pre-training, masking 300 random patches in time and frequency. At epoch 15, the model has a good starting point, better than in the previous steps since it has correctly learned which patches to reconstruct and how to do it, however, there are still red patches that have not been reconstructed.

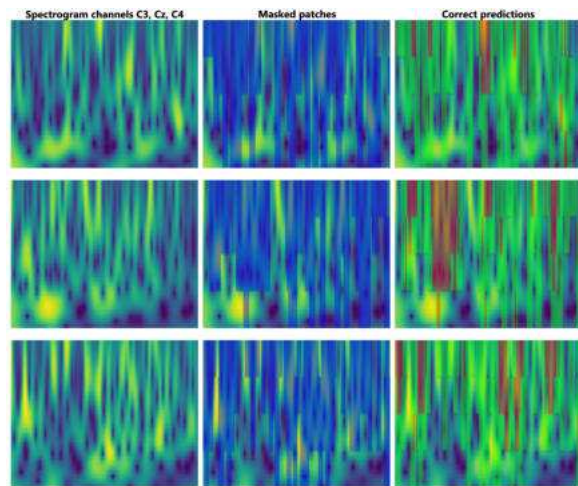


Figure 4.44: Hard step of the curriculum masked spectrogram pre-train, masking 300 random patches in both time and frequency. At epoch 90, there is some improvement, but it is not significantly different from the state at epoch 15.

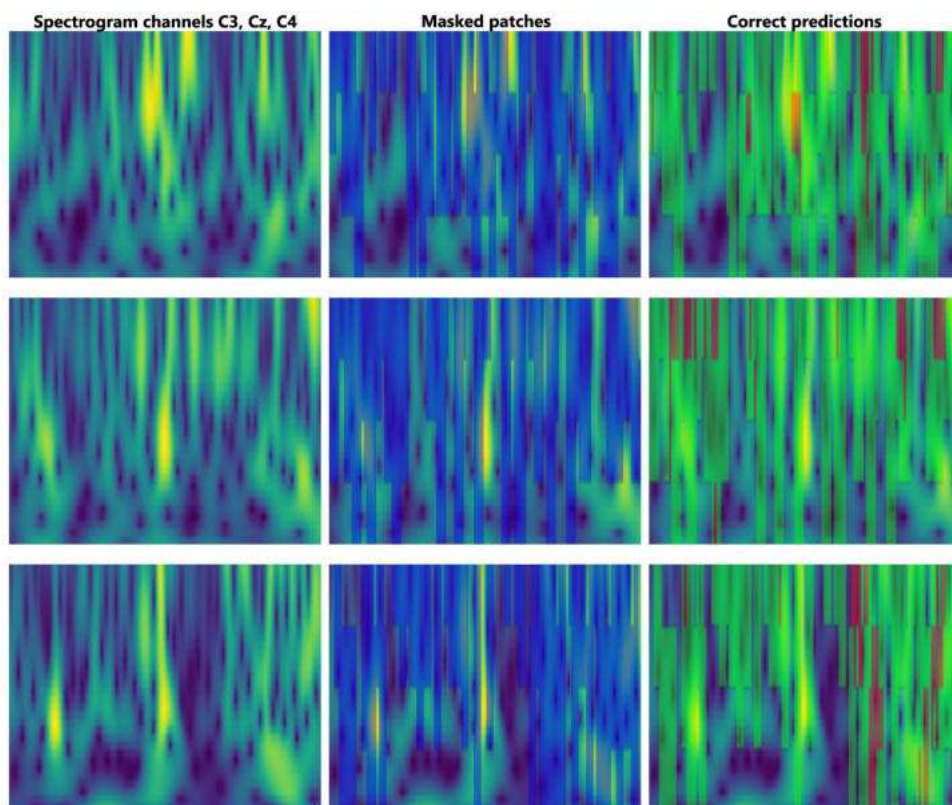


Figure 4.45: Hard step of the curriculum masked spectrogram pre-training, masking 300 random patches in time and frequency. By epoch 150, the model has understood which patches to reconstruct, and most of them are colored green. In this case, the difference compared to previous epochs is more evident. This marks the final step, completing the pre-training process.

The images clearly show that increasing the number of patches, and consequently the task difficulty, leads to a higher number of epochs needed for the model to learn how to correctly reconstruct the patches. This is further evidence that curriculum training is necessary. Upon completing the pre-training, the model is expected to generate features that improve performance, having better understood both the time-frequency structure of the spectrogram and the internal structure of EEG signals across different patients.

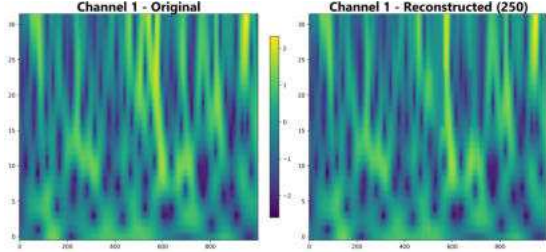


Figure 4.46: Comparison between the reconstructed and the original spectrogram of channel C3. The number of masked patches is  $N = 250$ .

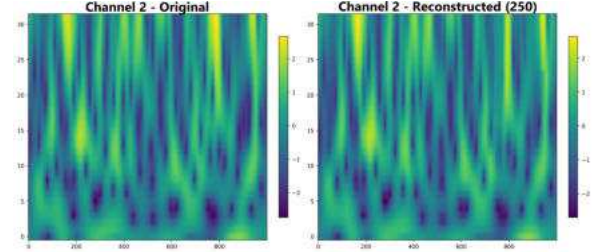


Figure 4.47: Comparison between the reconstructed and the original spectrogram of channel Cz. The number of masked patches is  $N = 250$ .

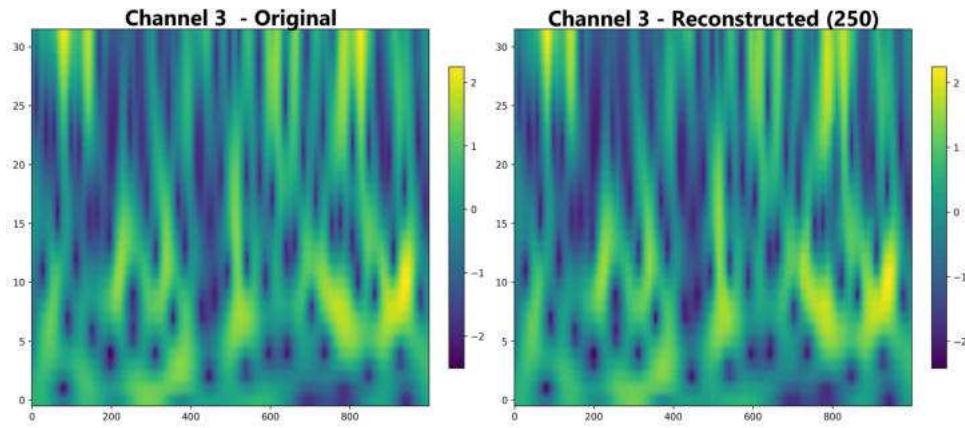


Figure 4.48: Comparison between the reconstructed and the original spectrogram of channel C4. The number of masked patches is  $N = 250$ .

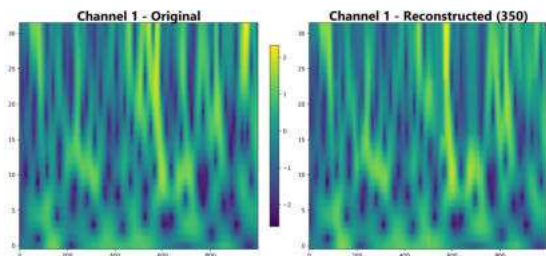


Figure 4.49: Comparison between the reconstructed and the original spectrogram of channel C3. The number of masked patches is  $N = 350$ .

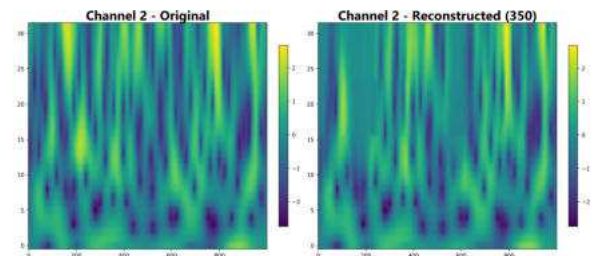


Figure 4.50: Comparison between the reconstructed and the original spectrogram of channel Cz. The number of masked patches is  $N = 350$ .

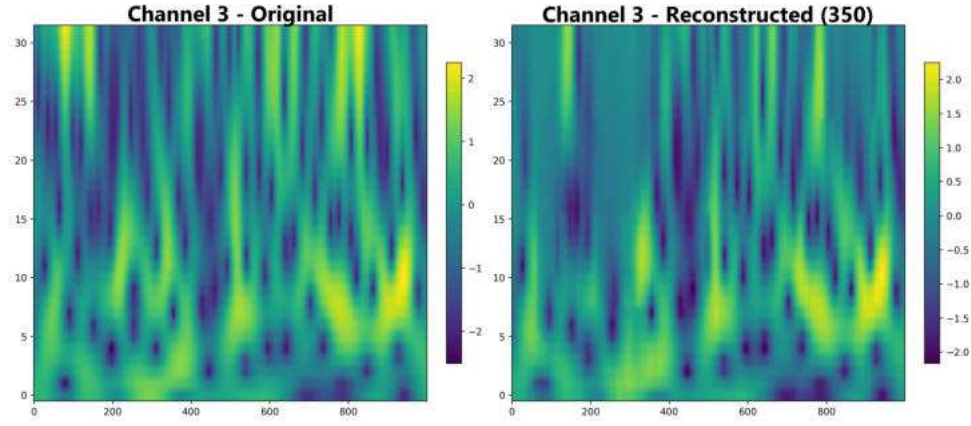


Figure 4.51: Comparison between the reconstructed and the original spectrogram of channel C4. The number of masked patches is  $N = 350$ .

From the images, it can be seen that the model is able to reconstruct the spectrograms. The sample shown belongs to a dataset that the model has never seen before, demonstrating that it has learned the underlying structure of a spectrogram. From the comparison, it appears that the reconstruction is nearly perfect up to 250 removed patches. Beyond 350 patches, exceeding the pre-training threshold, the model starts to show difficulties, particularly in the high-frequency regions. Nevertheless, this suggests that with more data, the reconstruction performance could improve further.

#### 4.4.2 Curriculum masked spectrogram pre-train cross-subject performance

The first test performed is the Leave One Subject Out, used to evaluate the improvement of the pre-trained model (pre-trained EEG-MVIT) in cross-subject generalization compared to the not pre-trained model (standard EEG-MVIT). Consequently, nine models are trained, each on the data of all patients except the  $n$ -th one, which is used as the actual test subject.

| Subject | Accuracy | Kappa |
|---------|----------|-------|
| 1       | 77.36    | 0.54  |
| 2       | 61.03    | 0.22  |
| 3       | 57.22    | 0.14  |
| 4       | 92.30    | 0.84  |
| 5       | 76.76    | 0.53  |
| 6       | 82.50    | 0.65  |
| 7       | 70.83    | 0.42  |
| 8       | 79.47    | 0.59  |
| 9       | 79.86    | 0.60  |

Table 4.22: Accuracy and kappa score of all subjects of leave one subject out on BCI Competition IV 2b using pre-trained EEG-MVIT

| Subject | Accuracy | Kappa |
|---------|----------|-------|
| 1       | 71.94    | 0.44  |
| 2       | 57.21    | 0.14  |
| 3       | 55.28    | 0.10  |
| 4       | 91.08    | 0.82  |
| 5       | 81.22    | 0.62  |
| 6       | 78.89    | 0.58  |
| 7       | 68.89    | 0.38  |
| 8       | 76.84    | 0.54  |
| 9       | 81.25    | 0.62  |

Table 4.23: Leave one subject out accuracy and kappa score of all subjects on BCI Competition IV 2b using standard EEG-MVIT

| Model                  | Dataset | Sub1  | Sub2  | Sub3  | Sub4  | Sub5  | Sub6  | Sub7  | Sub8  | Sub9  | Mean Acc $\pm$ std | Mean Kappa $\pm$ std | p-value   |
|------------------------|---------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--------------------|----------------------|-----------|
| EEG-MVIT (standard)    | 2b      | 71.94 | 57.21 | 55.28 | 91.08 | 81.22 | 78.89 | 68.89 | 76.84 | 81.25 | 73.62 $\pm$ 11.01  | 0.47 $\pm$ 0.22      | -         |
| EEG-MVIT (pre-trained) |         | 77.36 | 61.03 | 57.22 | 92.30 | 76.76 | 82.50 | 70.83 | 79.47 | 79.86 | 75.26 $\pm$ 10.20  | 0.51 $\pm$ 0.20      | 0.00099** |

Table 4.24: Comparison between the accuracy of all subjects with leave one subject out on BCI Competition IV 2b between the self supervised pre-trained EEG-MVIT and the standard EEG-MVIT. The last columns report the average accuracy, the average kappa-score and the p-value. In this case ( $p < 0.01$ ) for self supervised pre-trained EEG-MVIT, indicating that the performance improvement is statistically significant.

The average accuracy of the pre-trained model is about 2% higher than that of the non-pre-trained model, indicating that pre-training has enhanced the model’s generalization ability: subjects 1 and 2 show the most noticeable performance improvements, suggesting that pre-training may have exploited information from other EEG signals to boost performance.

However the pre-trained model could be considered to have an advantage during the comparison, as it may have been exposed to the training data of the  $n$ -th subject, unlike the non-pre-trained model. Nevertheless, it is important to note that the data processed during the pre-train are unlabeled, and the windowing procedure does not allow it to see exactly the same data used in the motor imagery task.

Noticing the performance increase on subjects 1 and 2, before implementing the final verification provided by intra-subject training, it was decided to continue exploring the cross-subject approach, trying to leverage data from multiple patients. Consequently, the previously pre-trained supervised model (which has therefore also seen the training data of all patients) is used and compared with its self supervised pre-trained version in a zero-shot setting on the BCI Competition IV 2b dataset, using the test data of the patients. This approach clearly demonstrates the model’s capability to generalize without having been trained on the individual subject.

| Subject | Accuracy | Kappa |
|---------|----------|-------|
| 1       | 78.12    | 0.56  |
| 2       | 61.07    | 0.22  |
| 3       | 56.25    | 0.12  |
| 4       | 94.69    | 0.89  |
| 5       | 80.00    | 0.60  |
| 6       | 90.94    | 0.82  |
| 7       | 77.81    | 0.56  |
| 8       | 90.00    | 0.80  |
| 9       | 87.81    | 0.76  |

Table 4.25: Accuracy and kappa score of all subjects of zero-shot on BCI Competition IV 2b using (supervised + self supervised) EEG-MVIT

| Subject | Accuracy | Kappa |
|---------|----------|-------|
| 1       | 77.19    | 0.54  |
| 2       | 53.93    | 0.08  |
| 3       | 57.19    | 0.14  |
| 4       | 94.37    | 0.89  |
| 5       | 84.38    | 0.69  |
| 6       | 88.12    | 0.76  |
| 7       | 75.00    | 0.50  |
| 8       | 89.06    | 0.78  |
| 9       | 85.94    | 0.72  |

Table 4.26: Accuracy and kappa score of all subjects of zero-shot on BCI Competition IV 2b using self supervised EEG-MVIT

| Model                                   | Dataset | Sub1  | Sub2  | Sub3  | Sub4  | Sub5  | Sub6  | Sub7  | Sub8  | Sub9  | Mean Acc $\pm$ std | Mean Kappa $\pm$ std | p-value   |
|-----------------------------------------|---------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--------------------|----------------------|-----------|
| EEG-MVIT (supervised)                   | 2b      | 77.19 | 53.93 | 57.19 | 94.37 | 84.38 | 88.12 | 75.00 | 89.06 | 85.94 | 78.35 $\pm$ 13.41  | 0.57 $\pm$ 0.26      | -         |
| EEG-MVIT (supervised + self supervised) |         | 78.12 | 61.07 | 56.25 | 94.69 | 80.00 | 90.94 | 77.81 | 90.00 | 87.81 | 79.63 $\pm$ 12.59  | 0.59 $\pm$ 0.25      | 0.00492** |

Table 4.27: Zero shot comparison between the accuracy of all subjects on BCI Competition IV 2b between the (supervised + self supervised) EEG-MVIT and the supervised EEG-MVIT. The last columns report the average accuracy, the average kappa-score and the p-value. In this case ( $p < 0.01$ ) for self supervised pre-trained EEG-MVIT, indicating that the performance improvement is statistically significant.

In this case, all the advantages brought by both the supervised pre-train and the self supervised pre-train can be observed, with the latter again contributing to a performance increase. From the analysis of the supervised pre-train, it was noted that some subjects benefit from the data of other subjects; here, it can further be seen that subject two in particular, also benefits from the self supervised pre-training.

#### 4.4.3 Curriculum masked spectrogram pre-train intra-subject performance

From the previous section, it was shown that the combination of supervised pre-training and self supervised pre-training improves the model's performance when tested in a zero-shot setting. At this point, an intra-subject training is carried out and the performances are compared with the baseline model, in order to definitively assess the effects of pre-training.

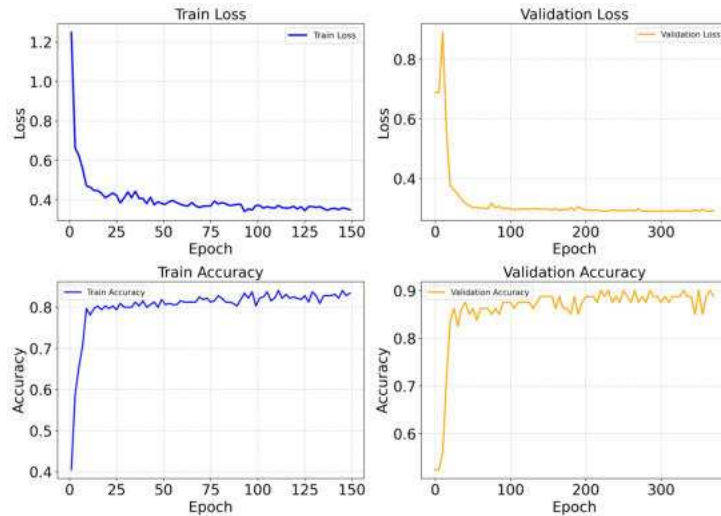


Figure 4.52: Loss of the training and validation sets of subject 1 from BCI Competition IV 2b using (supervised + self supervised) EEG-MVIT. After pre-training, it can be observed that intra-subject training across is very fast, as the model is able to leverage its ability to reconstruct the spectrogram to produce features for the MI task, thereby converging more quickly. Only the results for subject 1 are reported, since the plots for the other subjects are very similar.

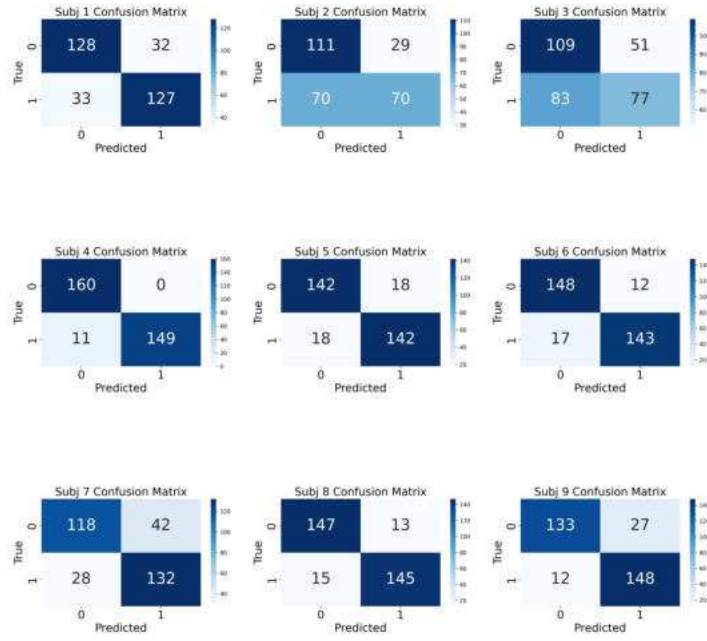


Figure 4.53: Confusion matrices of all subjects from BCI Competition IV 2b using (supervised + self supervised) EEG-MVIT.

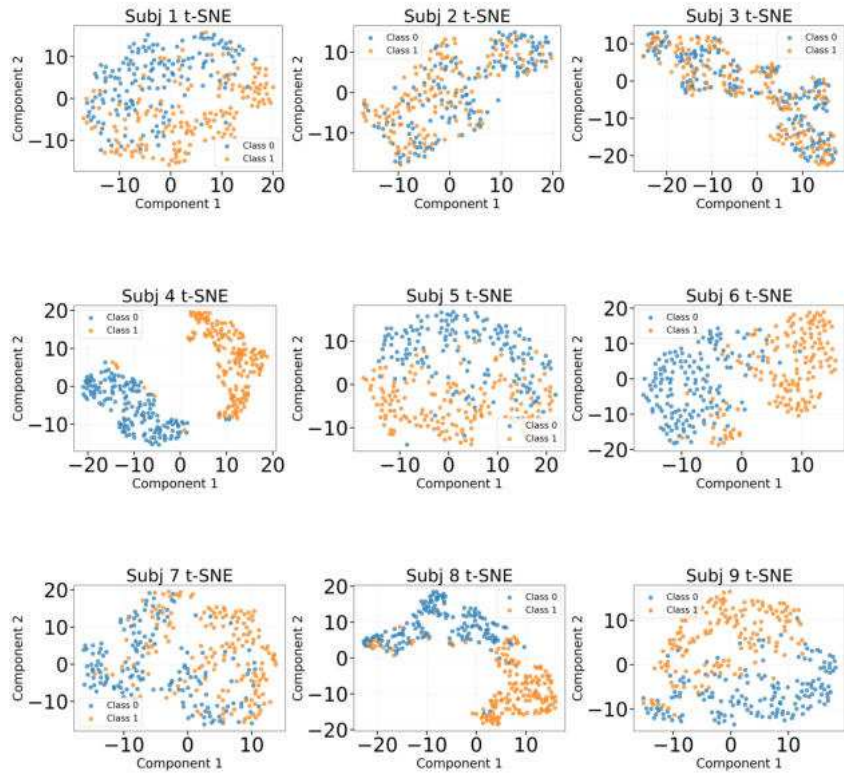


Figure 4.54: T-SNE of all subjects from BCI Competition IV 2b using (supervised + self supervised) EEG-MVIT: blue points represent left-hand features and orange represent right-hand features before classification.

| Subject | Accuracy | Kappa |
|---------|----------|-------|
| 1       | 79.69    | 0.59  |
| 2       | 64.64    | 0.29  |
| 3       | 58.13    | 0.16  |
| 4       | 96.56    | 0.93  |
| 5       | 88.75    | 0.77  |
| 6       | 90.94    | 0.82  |
| 7       | 78.12    | 0.56  |
| 8       | 91.25    | 0.82  |
| 9       | 87.81    | 0.76  |

Table 4.28: Accuracy and kappa score of all subjects on the test set of BCI Competition IV 2b using (supervised + self supervised) EEG-MVIT.

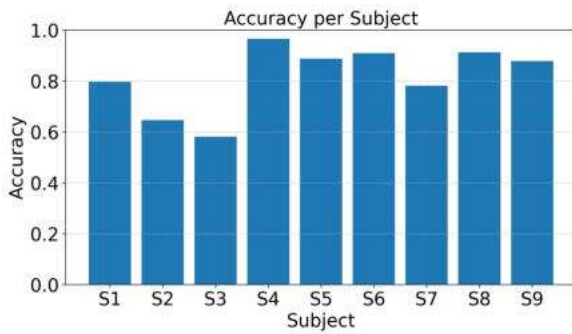


Figure 4.55: Accuracy bar chart for all subjects on the BCI Competition IV 2b test set using (supervised + self supervised) EEG-MVIT.

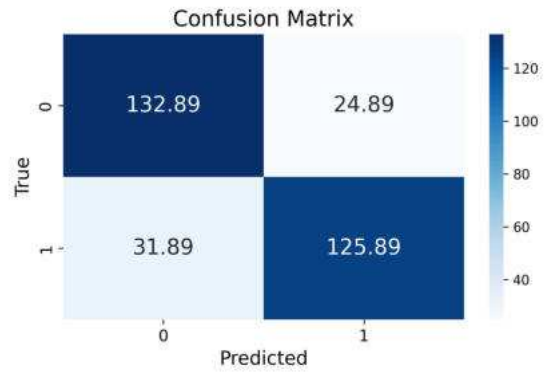


Figure 4.56: Mean confusion matrix on the BCI Competition IV 2b test set using (supervised + self supervised) EEG-MVIT. The classification is fairly balanced, a minor error is observed for the left hand.

| Model                                  | Dataset | Sub1  | Sub2  | Sub3  | Sub4  | Sub5  | Sub6  | Sub7  | Sub8  | Sub9  | Mean Acc $\pm$ std | Mean Kappa $\pm$ std | p-value   |
|----------------------------------------|---------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--------------------|----------------------|-----------|
| EEG-MVIT                               | 2b      | 76.56 | 57.14 | 55.94 | 93.75 | 87.50 | 87.81 | 75.31 | 85.95 | 83.44 | 78.15 $\pm$ 13.50  | 0.56 $\pm$ 0.27      | -         |
| EEG-MVIT(supervised)                   |         | 78.75 | 60.71 | 58.44 | 94.37 | 90.94 | 86.25 | 75.94 | 88.44 | 84.69 | 79.84 $\pm$ 12.81  | 0.60 $\pm$ 0.25      | 0.00187** |
| EEG-MVIT(supervised + self supervised) |         | 79.69 | 64.64 | 58.13 | 96.56 | 88.75 | 90.94 | 78.12 | 91.25 | 87.81 | 81.77 $\pm$ 12.98  | 0.64 $\pm$ 0.25      | 0.00001** |

Table 4.29: Comparison of accuracy across all subjects with intra-subject evaluation on BCI Competition IV 2b between EEG-MVIT, EEG-MVIT trained with the supervised pre-train, and EEG-MVIT trained with the supervised pre-training and self supervised pre-training. The last columns report the average accuracy, the average kappa-score and the p-value. In this case ( $p < 0.01$ ) for EEG-MVIT (supervised + self supervised), indicating that the performance improvement is statistically significant.

From the final accuracy comparison, it can be observed that performance has gradually improved, reaching a final accuracy about 4% higher than the initial model. Some subjects, such as 2 and 8, have gained significant benefits from pre-training, achieving +7% and +6% accuracy respectively compared to the baseline model. Overall, it can be concluded that all subjects experienced performance improvements, confirming the positive impact of pre-training. By comparing the confusion matrices of subjects 2 and 3 with those of the baseline model shown

in **figure 4.11**, it can be observed, in addition to the improvement in classification, that the error is now concentrated in the classification of the right-hand class for both patients. This may suggest that the features produced by the self supervised pre-trained model are more general and not tied to a specific subject. Looking at the t-SNE plots compared to those of the baseline model shown in **figure 4.9**, it can be observed that, for example, the plots of subjects 4 and 8 display clearly separated features, which allow for a clear distinction between the classes. In the case of subject 2, the features seem to be starting to split towards the upper and lower parts of the space. All these elements provide evidence that the self supervised pre-trained model has a different interpretation of the data compared to the baseline model and is able to exploit them more effectively to improve performance.

After proving the effectiveness of the model, a final experiment is conducted, consisting of an intra-subject training similar to TSFF [16]. The first two classes of the BCI Competition IV 2a dataset are used, corresponding to left and right hand movements, and only the C3, Cz, and C4 channels are selected instead of all 22, in order to show the model's ability to perform on a different dataset too.

| Model                                  | Dataset | Sub1  | Sub2  | Sub3  | Sub4  | Sub5  | Sub6  | Sub7  | Sub8  | Sub9  | Mean Acc $\pm$ std | Mean Kappa $\pm$ std | p-value  |
|----------------------------------------|---------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--------------------|----------------------|----------|
| EEG-MVIT                               | 2a      | 68.06 | 45.83 | 69.44 | 66.67 | 56.94 | 62.50 | 59.03 | 84.72 | 87.50 | 66.74 $\pm$ 13.11  | 0.33 $\pm$ 0.13      | -        |
| EEG-MVIT(supervised)                   |         | 70.83 | 45.83 | 77.08 | 70.83 | 53.47 | 63.89 | 65.28 | 86.81 | 85.42 | 68.83 $\pm$ 13.59  | 0.38 $\pm$ 0.27      | 0.12349  |
| EEG-MVIT(supervised + self supervised) |         | 78.47 | 53.47 | 86.11 | 67.36 | 54.17 | 64.58 | 61.81 | 85.42 | 90.28 | 71.30 $\pm$ 0.14   | 0.42 $\pm$ 0.28      | 0.00322* |

Table 4.30: Comparison of accuracy across all subjects with intra-subject evaluation on 2 class of 3 channels BCI Competition IV 2a between EEG-MVIT, EEG-MVIT trained with the supervised pre-train, and EEG-MVIT trained with the supervised pre-training and self supervised pre-training. The last columns report the average accuracy, the average kappa-score and the p-value. In this case ( $p < 0.05$ ) for EEG-MVIT (supervised + self supervised), indicating that the performance improvement is statistically significant.

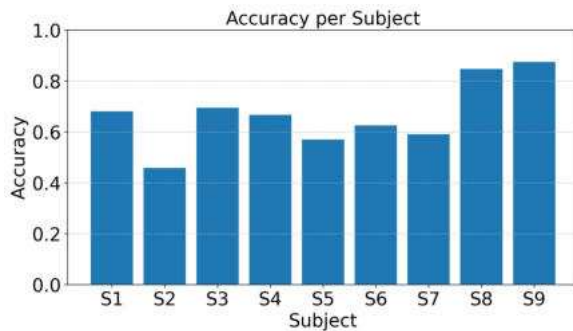


Figure 4.57: Accuracy bar chart for all subjects on the BCI Competition IV 2a test set using standard EEG-MVIT.

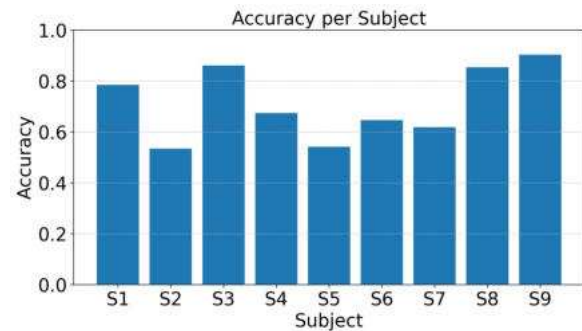


Figure 4.58: Accuracy bar chart for all subjects on the BCI Competition IV 2a test set using (supervised + self supervised) EEG-MVIT.

Accuracy increases by about 5% compared to the baseline model thanks to pre-training, with some subjects, such as 1 and 3, benefiting from an accuracy increase of +10%. From the plots showing the features obtained via t-SNE, the difference is clearly noticeable: in the pre-trained model, the features of the left-hand class are in the upper part of the space compared to those of the right hand; in the standard model, this separation is not as clear.

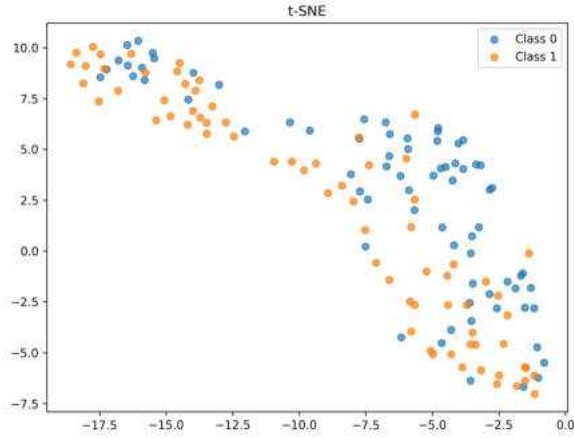


Figure 4.59: T-SNE of subject 3 on BCI Competition IV 2a using standard EEG-MVIT.

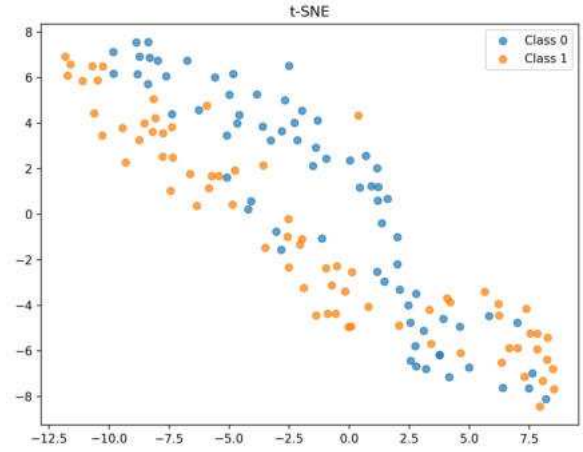


Figure 4.60: T-SNE of subject 3 on BCI Competition IV 2a using (supervised + self supervised) EEG-MVIT.

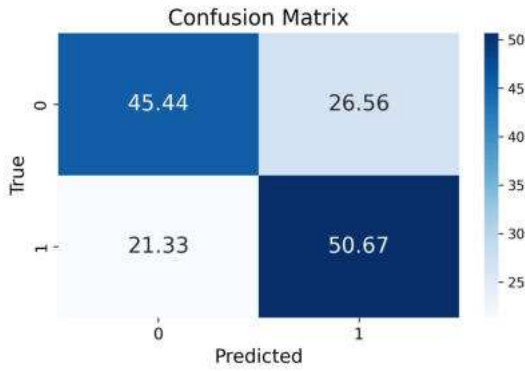


Figure 4.61: Mean confusion matrix on the BCI Competition IV 2a (2 class) test set using standard EEG-MVIT.

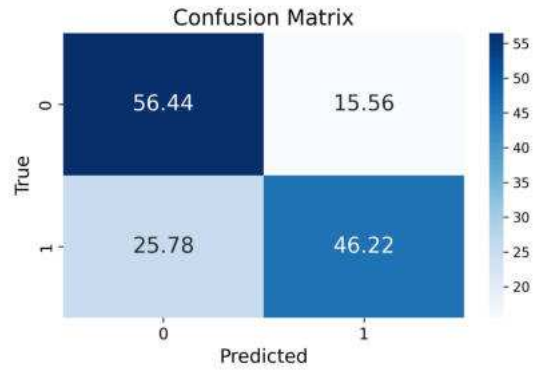


Figure 4.62: Mean confusion matrix on the BCI Competition IV 2a (2 class) test set using (supervised + self supervised) EEG-MVIT.

The comparison between the two confusion matrices before and after pre-train shows a 10% decrease in the classification error for the left hand, accompanied by a slight increase for the right hand. These findings further support the feature separation observed in the T-SNE visualizations. Furthermore, the supervised pre-training used was conducted with data from the 2b dataset, so all models have never seen data from this dataset, which demonstrates the generalization capability of a self supervised pre-trained model.

## 4.5 Conclusions and future improvements

| Dataset | Model       | Test                                                              | Sub1  | Sub2  | Sub3  | Sub4  | Sub5  | Sub6  | Sub7  | Sub8  | Sub9  | Mean Acc $\pm$ std                  | Mean Kappa $\pm$ std              |
|---------|-------------|-------------------------------------------------------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------------------------------------|-----------------------------------|
| 2a      | standard    | no augmentation intra-subj                                        | 68.06 | 46.53 | 79.86 | 49.65 | 36.81 | 48.26 | 63.54 | 69.44 | 72.92 | 59.45 $\pm$ 14.53                   | 0.46 $\pm$ 0.19                   |
|         |             | augmentation intra-subj                                           | 72.57 | 46.88 | 75.35 | 48.61 | 40.97 | 51.39 | 71.18 | 69.79 | 71.88 | 61.00 $\pm$ 13.63                   | 0.48 $\pm$ 0.18                   |
|         |             | supervised intra-subj                                             | 72.57 | 52.08 | 83.33 | 52.43 | 34.38 | 53.47 | 71.88 | 73.96 | 73.26 | 63.04 $\pm$ 15.61                   | 0.51 $\pm$ 0.20                   |
|         |             | <b>2 class 3 channels intra-subj</b>                              | 68.06 | 45.83 | 69.44 | 66.67 | 56.94 | 62.50 | 59.03 | 84.72 | 87.50 | <b>66.74 <math>\pm</math> 13.11</b> | <b>0.33 <math>\pm</math> 0.13</b> |
|         |             | 2 class 3 channels supervised intra-subj                          | 70.83 | 45.83 | 77.08 | 70.83 | 53.47 | 63.89 | 65.28 | 86.81 | 85.42 | 68.83 $\pm$ 13.59                   | 0.38 $\pm$ 0.27                   |
|         | pre-trained | <b>2 class 3 channels supervised + self supervised intra-subj</b> | 78.47 | 53.47 | 86.11 | 67.36 | 54.17 | 64.58 | 61.81 | 85.42 | 90.28 | <b>71.30 <math>\pm</math> 0.14</b>  | <b>0.42 <math>\pm</math> 0.28</b> |
| 2b      | standard    | no augmentation intra-subj                                        | 78.44 | 57.14 | 51.56 | 95.00 | 90.31 | 85.00 | 68.13 | 84.06 | 84.06 | 77.08 $\pm$ 14.95                   | 0.54 $\pm$ 0.29                   |
|         |             | <b>augmentation intra-subj</b>                                    | 76.56 | 57.14 | 55.94 | 93.75 | 87.50 | 87.81 | 75.31 | 85.94 | 83.44 | <b>78.15 <math>\pm</math> 13.50</b> | <b>0.56 <math>\pm</math> 0.27</b> |
|         |             | standard cross-subj                                               | 71.94 | 57.21 | 55.28 | 91.08 | 81.22 | 78.89 | 68.89 | 76.84 | 81.25 | 73.62 $\pm$ 11.01                   | 0.47 $\pm$ 0.22                   |
|         |             | supervised intra-subj                                             | 78.75 | 60.71 | 58.44 | 94.37 | 90.94 | 86.25 | 75.94 | 88.44 | 84.69 | 79.84 $\pm$ 12.81                   | 0.60 $\pm$ 0.25                   |
|         |             | supervised zero-shot                                              | 77.19 | 53.93 | 57.19 | 94.37 | 84.38 | 88.12 | 75.00 | 89.06 | 85.94 | 78.35 $\pm$ 13.41                   | 0.57 $\pm$ 0.26                   |
|         | pre-trained | self supervised cross-subj                                        | 77.36 | 61.03 | 57.22 | 92.30 | 76.76 | 82.50 | 70.83 | 79.47 | 79.86 | 75.26 $\pm$ 10.20                   | 0.51 $\pm$ 0.20                   |
|         |             | supervised + self supervised zero-shot                            | 78.12 | 61.07 | 56.25 | 94.69 | 80.00 | 90.94 | 77.81 | 90.00 | 87.81 | 79.63 $\pm$ 12.59                   | 0.59 $\pm$ 0.25                   |
|         |             | <b>supervised + self supervised intra-subj</b>                    | 79.69 | 64.64 | 58.13 | 96.56 | 88.75 | 90.94 | 78.12 | 91.25 | 87.81 | <b>81.77 <math>\pm</math> 12.98</b> | <b>0.64 <math>\pm</math> 0.25</b> |

Table 4.31: Summary table of results on EEG-MVIT. Shows how self supervised pre-training improved performance.

The objective was to analyze the typical problems of the Motor Imagery task with EEG signals. Specifically, it explored how to overcome data scarcity and the challenge that EEG signals, as oscillatory signals, do not exhibit stationary patterns, making generalization across subjects difficult. The study also aimed to better understand both the benefits and the limitations of inductive bias in a neural network model used for MI classification.

In order to investigate the effect of a Vision Transformer in the context of BCIs for the Motor Imagery task, attention was concentrated on an architecture designed to leverage input channels of a spectrogram: EEG-MVIT. The initial comparison involved the TSFF-net architecture, consisting of several CNN layers, and was tested on the BCI Competition IV 2a and 2b datasets to assess the performance of a Vision Transformer without additional convolutional layers, apart from the adaptive patch embedding, compared to a CNN network. Both models show similar performance on the 2b dataset, whereas the VIT outperforms by 5% on the 2a dataset. Even though the results do not match state-of-the-art performance, the investigation of EEG-MVIT proceeds through the development of multiple VIT models to examine how different layers of the architecture affect performance. The models compared are: ENC-MVIT, which merges the features at the patch embedding stage using a single encoder: it reaches 60% accuracy on dataset 2a and 79% on dataset 2b. Therefore, using a single encoder and concatenating all features from the different channels before feeding them into it does not provide benefits; consequently, it is worth trying a different input channel structure. DEPTH-VIT concatenates the channels along the depth dimension, it reaches 66% accuracy on dataset 2a and 79% on dataset 2b. Therefore, concatenating the channels in depth significantly improves performance on 2a but not on 2b. The improvement on 2a is due to the use of a single patch embedding layer per channel, which results in fewer parameters for a smaller amount of data compared to the EEG-MVIT model, this

conclusion is supported by the fact that on 2b the performance improvement is not as evident, however, for this reasons, it is worth further exploring the impact of the patch embedding layer. The architecture 3BRANCH-ViT, employs patch embedding layers with different patch sizes; it reaches 63% accuracy on dataset 2a and 80% on dataset 2b. Therefore, patches of different sizes provide benefits to the task, although not as pronounced as in the previous architecture. These benefits, besides arising from the reduced number of parameters, may also result from a patch division that emphasizes the frequency bands characteristic of the MI task. However, in the final comparison, the importance of the CNN within the adaptive patch embedding layer will be maximized to observe any potential performance changes. LAMBDA-ViT is similar to the previous model but includes an additional output derived from the network’s single convolutional layer, it reaches 67% accuracy on dataset 2a and 80% on dataset 2b. These results provide definitive evidence that a hybrid CNN+ViT architecture, when working with a limited amount of data, significantly improves performance by leveraging the inductive bias. It is enough to note that, by varying lambda from 0.4 to 0.7, performance increases. However, the inductive bias, advantageous with few data, imposes a performance ceiling when a large amount of data is available, which greatly limits generalization, one of the main objectives of BCI. This effect is already noticeable when comparing performance on dataset 2b, which shows only a small improvement over 2a, since it contains more data. To test the generalization capabilities of the EEG-MViT an additional pre-training procedure was also tested: the supervised pre-training., in order to assess its impact on performance and to further analyze the model’s behavior when exposed to larger amounts of data coming from different patients. It enhances performance, achieving 63% accuracy on 2a and almost 80% on 2b, demonstrating that a ViT architecture gains from having more data from multiple subjects, and thus from diverse distributions. The key results achieved show that:

- the network achieving the best performance is LAMBDA-ViT, which leverages the single convolutional layer to exploit inductive bias when only a limited amount of data is available;
- the performance across models on the 2b dataset is more stable compared to 2a, further supporting the hypothesis that, with a larger amount of data, the ViT model performs similarly to a CNN-based model
- The model benefits from supervised training and data augmentation, suggesting that pre-training could provide additional improvements.

The findings confirm that hybrid architectures exploit the inductive bias of CNNs to improve performance. Nonetheless, this inductive bias can become a drawback, since it relies on subject-specific training, which, when more data are available, imposes a performance ceiling. For an effective implementation of a BCI that performs well in general and fully exploits the architecture of a Vision Transformer, it is decided to pursue pre-training on general EEG signals, not

limited to motor imagery tasks, supported by the evidence from previous results, particularly those on supervised pre-training. The aim is to test the network’s ability to understand relationships between signals from different subjects and to pave the way for pre-training on large-scale EEG datasets covering a wide variety of brain activities. For this reason the curriculum masked spectrogram pretraining is then performed, a new framework proposed for the motor imagery task, on a dataset that is an extended version of 2b, including all EEG signals segmented by a temporal window. This allows the generation of more data, not limited to motor imagery, for pre-training purposes. In this scenario, the EEG-MVIT architecture is especially well-suited for pre-training, because its independent branches ensure that each channel operates separately from the others. Thus, each channel has its own masked patches, clarifying the objective of learning to detect and reconstruct patches for every single channel. Furthermore, the chosen signals include 3 channels, leading to a manageable number of branches in terms of computational cost, while keeping the network large enough and suitable for effective pre-training. The pre-training results are first compared using a LOSO validation on 2b against the standard EEG-MVIT model, showing an improvement in accuracy of approximately 2%. Subsequently, before proceeding with intra-subject validation, a supervised pre-training is performed on the self supervised pre-trained model and it is compared in a zero-shot setting with the same non-pre-trained model; the results indicate higher accuracy for the pre-trained model of 1.5%. The intra-subject training results on 2b confirm the findings obtained previously: thanks to pre-training, the model was able to understand the structure of the EEG signals by leveraging the large dataset provided from multiple subjects. Indeed, the self supervised pre-trained model, further pre-trained with multi-subject data, achieves approximately a 4% higher accuracy compared to the standard EEG-MVIT. These improvements are even more evident when examining the accuracies of individual subjects, which exceed 6% compared to the initial model, indicating that the model has learned to produce features that are general and can leverage the characteristics of EEG signals from multiple patients, not limited to the motor imagery task. As a final verification, it is decided to perform a test on the 2a dataset, which contains EEG signals different from those used in pre-training, with 22 channels reduced to the 3 considered. Even in this case, the pre-trained model achieves approximately 5% higher performance compared to the initial model, demonstrating the benefits of pre-training in generalization on a dataset that should be different from the one seen, while the model is still able to interpret the data, having improved its ability to understand EEG signals. In conclusion, using EEG signals from different subjects and activities beyond motor imagery, when combined with a pre-training framework like the one proposed, can improve the model’s performance, increasing from 78% to 81.7% on intra-subject, and from 73% to 75% on cross-subject. In this sense, this approach moves towards the development of general BCIs, addressing the problem of limited data for this task by leveraging EEG signals from any type of neurological activity of a subject. The results indeed show that, after pre-training, the model was able to generate more general features, effectively utilizing data from all patients.

The next step will be the implementation of pre-training on a significantly larger dataset than the one considered, increasing from approximately 60,000 samples to 600,000 samples contained in the TUH dataset [25], adapted for the EEG-MVIT under consideration. Employing this dataset may substantially enhance performance, but it is important to note that a dataset with a higher number of samples inevitably exhibits considerably more variability in the signals than the dataset used in the current study. Indeed, from the few epochs carried out during an implementation trial, it was observed that convergence takes much longer compared to the extended 2b dataset, in addition to the time required for a single epoch. However, starting from the self supervised pre-trained model presented rather than random weights, an increase in convergence speed was noticed. The results obtained in this thesis thus provide reason to expect a successful outcome from the pre-training procedure.

#### 4.5.1 Future improvements: Temple University Hospital dataset

The Temple University Hospital EEG corpus dataset (TUH) is a large collection of EEG signals of varying lengths, recorded in real clinical settings. It is divided into sections, namely TUAB, TUAR, TUEP, TUEV, TUSZ, and TUSL, as reported by [25], each representing specific aspects or clinical conditions. To facilitate a self-supervised pre-training, a variant of the TUH dataset was used, featuring a new structure with:

- a fixed number of channels including only the 20 most common channels according to the international 10-20 system notation;
- a uniform sampling frequency set at 250 Hz;
- initial pre-processing retaining frequencies in the [0.5, 80] Hz range using a 4th-order Butterworth band-pass filter to remove low-frequency components and non-physiological high frequencies;
- a notch filter applied to signals exhibiting a narrow, high peak at 50 or 60 Hz, representing a noise source;
- all signals segmented into 4-second windows to standardize them for EEG-based tasks.

In total, there are 69,652 recordings divided between healthy and epileptic patients, organized according to the type of EEG montage used: linked ears (LE), average reference (AR), and bipolar montage. Additionally, the files include the mean and standard deviation of the signals, allowing for channel-wise normalization. For future implementations, the data were adapted to fit the requirements of the model presented in this work: TUAB and TUEP signals from healthy patients with bipolar montage were selected, normalized per channel using the mean and standard deviation values provided in the dataset. Subsequently, the scalogram was computed for each file as described in **section 3.1**, considering only channels C3, Cz, and C4, and finally,

each scalogram was normalized. Each file contains over 10,000 samples, for a total of 58 files; consequently, the version to be used contains approximately 600,000 spectrograms for pre-training purposes.

Since the files are too large to be fully loaded into RAM, a structured approach is applied for pre-training: the full list of files is first shuffled, and mini-batches are then created from each selected file using the DataLoader class with chunks. Once a file is processed, it is closed and the next one is loaded, allowing for efficient randomization throughout. Several trials, as mentioned in the previous sections, were conducted, showing that the model takes longer to converge due to the high variability of the dataset. However, when using the pre-trained model discussed in this work, instead of initializing with random weights, the convergence speed appears to improve.

## 4.5.2 EEG web interface

A Streamlit web interface was developed to easily interact with the model in inference mode. This interface allows users to select the type of dataset to test, visualize the loaded signal through the different preprocessing stages, and perform the final inference. Moreover, the epoch inference mode enables the simulation of a real-world scenario, where the incoming brain signal is single.

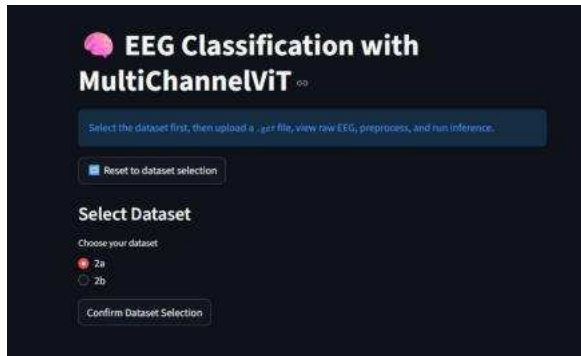


Figure 4.63: The first available option allows the user to choose between the two datasets.

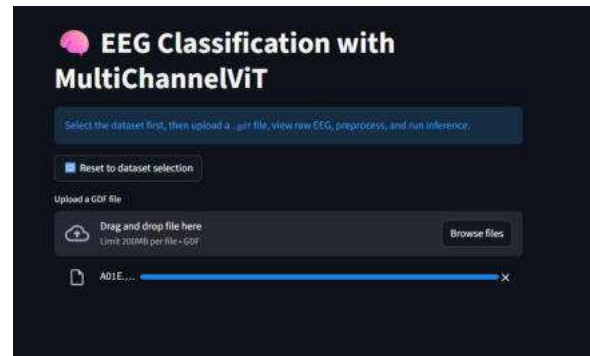


Figure 4.64: Once the dataset has been chosen, the user selects the .gdf file that holds the EEG signals. Here, the file corresponding to subject 1 from the BCI Competition IV 2a dataset is used.

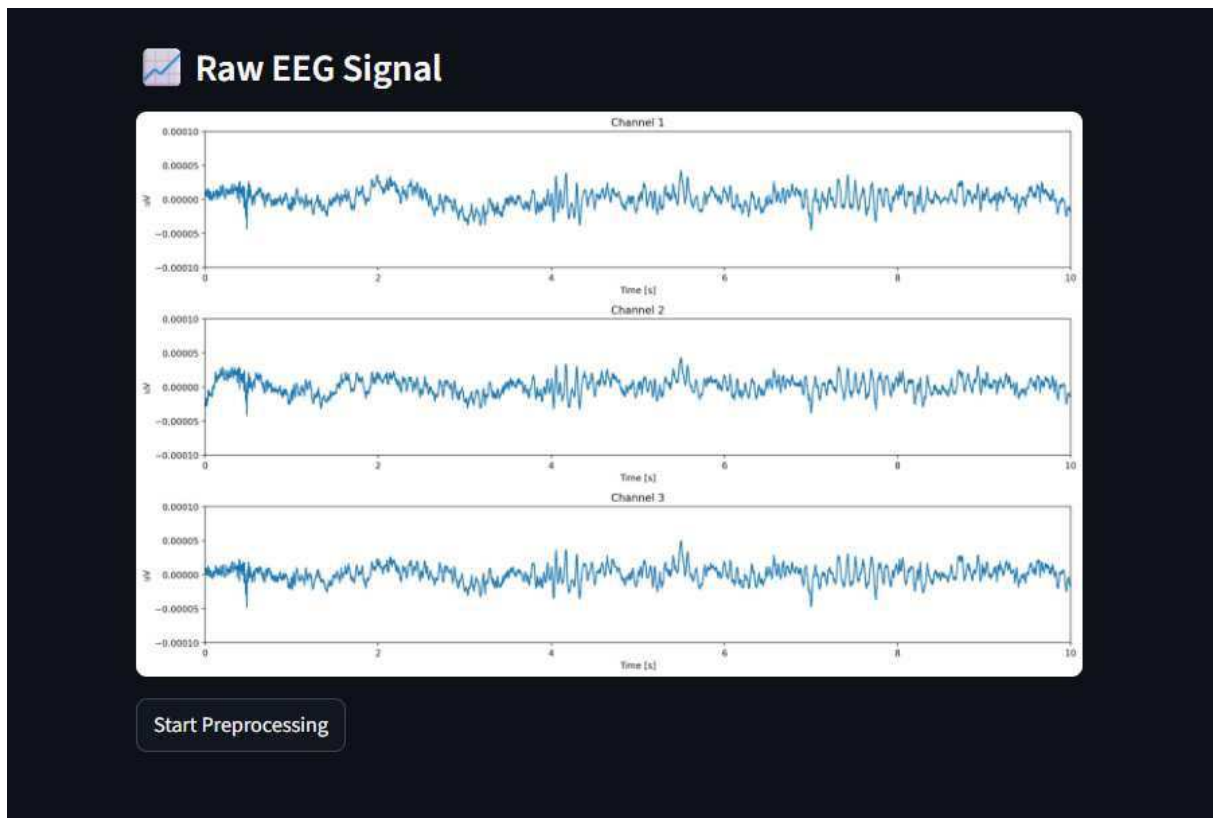


Figure 4.65: After uploading the file, the model loads automatically and the raw EEG signal is shown. The preprocessing pipeline can then be initiated by clicking the “Start Preprocessing” button.

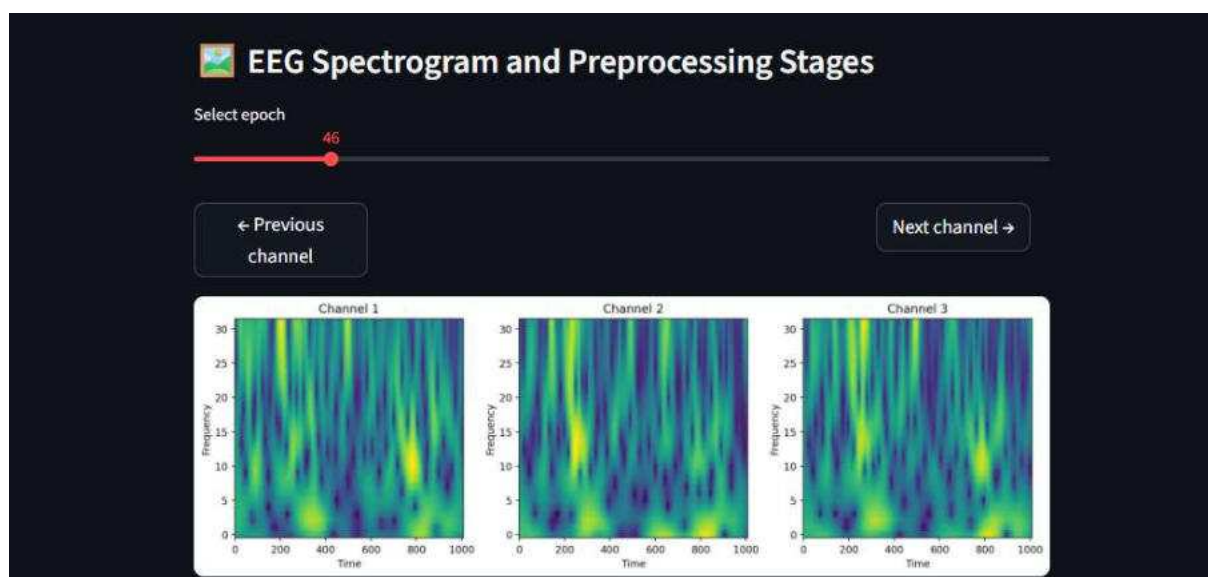


Figure 4.66: Once preprocessing is completed, the generated spectrograms are shown, and users can move between samples using a sidebar. For the 2a dataset, which includes more than three channels, navigation between channels is also available.



Figure 4.67: Users can also inspect the signal during the various preprocessing phases by pressing the appropriate buttons. In the figure, the signal is displayed after choosing the 4-second window corresponding to the MI task.

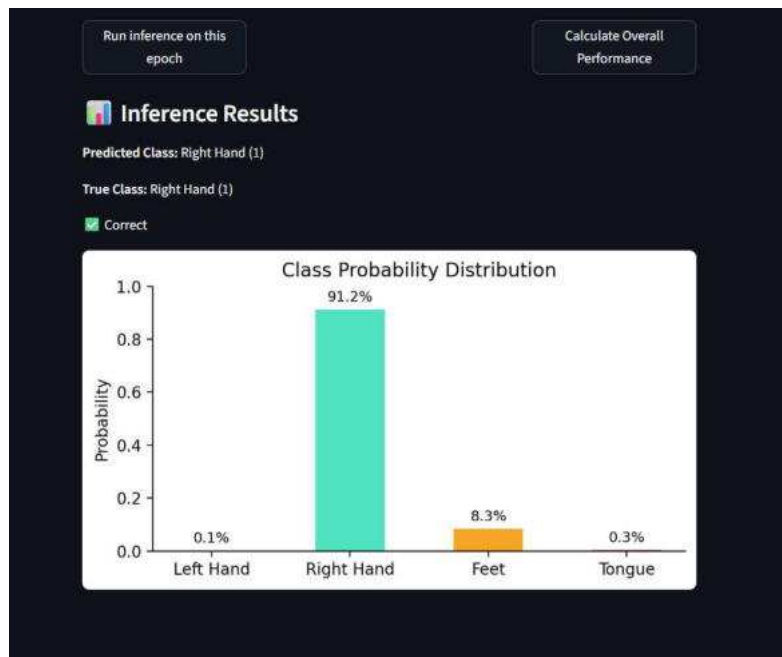


Figure 4.68: The “Run inference on this epoch” button enables users to display the classification output for an individual sample using a bar plot. By comparing it with the true label, one can check if the prediction would have been accurate in a real setting.



Figure 4.69: The “Calculate overall performance” button allows, in the presence of multiple signals, to compute the mean accuracy and visualize it using a confusion matrix, a bar graph, or a pie chart displaying the frequency with which the model predicts each class.

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